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In-Cd-Ga enrichment in sphalerite from the Gaocheng skarn Pb-Zn-Ag deposit, Yunkai Domain, South China

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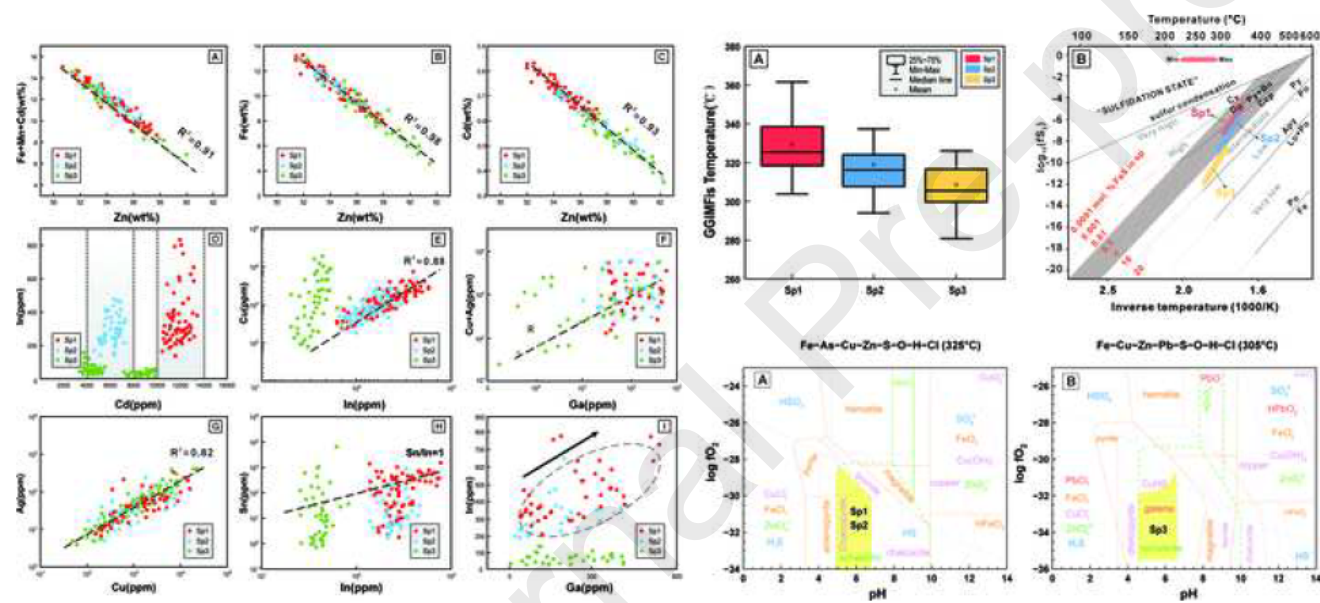
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Highlights

- The Gaocheng deposit represents a type of Pb-Zn-Ag skarn deposit in the Yunkai Domain of South China.
- The sphalerite in the Gaocheng deposit contains highly enriched key metallic elements, such as In, Ga, Cd and so on.
- Trace element analysis and thermodynamic calculations revealed an acidic metallogenic fluid that gradually decreased in temperature from high to medium, along with a reduction in sulfur and oxygen fugacity.
- The Gaocheng sphalerite is a novel type of high In, Ga and Cd sphalerite, originating from a high-temperature skarn deposit.

Abstract

Indium (In), Gallium (Ga), and Cadmium (Cd) are strategic critical metals with widespread applications in high-tech industries. The growing demand and global supply limitations have led to widespread attention on the genesis and exploration of deposits rich in In, Ga, and Cd. The

Yunkai Domain (Cathaysia Block, South China) contains numerous Zn-Pb polymetallic deposits, but the potential of In, Ga, and Cd in these deposits has been insufficiently studied. To fill this gap, we selected the Gaocheng deposit, a representative skarn-type Zn-Pb-Ag deposit, to investigate the content, spatial distribution, and incorporation behavior of In, Ga, and Cd in sphalerite, which serves as their major mineral host. Field surveys and microscopic observations revealed that the Gaocheng is featured with a variety of skarn minerals such as diopside, epidote, and garnet. Three generations of sphalerite have been identified with distinct mineral assemblages and colors. Electron probe microanalysis (EPMA) and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) analyses revealed that each sphalerite has distinct compositions, in particular In, Ga, and Cd concentrations. In Sp1 (n=56), In (mean = 414 ppm), Ga (mean = 177 ppm), and Cd (mean = 11517 ppm) are the most enriched. The concentrations in the subsequent two generations of sphalerite show a decreasing trend, with Sp2 (n=52) having In (mean = 271 ppm), Ga (mean = 119 ppm), and Cd (mean = 6196 ppm), and Sp3 (n=49) having In (mean = 48 ppm), Ga (mean = 73 ppm), and Cd (mean = 5732 ppm).

Binary diagrams indicate that the substitution mechanism of In into the sphalerite lattice is: Cu^+ (or Ag^+) + $\text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, and Ga follows a similar mechanism: Cu^+ (or Ag^+) + $\text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, while the substitution mechanism for Cd is: $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$. It is noticeable that the effect of the indium window, the highly enriched In, correlating with the specific Cd concentrations (4000-8000ppm, 10000-14000ppm), is observed in the

Gaocheng sphalerite. Then, we use the GGIMFis geothermometer to calculate the crystallization temperatures of three sphalerite generations that exhibit a clear decreasing trend from 331°C (mean) for Sp1 through 320°C for Sp2, to 309°C for Sp3. Based on temperature and thermodynamic calculations, the sulfur fugacity (f_{S_2}) shows a gradual decreasing trend, with Sp1 (-8 to -5), Sp2 (-10 to -6), and Sp3 (-13 to -9). Meanwhile, the oxygen fugacity (f_{O_2}) of Sp1 and Sp2 in the earlier stages ranges from -34 to -28.5, with pH values between 4.9 and 6.7, while in the later stages, the oxygen fugacity (f_{O_2}) of Sp3 ranges from -36 to -31.5, with pH values between 4.6 and 6.6. Collectively, we conclude that high temperature, elevated sulfur fugacity, higher oxygen fugacity, and weakly acidic physicochemical conditions are favorable for the incorporation of In, Ga, and Cd into sphalerite, with temperature and sulfur fugacity playing a more dominant role. This novel finding helps establish key exploration indicators for skarn Zn-Pb deposits in the future, enhancing our understanding of the enrichment of In, Ga, and Cd elements.

Keywords: Sphalerite; Indium; Gallium; Cadmium; Enrichment mechanism; Pb-Zn-Ag deposit; Skarn; Yunkai Domain; Gaocheng

1 Introduction

Indium (In), gallium (Ga), and cadmium (Cd) are important, rare, and scattered critical metals. Among them, In and Ga are widely used in

high-tech fields such as semiconductors, solar cells, light-emitting diodes (LEDs), and touchscreens (Schwarz-Schampera and Herzig, 2002; Zheng et al., 2023). Although Cd was previously defined as a well-known heavy metal, causing serious environmental pollution and human health hazards (Ye et al., 2012), recent studies have revealed that Cd plays an irreplaceable role in high-tech industries such as alloys, electroplating, and nuclear industries (Zhu et al., 2013). Against the backdrop of rapid global technological development, the increasing demand for In, Ga, and Cd, coupled with the limited global supply, has attracted widespread attention. Among them, In and Ga have been listed as some of the most important strategic metals globally (Frenzel et al., 2017; Werner et al., 2017). In nature, In, Ga, and Cd are dispersed (rarely forming independent minerals), scarce, and have a low abundance in the Earth's crust (Werner et al., 2017; Zheng et al., 2023). All three are primarily enriched in various lead-zinc deposits, including skarn-type, volcanic-hosted massive sulfide (VMS), hydrothermal, porphyry, and sedimentary-exhalative (SEDEX) deposits (Schwarz-Schampera and Herzig, 2002). In these deposits, sphalerite is the main host, with the indium content accounting for approximately 95% of global indium production (Werner et al., 2017). Therefore, studying the trace element characteristics in sphalerite (ZnS) is essential, as it helps in understanding their spatiotemporal distribution, enrichment mechanisms, and genesis. (Cook et al., 2012).

In related studies, Seifert and Sandmann (2006) identified an "indium stage" of indium enrichment in the Zn-Sn-Cu mineral assemblage of the Erzgebirge deposit in Germany. Dill et al. (2013) discovered an "indium window," where In precipitates in large quantities under specific

conditions, with Cd content ranging from 0.2% to 0.6% in sphalerite. Xu et al. (2021a) confirmed significant In enrichment during a brief period in the entire ore-forming stage at the Dulong deposit in China. In studies on Ga, Wu et al. (2024) proposed that Ga is enriched in light-colored sphalerite crystallized in the later stages at lower temperatures, while Moller (1987) suggested that Ga is enriched in high-temperature stages. Meanwhile, Meng et al. (2024) argued that Ga may accumulate at any stage of mineralization, without a clear enrichment pattern. In the latest studies on Cd, Shu et al. (2024) found that low-temperature hydrothermal fluids in an oxidizing environment promoted cadmium enrichment in the Kangjiawan deposit, Wang et al. (2024) observed that high-temperature sphalerite crystallized in the early stages of the Yiliu deposit was richer in Cd, and Chen et al. (2024) discovered that early crystallized high-temperature sphalerite in the Pangxidong deposit was also richer in Cd. Overall, studies on the enrichment of In, Ga, and Cd are filled with complex and variable cases, some of which are controversial or lack universal applicability. Many key issues remain unresolved, particularly the lack of sufficient exploration of the physicochemical conditions that control the high enrichment of these three elements during mineralization (Frenzel, 2022). Therefore, further case studies are needed to deepen the understanding of the complex enrichment patterns of these three elements.

This study focuses on the Gaocheng Pb-Zn-Ag deposit, located in the Yunkai Domain of South China. The region has undergone multiple tectonic-magmatic events, including the Caledonian, Hercynian-Indosinian, and Yanshanian periods, resulting in widespread skarn-related

mineralization (Mao et al., 2013). Previous studies have revealed significant reserves of Pb, Zn, Ag, and Sn in the Gaocheng deposit, making it a prime target for investigating the metallogensis of In, Ga, and Cd in skarn systems (Huang et al., 2021). The deposit features typical skarn mineral assemblages, such as garnet, diopside, and epidote, and is characterized by complex hydrothermal mineralization processes. However, systematic research on the distribution of In, Ga, and Cd, their enrichment mechanisms, and the physicochemical conditions controlling their enrichment at different ore-forming stages has not yet been conducted.

To address this research gap, this study combines field geological surveys, petrographic and geochemical analyses, along with advanced techniques such as EPMA, LA-ICP-MS, and thermodynamic modeling methods, to investigate sphalerite from the Gaocheng deposit. This study provides the trace element concentrations in three generations of sphalerite, describes the mechanisms and distribution characteristics of these elements, and elucidates the physicochemical conditions controlling the enrichment of In, Ga, and Cd. This study, by combining these results with the geological framework of the Gaocheng deposit, aims to refine the enrichment and distribution characteristics of In, Ga, and Cd in skarn-type deposits and provide new insights for the exploration of similar polymetallic deposits worldwide.

2 Regional geology

The Gaocheng lead-zinc-silver deposit is located at the north-central margin of the Yunkai Domain within the South China Plate (Fig. 1A,1B),

in the rotational structure where the Wuchuan-Sihui fault zone intersects with the Songgui fault zone, and at the intersection of the Daganshan rotational structure and the Gaocheng fault zone (Wu et al., 2012). The South China Plate was formed by the tectonic amalgamation of the Cathaysia and Yangtze blocks during the Paleozoic, and subsequently experienced at least three tectonic thermal events during the Paleozoic, Triassic, and Jurassic-Cretaceous periods, resulting in complex tectonic evolution in the region (Wang et al., 2011; Zheng et al., 2015; Zheng et al., 2022a). The Yunkai Domain is located in the southern segment of the Qin-Hang metallogenic belt, the tectonic junction between the Cathaysia and Yangtze plates. It has undergone multiple phases of tectonic-magmatic activities, including deformation, metamorphism, and mineralization (Yu et al., 2018; Huang et al., 2021; Chen et al., 2024).

The Yunkai Domain is composed of an ancient metamorphic basement and a sedimentary cover (Fig. 1B; Wang et al., 2007a, 2007b). The basement in this area consists of a Proterozoic to Early Paleozoic metamorphic basement, while the cover includes a series of shallow marine sedimentary strata from the Late Paleozoic to the Middle Triassic and terrestrial basin sedimentary strata from the Mesozoic to Cenozoic (Yuan et al., 2015; Zheng et al., 2015a; Zhang et al., 2018a). The sedimentary cover has been influenced by multiple phases of tectonic deformation, magmatic activities, and metamorphism (Zheng et al., 2015b; Liu et al., 2018). The Yunkai area has undergone multiple phases of magmatic activity, including the Caledonian, Hercynian-Indosinian, and Yanshanian periods, making it one of the most tectonically and magmatically active

regions of the South China Plate, with extensive development of igneous rocks (Chen et al., 2012). Granites from the Caledonian period are widely exposed in the area, with some developing gneissic structures and migmatization due to high-grade metamorphism, resulting in banded migmatites, orbicular-banded granitic gneiss, and gneissic granite (Shu et al., 2014). During the Hercynian-Indosinian period, biotite monzonitic granite was mainly formed, often intruding sedimentary strata along fault zones (Chen et al., 2012), controlled by the Bobai-Cenxi fault zone and the Lingshan-Tengxian fault zone in the region. Magmatic activity during the Yanshanian period led to the development of intermediate-felsic igneous rocks, with strong magmatic activity and a diagenetic age of mainly 88~113 Ma (Zhang et al., 2018a, 2018b). These rocks are primarily distributed along NE-trending fault zones, mostly in the southwestern margin of the Yunkai area, and many mineralization processes in the region are associated with this magmatic activity (Cai et al., 2002; Peng et al., 2006).

The Yunkai Domain is characterized by widespread fold and fault development. The Bobai-Cenxi fault zone is one of the major fault zones in the area, controlling the distribution of Mesozoic sedimentary clastic rocks and the development of faulted basins (Fig. 1B; Qin et al., 2018; Xiang et al., 2020). Additionally, ductile shear zones are well developed in the region, such as the Lucuan-Cenxi fault zone within the Bobai-Cenxi belt, which exhibits strong ductile deformation (Zhao et al., 2012; Wu et al., 2013; Zhang et al., 2018a). The Wuchuan-Sihui fault zone is one of the key ore-controlling and ore-guiding structural zones in the region, featuring brittle-ductile fault deformation and intense magmatic

activity along the fault zone (Yuan et al., 2015). The Yunkai Domain is influenced by the interaction of the Yangtze, Cathaysia, and Indosinian blocks, controlled by both the Pacific and Tethyan tectonic domains (Mao et al., 2011; Lin et al., 2015). It has undergone multiple stages of tectonic and magmatic evolution, resulting in the formation of various types of ore deposits, including volcanogenic sedimentary, orogenic hydrothermal, and Yanshanian magmatic-hydrothermal systems. These deposits are predominantly enriched in Pb, Zn, Ag, and Au, and are spatially distributed along major fault zones, exhibiting well-defined metallogenic zoning (Liu et al., 2018). Among them, skarn-type Pb-Zn-Ag deposits formed during the Yanshanian period are the most representative, typically occurring at the contact zones between intermediate-felsic intrusions and carbonate rocks, and reflecting a classic model of magmatic fluid–host rock metasomatism (Zheng et al., 2015). The Gaocheng deposit was formed under such a geological setting. It is structurally controlled and hosted in Devonian carbonate rocks. Although no intrusive bodies are exposed at the surface, the deposit shows strong indications of influence from concealed Yanshanian magmatism. It features garnet-, diopside-rich skarn assemblages and a typical precipitation sequence of oxides, sulfides, and carbonates. The trace element characteristics of sphalerite are consistent with those of other regional skarn-type deposits, indicating a shared mineralization mechanism driven by fluid–rock interaction (Zhuang et al., 2018a). As one of the largest Pb-Zn-Ag deposits in the Yunkai Domain, Gaocheng is representative in terms of genesis, mineralization evolution, and geochemical characteristics, and serves as a key example for understanding the regional polymetallic metallogenic system.

3 Deposit geology

The Gaocheng deposit is located on the northwest side of the Wuchuan-Sihui major fault trending northeast-east, within the secondary northwest-trending Gaocheng fault zone (Fig. 1B; Qin et al., 2018). A total of 8 industrial orebodies have been delineated, among which V2, V5, and V7 are the main orebodies (Fig. 2A). The deposit's proven reserves are: Ag (1516 tons, with an average grade of 140.96×10^{-6}); Pb (173,829 tons, with an average grade of 1.72%); Zn (376,617 tons, with an average grade of 3.06%); Sn (16,161 tons, with an average grade of 0.18%) (Wu et al., 2012). It is one of the major polymetallic enrichment zones in the South China Plate, and the region still holds great potential for further mineral exploration.

3.1 Host rocks

The exposed strata in the mining area are relatively simple (Fig. 2), ranging from the Sinian to the Triassic, Cretaceous, and Quaternary periods, with the ore-bearing layer belonging to the Cretaceous Luoding Formation (Shu et al., 2014). The Sinian strata are mainly distributed in the western part of the mining area, with a small exposure area, and are characterized by turbidite-like deposits from the Daganshan Formation, with lithologies including mica quartz schist, calcareous quartzite, and argillaceous crystalline limestone (Chen et al., 2012; Zheng et al., 2015b). The Triassic Xiaoyunwushan Formation has a small exposure area within the mining area, mostly distributed in the southern part, near Dakeng

and Datian-Huangwu, with lithologies primarily consisting of shale and quartz sandstone (Peng et al., 2006). The Cretaceous Luoding Formation is distributed in the western part of the mining area, with lithologies comprising red sandstone, conglomerate, and other similar rocks. Quaternary loose sediments are widely distributed in the mining area and are mainly composed of sand, gravel, and other loose deposits. (Zhao et al., 2012; Wu et al., 2013)

3.2 Ore-controlled structures

The mining area exhibits well-developed fault structures, consisting of two sets of fault zones trending NWW and NE (Fig. 2), with the NWW-trending faults being dominant and the NE-trending ones secondary (Zhao et al., 2021). The NWW-trending extensional shear faults are located within the Gaocheng fault zone, developed in Caledonian biotite granite, including the F2-F9 faults, which are arranged in a nearly parallel oblique manner, with an overall trend of 290°, dipping SSW at angles of 60° to 80°, serving as the primary ore-bearing and controlling structural zone that hosts most of the orebodies (Fig. 2; Wu et al., 2012; Zhao et al., 2012). The fault zone is located near Datian-Bai Mei, with a width of approximately 2 km and a length of about 4.8 km. The migration of mineralizing fluids and the mineralization process in the mining area are primarily controlled by this set of fault zones (Qin et al., 2018). The secondary ore-bearing structural zone in the mining area is situated within the NE-trending faults, which have a lower control degree compared to the NWW-trending faults, mainly represented by the F7 fault, with an overall

trend of 30° to 45° and dipping SE at angles of 60° to 75° (Zhao et al., 2012). The NWW and NE-trending fault zones intersect within the mining area, showing no significant cutting or displacement, suggesting they may have formed simultaneously (Zhao et al., 2021).

3.3 Magmatism

The mining area exhibits significant magmatic activity, with the presence of Caledonian gneissic granite, biotite granite, potassium feldspar granite, and Yanshanian dikes (Fig. 2). The distribution of orebodies is closely related to the magmatic rocks. Biotite granite is widely distributed in the mining area, with most orebodies developed within this type of granite. The main minerals of biotite granite include potassium feldspar (30%–40%), quartz (25%–30%), plagioclase (20%–25%), and biotite (~10%), with accessory minerals such as zircon and localized alteration phenomena including chloritization and sericitization (Zhao et al., 2012). The Caledonian gneissic granite is found in the northern part of the mining area, which has undergone metamorphism and mixing and is primarily located in the Liukuakeng-Xinwu area. Potassium feldspar granite mainly occurs in the central part of the mining area, with a small number of orebodies also developed within it. A variety of Yanshanian dikes, including quartz porphyry and granitic porphyry, are exposed in the mining area, and the formation of these dikes is related to mineralization. The NE- and NW-trending quartz porphyry dikes, intruding during the Late Cretaceous (Zhao et al., 2021), penetrate the Luoding Formation, where disseminated pyrite is developed.

3.4 Features of alteration and mineralization

The main types of wall rock alteration in the mining area include silicification, chloritization, sericitization, carbonatization, and feldspathization (Fig. 3). The alteration exhibits zonation due to the variation in the wall rocks, with alteration centered around the ore veins and symmetrically distributed in bands on both sides. The alteration is most intense at the center, where silicification predominates, and gradually weakens towards the periphery. The width of the alteration zones varies considerably. The alteration zones do not have uniform boundaries and often overlap and coexist. Silicification is the most common alteration in wall rocks, typically developed in the fracture zones of biotite granite, and is closely associated with mineralization. Chloritization commonly occurs in skarn and granite, while feldspar in granite is often sericitized. Carbonatization typically results in the formation of calcite veins (Fig. 3B, 3D). The orebodies of the Gaocheng deposit are primarily vein-type and disseminated, with local development of stockwork mineralization. Vein-type orebodies typically range from 1 to 3 meters in thickness and are distributed along major NWW-trending faults (e.g., F5 and F6), dipping toward the SSW, indicating that their occurrence is controlled by regional tensional-shear fault systems. Disseminated mineralization is mainly hosted within skarn zones and silicified alteration zones, where it occurs in stratiform or lenticular forms, showing a close spatial association with alteration halos. (Fig. 3) The main ore minerals include pyrite, sphalerite, chalcopyrite, magnetite, and galena, with secondary minerals such as arsenopyrite, cassiterite, and rhodochrosite (Fig. 3, 4, 5). The

gangue minerals mainly include quartz, plagioclase, potassium feldspar, muscovite, calcite, fluorite, and sericite, while skarn minerals include garnet, chlorite, and diopside (Figs. 3, 4, 5).

4 Sampling and analytical methodology

4.1 Sampling

A total of 33 ore samples were collected from the V2, V5, and V7 orebodies of the Gaocheng Pb-Zn-Ag deposit, within the 200 m to 600 m mining levels. These samples, obtained from different burial depths, include high-grade ores, fresh host rocks, and hydrothermally altered host rocks. They were prepared into 61 thin sections for further mineralogical studies and observations. Through microscopy and electron microscopy, we delineated the mineral formation sequences. We further selected 44 representative thin sections for electron probe microanalysis (EPMA) and laser ablation-inductively coupled plasma mass spectrometry (LA-ICP-MS) testing.

4.2 Electron probe microanalysis

The EPMA compositional data for sphalerite were analyzed using a JEOL JXA-iSPA00 electron probe micro-analyzer at Guangzhou Tuoyan Analytical Technology Co., Ltd., China. Quantitative analyses were conducted under the condition of an accelerating voltage of 20 kV, a probe current of 20 nA, and a beam size of 5 μm . Peak counting times were 10 seconds for elements S, Zn, and Fe, and 20 or 50 seconds for Mn, Cd,

Ga, Ge, In, Cu, Ag, and Sn. Background counting times were set to half of the respective peak counting times for both high- and low-energy background positions. The ZAF correction method was applied to refine the analytical results. Calibration standards included sphalerite for Zn, pyrite for Fe and S, chalcopyrite for Cu, greenockite for Cd, manganese for Mn, germanium for Ge, indium phosphide for In, tin for Sn, and GaAs alloy for Ga.

4.3 Laser ablation inductively coupled plasma mass spectrometry

Trace element analysis of sphalerite was conducted using LA-ICP-MS at Guangzhou Tuoyan Analytical Technology Co., Ltd., employing an NWR 193 nm ArF Excimer laser-ablation system and an iCAP RQ ICP-MS. The ICP-MS was calibrated with NIST 610 standard glass to minimize oxide formation. A helium carrier gas flow of 0.7 L/min was introduced, followed by the addition of 0.89 L/min Ar make-up gas. The laser operated with a fluence of 3.5 J/cm² and a spot size of 30 µm, performing a 45-second analysis followed by a 40-second background measurement. The sphalerite analysis included 15 elements: ³⁴S, ⁵⁵Mn, ⁵⁷Fe, ⁶⁵Cu, ⁶⁶Zn, ⁷¹Ga, ⁷²Ge, ⁷⁵As, ¹⁰⁷Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ²⁰²Hg and ²⁰⁸Pb. The trace element compositions of sphalerite were calibrated using NIST SRM610 and MASS-1 reference materials, without an internal standard (Liu et al., 2008). To assess data quality, sulfide pellet standards SRM 612 and Ge8 were analyzed as unknown samples (Wilson et al., 2002). Raw isotope data were processed using the 3D Trace Element scheme in IOLITE software (Paton et al., 2011). Detailed procedures for the LA-ICP-MS

analytical techniques used in this study are provided by Liu et al. (2008) and Chu et al. (2023).

4.4 Thermodynamic calculations based on CHNOSZ

We employed the CHNOSZ R package to construct logfO₂-pH diagrams for potential ore-forming minerals, a widely recognized tool in geochemistry for conducting thermodynamic calculations (Dick, 2019). This open-source package is equipped with the OBIGT thermodynamic database and integrates well-established mineral and aqueous species models, including those by Berman and Brown (1985) and the updated equations from Helgeson, Kirkham, and Flowers (1981). With these advanced features, we were able to conduct a thorough investigation of the physicochemical conditions in ore-forming fluids.

To focus our study, based on petrographic results, we selected mineral assemblages from both the early and middle sulfide stages, reflecting critical phases of ore formation. For the early sulfide stage, we defined the mineral system as Fe-As-Cu-Zn-S-O-H-Cl, while for the middle sulfide stage, we used the Fe-Cu-Zn-Pb-S-O-H-Cl system. In terms of calculation parameters, the temperature was set within a range of 300°C to 325°C, consistent with the conditions of the skarn system we are studying. The pressure was maintained at 1,000 bar, with a sulfur concentration (ΣS) of 0.1 m and NaCl concentration fixed at 1 M. pH was varied from 0 to 14 to capture the full range of fluid conditions. And the logfO₂ range was determined by combining previous research findings (Zhao et al., 2021).

5 Result

5.1 Paragenetic sequence

The ore minerals at the Gaocheng mining area include sphalerite, galena, pyrrhotite, arsenopyrite, chalcopyrite, pyrite, and minor amounts of magnetite and cassiterite. The gangue minerals are mainly composed of skarn minerals, including garnet, diopside, tremolite, and chlorite, along with quartz, calcite, and fluorite (Fig. 3, 4, 5). Based on ore textures, mineral assemblages, cross-cutting and replacement relationships, we divided the ore-forming stages of this deposit into two periods and five stages: the skarn period and the sulfide period (Fig. 6). The skarn period is further subdivided into an earlier dry skarn stage, characterized by abundant garnet and diopside, and a later wet skarn stage, composed of tremolite and chlorite. The subsequent oxide stage consists of small amounts of magnetite and quartz. The early sulfide stage is characterized by pyrite, arsenopyrite, and pyrrhotite, while the late sulfide stage is dominated by sphalerite, galena, and chalcopyrite (Fig. 5, 6).

Three generations of sphalerite with distinct morphologies and mineral associations were identified: (1) the earliest generation, Sp1, appears black under transmitted light, replaces pyrrhotite, and also contains chalcopyrite inclusions. Additionally, chalcopyrite precipitates in the solid solution, forming a solid solution separation structure in sphalerite (Fig. 4, 5A, 5B, 5C); (2) Sp2 is abundant in the mining area, with a core of dark brown and an outer rim of dark red, containing chalcopyrite inclusions (Fig. 5D, 5E, 5F); (3) the latest generation Sp3, which is pure, inclusion-

free, and has a well-preserved crystal form with a bright red color (Fig. 5J, 5K, 5L)

5.2 EPMA data of sphalerite

A total of 111 points were selected for the analysis of major elements in sphalerite, with 41 points from Sp1, 38 points from Sp2, and 32 points from Sp3. The data show that the major element contents in the three different types of sphalerite exhibit minimal differences. Among the tested elements, S, Fe, and Zn have good detection results, with concentrations greater than 1%. Some samples have Mn concentrations above 1%, while Cd, In, Sn, and Cu contents are all less than 1% (Table 1).

The S content in Sp1 (mean = 33.06 wt.%) ranges from 32.36% to 33.56%; Zn (mean = 53.75 wt.%) ranges from 51.31% to 56.86%; Fe (mean = 11.29 wt.%) ranges from 8.54% to 12.63%; Mn (mean = 0.56 wt.%) ranges from 0.10% to 1.13%; Cd (mean = 0.60 wt.%) ranges from 0.51% to 0.81%.

The S content in Sp2 (mean = 33.13 wt.%) ranges from 32.55% to 33.57%; Zn (mean = 54.13 wt.%) ranges from 51.69% to 61.30%; Fe (mean = 10.95 wt.%) ranges from 4.40% to 12.78%; Mn (mean = 0.64 wt.%) ranges from 0.11% to 1.97%; Cd (mean = 0.52 wt.%) ranges from 0.36% to 0.71%.

The S content in Sp3 (mean = 33.14 wt.%) ranges from 32.26% to 33.49%; Zn (mean = 53.95 wt.%) ranges from 50.26% to 61.76%; Fe

(mean = 10.59 wt.%) ranges from 3.58% to 13.19%; Mn (mean = 1.12 wt.%) ranges from 0.12% to 2.47%. Cd (mean = 0.46 wt.%) ranges from 0.33% to 0.64%.

In three generations of sphalerite, there is no significant difference in sulfur (S) content, with a relatively small range of variation, but the range of zinc (Zn) varies significantly, and the average content of iron (Fe) is as high as 10wt.%. In addition, the average content of cadmium (Cd) gradually decreases in Sp1(0.60 wt.%), Sp2(0.52 wt.%), and Sp3(0.46 wt.%).

5.3 LA-ICP-MS data of sphalerite

A total of 157 points were selected for trace element testing of sphalerite by LA-ICP-MS to determine the concentrations of selected elements in sphalerite. Among these, 56 points were selected for Sp1, 52 points for Sp2, and 49 points for Sp3. The corresponding data are listed in Table 2, and the statistical box-and-whisker plots are presented in Figure 7. The trace element data results (see Table 2) show that the concentrations of Ge, As, Sb, and Tl in sphalerite are relatively low, with the majority being below the detection limit, while the test results for the other elements are satisfactory. Besides, to visually identify the occurrences of trace elements, the representative time-resolved depth profile corresponding to three sphalerite generations is displayed in Figure 8.

5.3.1 Sp1 compositions

In the laser ablation analysis, only a limited number of elemental signals can be detected, including Zn, Mn, In, Sn, Cu, Ga, and Ag (see Fig. 7). The signal lines of these elements are parallel to the Zn signal and show a relatively flat pattern, indicating lattice substitution within the sphalerite structure.

Manganese (Mn) concentrations are relatively high, ranging from 1826 to 20881 ppm, with an average of 8821 ppm. Iron (Fe) concentrations are relatively concentrated, ranging from 90415 to 221151 ppm, with an average of 115227 ppm. Cadmium (Cd) concentrations are very concentrated, ranging from 8959 to 13234 ppm, with an average of 11517 ppm. Other important trace elements include: Indium (In) (average: 414 ppm, range: 224 to 832 ppm), Gallium (Ga) (average: 177 ppm, range: 25 to 521 ppm), Silver (Ag) (average: 81 ppm, range: 7 to 962 ppm), Copper (Cu) (average: 2192 ppm, range: 201 to 7088 ppm), and Stannum (Sn) (average: 279 ppm, range: 12 to 1693 ppm).

5.3.2 Sp2 compositions

In the laser ablation analysis, only a limited number of elemental signals can be detected, including Zn, Mn, In, Sn, Cu, Ga, and Ag (see Fig. 7). The signal lines of elements such as Mn, Fe, In, and Ga are parallel to the Zn signal, showing relatively flat profiles, indicating lattice substitution within the sphalerite structure. Elements like Sn, Cu, and Ag show lower and more fluctuating signals, suggesting their presence as minor inclusions.

Manganese (Mn) concentrations are relatively high, ranging from 5596 to 11757 ppm, with an average of 8087 ppm. Iron (Fe) concentrations are relatively concentrated, ranging from 91598 to 176969 ppm, with an average of 106910 ppm. Cadmium (Cd) concentrations are also very concentrated, ranging from 3811 to 7525 ppm, with an average of 6196 ppm. Other important trace elements include: Indium (In) (average: 271 ppm, range: 140 to 662 ppm), Gallium (Ga) (average: 119 ppm, range: 12 to 441 ppm), Silver (Ag) (average: 41 ppm, range: 4 to 241 ppm), Copper (Cu) (average: 1846 ppm, range: 417 to 9177 ppm), and Stannum (Sn) (average: 110 ppm, range: 2 to 882 ppm).

5.3.3 Sp³ compositions

In the laser ablation analysis, only a limited number of elemental signals can be detected, including Zn, Mn, Fe, Cd, and Ag (see Fig. 7). The signal lines of these elements are parallel to the Zn signal, with relatively flat profiles, indicating lattice substitution within the sphalerite structure.

Manganese (Mn) concentrations are relatively high, ranging from 5247 to 40371 ppm, with an average of 19278 ppm. Iron (Fe) concentrations are relatively concentrated, ranging from 70610 to 124707 ppm, with an average of 98814 ppm. Cadmium (Cd) concentrations are also very concentrated, ranging from 3380 to 9443 ppm, with an average of 5732 ppm. Other important trace elements include: Indium (In) (average: 48 ppm, range: 6 to 177 ppm), Gallium (Ga) (average: 73 ppm, range: 1 to 374 ppm), Silver (Ag) (average: 169 ppm, range: 9 to 1313 ppm), Copper (Cu) (average: 2465 ppm, range: 81 to 9777 ppm), and Stannum (Sn) (average: 60 ppm, range: 1 to 476 ppm).

6 Discussion

6.1 Occurrence and substitution mechanism of trace elements in sphalerite

Sphalerite hosts a variety of trace elements such as Fe, Mn, Cd, Cu, In, Ga, and Ge, and these elements occur in three main forms: (1) individual micro-mineral inclusions, (2) direct stoichiometric substitution, and (3) coupled stoichiometric substitution into the lattice. Individual micro-mineral inclusions can be directly observed under high-resolution optical or electron microscopes (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; Bonnet et al., 2016; Zhuang et al., 2019; Yang et al., 2022). When nano/micro-mineral inclusions are present in sphalerite, time-resolved depth profiles obtained by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) exhibit significant heterogeneity, with the relevant elements displaying fluctuating signal curves (Cook et al., 2009; Cugerone et al., 2018; Frenzel et al., 2016; Ye et al., 2011). Based on this criterion, we classify the trace elements in the Gaocheng sphalerite into two categories. The first group includes Mn, Cd, Fe, In, and Ga, which substitute for some of the original atoms in the sphalerite crystal lattice. In contrast, the second group includes Cu, Ag, and Sn, which occur both as micro-mineral inclusions and as lattice substitutions (Fig. 8). These findings align with our micropetrographic observations, which revealed widespread micro-mineral inclusions of chalcopyrite, tetrahedrite, and argentite in the sphalerite of the Gaocheng deposit.

It is widely recognized that divalent cations such as Fe^{2+} , Mn^{2+} , and Cd^{2+} , with ionic radii similar to that of Zn^{2+} , can directly substitute into

the sphalerite lattice, with substitution concentrations reaching several weight percent (Cook et al., 2009; Lockington et al., 2014). The data in Fig. 9a clearly show a significant negative correlation between the concentrations of Fe, Mn, and Cd and Zn, supporting the idea that elements with ionic radii similar to Zn^{2+} can effectively replace Zn in the sphalerite lattice. In the three generations, the substitution mechanisms between Fe, Mn, Cd, and Zn show strong correlations, such as ($\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$, $R^2 = 0.98$) and ($\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$, $R^2 = 0.93$), indicating that these are the primary substitution reactions in sphalerite. Additionally, another significant substitution reaction involving multiple elements is expressed as $(n + m + p) \text{Zn}^{2+} \leftrightarrow n\text{Fe}^{2+} + m\text{Cd}^{2+} + p\text{Mn}^{2+}$ (Bonnet et al., 2016; Chen et al., 2024), with a correlation of $R^2 = 0.91$ (Fig. 9A, 9B, 9C). Therefore, Fe, Mn, and Cd incorporate into the sphalerite lattice through direct substitution, with the reactions $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\text{Mn}^{2+} \leftrightarrow \text{Zn}^{2+}$, and $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$.

Our study investigates the correlation between In and Cd in sphalerite (Fig. 9D). There are two significant enrichments of indium that appear in the binary plot. The first significant enrichment of indium arises when the Cd content was between 10,000 and 14,00 ppm. When the Cd content was between 4,000 and 8,000 ppm, the second enrichment of indium was exhibited in Sp2. This phenomenon is consistent with the "Indium window" proposed by Dill et al. (2013), providing a clear exploration indicator for indium-rich mineralization in polymetallic deposits.

For coupled stoichiometric substitution, cations such as tetravalent Sn^{4+} and Ge^{4+} , trivalent Sn^{3+} , Ga^{3+} , and In^{3+} , and monovalent Cu^+ and Ag^+ have significantly different ionic radii and charges compared to Zn^{2+} (Cook et al., 2009, 2012; Belissont et al., 2014, 2016). To satisfy valence

equilibrium, they commonly couple with each other in the sphalerite lattice, such as $\text{Cu}^+ + \text{Sb}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, and $2(\text{Ag}, \text{Cu})^+ + \text{Sn}^{4+} \leftrightarrow 3\text{Zn}^{2+}$ (Makovicky and Topa, 2018; Lockington et al., 2014; Ye et al., 2011; Liu et al., 2022). We observed a strong positive correlation ($R^2 = 0.88$) between the In and Cu concentrations in Sp1 and Sp2, confirming that one of the coupled substitution mechanisms for In is $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, which is also the substitution mechanism in most deposits (Fig. 9E). The relationship of $\text{Cu}^+ + (\text{or Ag}^+) + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ is relatively less pronounced (Fig. 9F). Meanwhile, we observed a strong positive correlation ($R^2 = 0.82$) between Ag and Cu, indicating that Ag also contributes significantly to the incorporation of In into the sphalerite lattice (Fig. 9G). This suggests that $(\text{Cu}, \text{Ag})^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ could be another coupled substitution mechanism in the Gaocheng sphalerite deposit. In this study, we did not observe a clear coupling stoichiometric substitution of In with Sn to replace Zn in sphalerite ($\text{In}^{3+} + \text{Sn}^{4+} \leftrightarrow 3\text{Zn}^{2+}$) (Fig. 9H). We speculate that small mineral inclusions may have affected the data points. In the Gaocheng deposit, Ga also shows significant enrichment, but no specific correlation between the simultaneous enrichment of Ga and In was found solely through the projection of data points. However, it is not difficult to observe a certain degree of positive correlation between Ga and In in the early stages of the two types of sphalerites (Fig. 9I). The cause of this phenomenon may be that both elements enter the sphalerite in a similar way, relying on Cu^+ and Ag^+ for coupled substitution. During their individual reactions with Ga and In, the activity of Cu^+ and Ag^+ is enhanced, which promotes each other's reaction process and leads to a positive correlation between the

two.

6.2 Physicochemical conditions of mineralization

Physical and chemical parameters such as temperature, pressure, redox conditions, and pH environment profoundly influence the concentration of trace elements during mineral formation. Therefore, we can infer the physicochemical environmental conditions at the time based on the concentration and distribution characteristics of trace elements in the minerals. Sphalerite has demonstrated significant value in explaining the relationship between its elemental composition and the physicochemical environment (Lockington et al., 2014; Frenzel et al., 2021).

In the Gaocheng lead-zinc-silver deposit, Sp1 is enriched in Fe, Mn, and In, with a Ga/In ratio of 0.44 (based on average content); Sp2 is also enriched in Fe, Mn, and In, with a Ga/In ratio of 0.46 (based on average content); Sp3 has higher concentrations of Fe and Mn, but lower In content, with Ag being relatively enriched compared to Sp1, and a Ga/In ratio of 1.35 (based on average content) (Table. 2). Furthermore, the formation temperature of sphalerite is positively correlated with Fe content. The higher the Fe/Zn ratio, the higher the corresponding mineralization temperature of sphalerite (Yu et al., 1987; Han, 1994). The Fe/Zn ratios (based on average content) for Sp1, Sp2, and Sp3 are 0.23, 0.19, and 0.18, respectively, with the Fe/Zn ratio decreasing progressively (Table 2). These two indicators jointly suggest that the Gaocheng deposit formed under intermediate to high-temperature conditions, with the mineralization temperature decreasing from Sp1 to Sp2 to Sp3, which is similar to the

mineralization evolution of skarn deposits.

To further constrain the crystallization temperature of sphalerite, trace element concentrations can be used for quantitative calculations (Frenzel et al., 2022). Research by Frenzel et al. suggests that elements such as Mn, Fe, Ga, Ge, and In in sphalerite can be used to estimate its crystallization temperature, referred to as the GGIMFis thermometer. Using the GGIMFis geothermometer proposed by Frenzel et al. (2022), the results show that we determined the crystallization temperature of Sp1 to be 331°C (mean value) with a range of 303-361°C, Sp2 to be 320°C (mean value) with a range of 293-338°C, and Sp3 to be 309°C (mean value) with a range of 281-328°C, all within the intermediate to high-temperature range (Figure 10A, Table 2). This calculation result is higher than the fluid inclusion temperatures (240°C-320°C) of minerals such as fluorite during the sulfide mineralization stage measured by Zhao et al (2021), which is consistent with geological facts. Both the Ga/In ratio and the GGIMFis thermometer calculations indicate that the formation temperature decreases from Sp1 to Sp2 to Sp3, and the degree of In enrichment also gradually decreases (Fig. 7), suggesting that high-temperature conditions are more favorable for In enrichment. In conclusion, the enrichment of In in the Gaocheng lead-zinc-silver deposit is influenced by the mineralization temperature, with In primarily enriched in the early high-temperature Sphalerite.

Sulfur fugacity (fS_2) is an important parameter in deposit geochemistry, used to describe the chemical activity of sulfur in the mineralization

environment. The concentration of iron (Fe) in sphalerite is directly related to the temperature and sulfur fugacity at the time of formation. Therefore, by analyzing the Fe content in sphalerite, its formation conditions can be inferred (Frenzel et al., 2021). In the Gaocheng deposit, both Sp1 and Sp2 are associated with abundant pyrite, with more pyrite appearing in the Sp1 stage than in Sp2, while Sp3 shows little to no pyrite. Using the formula proposed by Frenzel et al. (2021), the sulfur fugacity (fS_2) for the three stages was calculated (Fig. 10B), with the (FeS)_{mol} content in sphalerite ranging from 0.01% to 1.0%, which falls within the intermediate sulfide range. The Log fS_2 values for Sp1 range from -8 to -5, for Sp2 from -10 to -6, and for Sp3 from -13 to -9 (Figure 10B). In summary, sphalerite from all three generations formed under high sulfur fugacity conditions, with sulfur fugacity gradually decreasing from Sp1 and Sp2 to Sp3. These results are consistent with the cooling process of the mineralizing fluid, indicating that sulfur fugacity decreases with temperature and reflecting the reduced availability of reducible sulfur in the hydrothermal fluid system during mineralization.

The oxygen fugacity and pH, as important indicators of ore-forming environments, should be emphasized when studying the enrichment of In, Ga, and Cd in sphalerite. Based on mineral paragenesis and the formation temperature of sphalerite, we simulated two distinct stages. One is the early sulfide stage, which is closely related to the relative enrichment of In, Ga, and Cd in Sp1 and Sp2, under the Fe-As-Cu-Zn-S-O-H-Cl system, with a mineralization temperature of approximately 325°C. The other is the late sulfide stage formed by the relatively barren Sp3 in the

Fe-Cu-Zn-Pb-S-O-H-Cl system, with a temperature of about 300°C (Fig. 11). The results in Figure A show that sphalerite can gradually precipitate under pH (4.9–9.9) and fO_2 (-34–-28.5) conditions, during which chalcopyrite, bornite, arsenopyrite, pyrite, and chalcocite may also be present. However, no bornite was observed in fieldwork or petrographic analysis. Combined with microscopic observations, the yellow region formed by closed contour lines was identified as the suitable condition range for the early-stage formation of Sp1 and Sp2. The oxygen fugacity (fO_2) of Sp1 and Sp2 in the early stages ranges from -34 to -28.5, with pH values between 4.9 and 6.7. As the temperature decreases (Fig. 11B), the chalcopyrite region expands laterally and contracts vertically, while the area of galena has the largest coverage, indicating that precipitation is very easy during this stage. The stable regions of sphalerite and pyrite also expand accordingly. In this stage, the precipitation of Sp3 is limited by the influence of copper-containing compounds, but is even more constrained by sulfur-bearing substances (H_2S , HS^-). The oxygen fugacity (fO_2) of Sp3 ranges from -36 to -31.5, with pH values between 4.6 and 6.6. In summary, the Gaocheng sphalerite precipitates in a weakly acidic fluid environment with a relatively high oxygen fugacity.

6.3 Ore deposit genesis and types

The formation of the Gaocheng deposit is closely related to the contact metasomatism between magma (moderately felsic) and carbonate rocks (Qin et al., 2018; Zhao et al., 2021). The surrounding rocks of the deposit are mainly composed of Caledonian biotite granite. The

mineralization age of sphalerite in the deposit is approximately 90 ± 6 Ma (Yu et al., 2018; Zhao et al., 2021), which is clearly later than the formation of the surrounding rocks, indicating that the mineralization is closely related to magmatic activity during the Yanshanian period. The widespread exposure of Devonian carbonate rocks in the region provided the basic conditions for mineralization, while the extensive fault zones provided favorable pathways for the migration and enrichment of magmatic hydrothermal fluids (Wu et al., 2012, 2013). Magmatic hydrothermal fluids, which are rich in metallogenic elements, are an important source of materials for mineralization.

Previous studies have shown that the temperature of the fluids during mineralization gradually decreases with stages. H-O isotope data indicate that the mineralizing fluids in the early stage primarily originated from magma, while in the later stages, atmospheric water was involved (Zhao et al., 2012, 2021). At the end of the Silurian period of the Caledonian, intense tectonic and magmatic activities led to the large-scale formation of biotite granite in the region (Yu et al., 2018; Chen et al., 2024). During the Late Hercynian to Indosinian and into the middle Yanshanian, regional tectonic activity intensified, resulting in the formation of numerous fault systems. However, magmatic activity around the Gaocheng deposit is relatively weak and is mainly composed of sedimentary strata (Fig.1, 2). By the Late Cretaceous (Yanshanian), regional magmatic activity significantly intensified, and the tectonic environment transitioned from compression to extension, forming multiple extensional fault zones (Wu et al., 2012; Cai et al., 2022). These fault zones provided important migration pathways for the mineralizing fluids and potential spaces for ore

body formation. During this period, sulfur-, lead-, and zinc-rich acidic magmatic hydrothermal fluids ascended along the fault zones, undergoing intense metasomatic reactions with the contact zones of Devonian carbonate rocks, promoting the formation of skarn (Yu et al., 2012; Zhao et al., 2021). Driven by magmatic hydrothermal fluids, the fluids migrated upwards along the extensional fault zones, simultaneously capturing a large amount of metallogenic elements from the surrounding rocks (Chen et al., 2012; Shu et al., 2014). As the hydrothermal fluids mixed with atmospheric precipitation during migration, physicochemical conditions changed, leading to a gradual decrease in temperature and salinity. These changes led to the hydrothermal fluids beginning to precipitate and concentrate different metal minerals in stages, ultimately forming the Gaocheng lead-zinc-silver deposit. This mineralization process not only reflects the influence of regional geological activity on ore formation but also reveals the key role of geological environmental changes in the mineralization process.

The trace element composition of sphalerite varies in different genetic types of ore deposits, and the ratio relationships between elements can be used to indicate the genesis of the deposit (Cook et al. 2009; Ye et al. 2011). For example, in the Gaocheng deposit, sphalerite formed in three different stages was analyzed, and it was found that the Fe/Cd ratio was consistently below 250, which is typically associated with skarn-type or volcanic-hydrothermal deposits. In addition, the Gaocheng deposit exhibits typical skarn mineral assemblages, and its mineralizing fluids have a distinct NaCl-KCl brine composition. Fluid inclusion analyses show that from the early to late stages of mineralization, the temperature and salinity

of the mineralizing fluids continuously decreased. The precipitation of metals followed the sequence from oxides to sulfides to carbonates, consistent with the mineralization sequence of magnetite-pyrite-lead-zinc ores-calcite, which clearly aligns with the mineralization evolution model of skarn-type deposits.

After a comparative analysis of the trace element compositions of skarn-type deposits worldwide, the trace element data of the Gaocheng deposit mostly fall within the range typical of skarn-type deposits (Fig. 12, Cook et al. 2009; Ye et al. 2011; Lockington et al. 2014; Wu et al. 2023; Chen et al. 2024). Furthermore, the comprehensive study of the trace element composition of sphalerite, detailed fluid inclusion analysis, and the geological context of the deposit all consistently support the conclusion that the Gaocheng lead-zinc-silver deposit highly conforms to the mineralization evolution model of skarn-type deposits. This evidence also provides an important geological basis for exploration and exploitation in the Gaocheng deposit.

7 Conclusions

(1) Three distinct types of sphalerite were identified in the Gaocheng deposit: (1) early-stage dark brown Sp1 replacing pyrrhotite; (2) mid-stage brown-red Sp2 with a zoned structure (dark core, red rim) coexisting with chalcopyrite; and (3) late-stage bright red Sp3, pure and enclosed by galena.

- (2) In sphalerite, divalent cations (Fe^{2+} , Mn^{2+} , Cd^{2+}) directly substitute Zn^{2+} , while monovalent/trivalent cations (Cu^+ , Ag^+ , In^{3+} , Ga^{3+}) form coupled substitutions, such as Cu^+ (or Ag^+) + Ga^{3+} (or In^{3+}) $\leftrightarrow$ 2Zn^{2+} .
- (3) GGMFI's geothermometry reveals a decreasing trend in ore-forming temperatures: Sp1 (331°C), Sp2 (320°C), and Sp3 (309°C). Correspondingly, sulfur fugacity ($f\text{S}_2$) declines from Sp1 (-8 to -5), Sp2 (-10 to -6), to Sp3 (-13 to -9). Oxygen fugacity ($f\text{O}_2$) and pH also decrease from early stages ($f\text{O}_2$: -34 to -28.5; pH: 4.9–6.7) to late stage ($f\text{O}_2$: -36 to -31.5; pH: 4.6–6.6).
- (4) Sp1 hosts the highest concentrations of In (mean: 414 ppm; max: 832 ppm), Ga (mean: 177 ppm; max: 521 ppm), and Cd (mean: 11517 ppm; max: 13234 ppm), enriched under high-temperature, high- $f\text{S}_2$, high- $f\text{O}_2$, weakly acidic fluids. Among them, In enrichment is more temperature-dependent, while Ga and Cd are less sensitive.
- (5) The mineral assemblage, sphalerite compositions, and ore-forming conditions suggest that the Gaocheng deposit shares key features with typical skarn-type systems, providing new evidence supporting its skarn classification.

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Table Captions

Table 1. Summary statistics of EPMA data for the Gaocheng sphalerite.

Table 2. Summary statistics of LA-ICP-MS element content and estimated temperature for the Gaocheng sphalerite.

Figure Captions

Fig. 1. (A) Tectonic sketch map of China, showing the location of the Gaocheng (modified after Chen et al. 2024); (B) Regional geological map of Yunkai Domian (modified after Peng et al. 2006).

Fig. 2. (A) Geologic map of the Gaocheng deposit (modified after Yu et al., 2018). (B) Geological profile of Gaocheng deposit from A to B (modified after Yu et al., 2018).

Fig. 3. Ore mineral assemblage and surrounding rock alterations from the Gaocheng deposit. (A) Chloritization; (B) Pyritization occurred on both sides of the calcite vein; (C) Veins with K-Feldspathization; (D) Large blocks of fluorite contain calcite veins; (E) Interwoven pyrite veins and

sphalerite veins discovered inside the mine.

Fig. 4. Reflected light (F), plane polarized light (A, C, E) and cross polars light (B, D) photomicrographs showing alteration and mineral assemblages in Gaocheng deposit. (A) Muscovite, chlorite, and sericite; (B) The presence of diopside under cross polars light; (C) Large grains of garnet are present, and biotite transforms into chlorite; (D) Fig. C under cross polars light; (E) Large grains of quartz and epidote; (F) Fig. E under cross polars light and first generation sphalerite (Sp1).

Abbreviations: Bt = biotite, Chl=Chlorite, Di = diopside, Ep = epidote, Grt = garnet, Ms = muscovite, Qtz = quartz, Ser = sericite, Sp =sphalerite.

Fig. 5. fig. A, B, and C form a set of images, representing the same field of view in sequence: reflected light photomicrograph, transmitted light photomicrograph, and hand-drawn illustration. And so on, fig D, E, and F form the second set; fig G, H, and I form the third set; and fig. J, K, and L form the fourth set. (A) (B) (C) The first generation sphalerite exhibits a black-brown color, with chalcopyrite inclusions, and has replaced early magnetite. The second generation sphalerite forms adjacent to the first generation sphalerite and is replaced by galena.; (D) (E) (F) The second generation sphalerite exhibits a distinct nucleation structure, with a deep red core and lighter-colored edges. Stannine is coexistent with the second generation sphalerite.; (G) (H) (I) Well-formed early-generated arsenopyrite crystals and extensive pyrrhotite. (J) (K) (L) The third generation sphalerite is pure and exhibits a bright crimson color, coexisting with galena.

Abbreviations: Apy = Arsenopyrite; Ccp = Chalcopyrite, Gn = Galena, Py = Pyrite, Po = Pyrrhotite; Sp = Sphalerite, Stn= Stannine.

Fig. 6. Mineral paragenetic sequence of the Gaocheng deposit emphasizing three generations of sphalerite.

Fig. 7. Box and whisker plot for trace element concentrations for three generations of sphalerite from the Gaocheng deposit.

Fig. 8. Representative time-resolved LA-ICP-MS depth profiles for three generations of sphalerite in the Gaocheng deposit.

Fig. 9. Minor elements analyzed by EPMA and LA-ICP-MS show the concentrations and possible substitution mechanisms in sphalerite. (A) Zn vs Fe + Mn + Cd; (B) Zn vs Fe; (C) Zn vs Cd; (D) Cd vs In; (E) In vs Cu; (F) Ga vs Cu + Ag; (G) Cu vs Ag; (H) In vs Sn; (I) Ga vs In.

Fig.10. (A) Box plots of temperature calculated by the GGIMFis thermometer; (B) Temperature vs. $\log fS_2$ diagram adapted from Frenzel et al. (2021). The different shadow fields represent the range of the three types of sphalerite sulfur fugacity.

Fig.11. (A) and (B) show the pH vs. $\log fO_2$ diagrams for the Fe-As-Cu-Zn-S-O-H-Cl (325°C) system and the Fe-Cu-Zn-Pb-S-O-H-Cl (300°C) system, respectively. In these diagrams, the Fe-As-S-O-H-Cl system is represented by the orange line, the Cu-S-O-H-Cl system by the purple line, the Zn-S-O-H-Cl system by the green line, the Pb-S-O-H-Cl system by the red line, sulfur-containing ions by the blue line, and the oxygen fugacity and pH conditions of three generations of sphalerite (yellow area). Both diagrams were calculated based on $\Sigma S = 0.1$ m, $\Sigma Fe = 0.0001$ m, $\Sigma As = 0.0001$ m, $\Sigma Cu = 0.0001$ m, $\Sigma Zn = 0.0001$ m, $\Sigma Pb = 0.0001$ m, $\Sigma NaCl = 1$ m, and $P = 1000$ bar. The thermodynamic data for other species were taken from the CHNOSZ OBIGT thermodynamic database.

Fig.12. Trace element relationships of sphalerite in lead-zinc deposits with different genesis (Cook et al. 2009; Ye et al. 2011; Lockington et al. 2014; Wu et al. 2023; Chen et al. 2024). (A) Fe/Mn vs Mn/Ga; (B) In vs Cd; (C) Mn vs Cd; (D) Mn vs Fe.

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Table 1. Summary statistics of electron probe micro-analyzer (EPMA) data for sphalerite (wt.%)

Samples No.	S	Zn	Fe	Cd	In	Sn	Mn
Gaocheng deposit							
Sp1 n=41							
GC-1-1A-1-1	33.02	52.46	12.57	0.59	0.04	0.12	1.10
GC-1-1A-1-2	33.07	52.23	12.44	0.60	0.06	0.03	0.98
GC-1-1A-1-3	33.04	52.52	12.00	0.62	0.03	0.01	1.08
GC-1-1A-1-4	32.74	56.57	8.80	0.64	0.02	0.00	0.27
GC-1-1A-1-5	32.98	52.21	12.35	0.61	0.02	0.02	0.98
GC-1-1A-2-1	32.75	52.99	11.93	0.64	0.06	0.18	0.52
GC-1-1A-2-2	33.05	52.19	12.20	0.60	0.03	0.00	1.06
GC-1-1A-2-3	32.36	55.11	9.60	0.62	0.02	0.00	0.23
GC-1-1A-2-4	33.08	54.25	10.73	0.72	0.02	0.00	0.31
GC-1-1B-1-3	33.33	52.23	12.30	0.56	0.05	0.00	0.97
GC-1-1B-1-4	33.18	56.17	9.01	0.81	0.08	0.01	0.10
GC-1-1B-2-1	33.17	52.27	12.36	0.54	0.03	0.20	0.96
GC-1-1B-2-2	33.17	55.62	8.54	0.64	0.05	0.25	0.15
GC-1-1B-2-3	33.41	55.25	9.52	0.60	0.04	0.01	0.21
GC-1-1B-2-4	33.24	53.59	11.47	0.62	0.02	0.04	0.37
GC-1-1B-2-5	33.32	54.36	11.03	0.59	0.04	0.04	0.50

GC-1-1B-3-1	33.56	52.56	12.27	0.59	0.00	0.00	0.91
GC-1-1B-4-1	33.22	53.49	12.02	0.59	0.03	0.00	0.66
GC-1-2C-1-1	33.18	56.86	8.93	0.61	0.02	0.03	0.15
GC-1-2C-1-3	32.83	55.49	10.54	0.58	0.01	0.00	0.13
GC-1-2C-2-1N	33.18	53.33	10.50	0.57	0.03	0.00	0.46
GC-1-2C-3-1	32.92	56.00	10.70	0.61	0.04	0.00	0.11
GC-1-2C-3-3	32.84	52.12	12.52	0.54	0.06	0.02	1.11
GC-1-2C-3-4	33.30	52.75	12.56	0.57	0.01	0.00	0.64
GC-1-2D-1-2	32.82	56.23	9.62	0.68	0.05	0.01	0.13
GC-1-2D-2-1	32.83	54.50	10.94	0.63	0.03	0.00	0.27
GC-1-2D-3-3	33.30	53.40	12.63	0.61	0.03	0.01	0.42
GC-1-2D-3-4	33.11	54.56	10.66	0.65	0.06	0.02	0.37
GC-1-3A-1-1	33.12	51.93	12.25	0.55	0.05	0.04	1.13
GC-1-3A-2-1	33.19	53.93	11.44	0.51	0.01	0.00	0.50
GC-1-3A-2-2	32.92	53.37	12.06	0.53	0.02	0.01	0.65
GC-1-3A-2-3	33.15	52.25	12.39	0.51	0.03	0.02	0.83
GC-1-3A-3-2	32.65	54.78	10.84	0.55	0.03	0.00	0.41
GC-1-3A-3-3	33.31	51.31	12.60	0.55	0.01	0.28	0.98
GC-1-3B-1-2	33.20	53.81	11.11	0.62	0.03	0.00	0.39
GC-1-3B-1-3	32.72	54.31	11.42	0.58	0.03	0.00	0.23
GC-1-3B-2-2N	32.80	53.03	11.58	0.58	0.05	0.01	0.44
GC-1-3B-2-3	33.17	51.88	12.30	0.57	0.03	0.01	0.93
GC-1-3B-3-1	33.25	52.45	12.59	0.51	0.01	0.00	0.88
GC-1-3B-4-1	33.11	54.16	11.49	0.67	0.02	0.00	0.39
GC-1-3B-4-2	32.95	55.18	10.17	0.55	0.03	0.00	0.17
Max	33.56	56.86	12.63	0.81	0.08	0.28	1.13

Min	32.36	51.31	8.54	0.51	0.00	0.00	0.10
Median	33.11	53.49	11.49	0.59	0.03	0.01	0.46
Mean	33.06	53.75	11.29	0.60	0.03	0.03	0.56
Sp2 n=38							
GC-1-1C-1-1	33.11	52.76	12.64	0.61	0.01	0.00	0.75
GC-1-1C-1-2	32.77	55.53	9.30	0.71	0.02	0.00	0.18
GC-1-1C-1-3	33.42	54.15	11.06	0.63	0.04	0.01	0.41
GC-1-1C-1-4	33.16	51.72	12.71	0.52	0.04	0.08	1.29
GC-1-1C-2-1	33.18	51.69	12.32	0.54	0.04	0.00	1.14
GC-1-1C-3-1	33.50	51.80	12.72	0.61	0.06	0.05	1.07
GC-1-1C-4-1	33.08	52.76	12.78	0.61	0.02	0.00	0.65
GC-1-1C-4-2	32.91	55.02	11.08	0.57	0.03	0.00	0.11
GC-1-1C-4-3	33.35	52.76	11.97	0.54	0.09	0.00	1.15
GC-1-2B-1-1	32.95	55.59	10.48	0.60	0.01	0.00	0.15
GC-1-2B-2-1	32.78	57.05	9.23	0.53	0.03	0.00	0.14
GC-1-2B-2-3	33.23	54.33	11.39	0.63	0.01	0.00	0.37
GC-1-2B-3-2	33.00	52.97	11.62	0.60	0.03	0.00	0.56
GC-1-3D-1-2	33.08	54.05	10.91	0.59	0.03	0.00	0.31
GC-1-3D-1-4	33.10	55.26	10.43	0.60	0.02	0.00	0.16
GC-1-3D-1-5	32.55	53.95	11.33	0.57	0.02	0.00	0.36
GC-1-3D-2-2	33.34	54.63	10.02	0.64	0.05	0.00	0.11
GC-1-3D-3-2	33.57	52.68	12.25	0.55	0.03	0.03	1.08
GC-2-1B-1-2	33.24	52.22	11.61	0.36	0.05	0.00	1.97
GC-2-1B-1-3	33.03	60.09	5.46	0.43	0.01	0.00	0.36
GC-2-1B-2-1	33.13	56.91	7.83	0.44	0.00	0.00	0.38
GC-2-1B-2-3	33.06	61.30	4.40	0.45	0.01	0.01	0.33

GC-3-1B-1-1	33.11	54.56	10.93	0.47	0.00	0.00	0.62
GC-3-1B-3-1	33.20	52.92	11.85	0.44	0.04	0.00	1.12
GC-3-1B-4-1	33.04	55.11	10.37	0.49	0.00	0.00	0.31
GC-3-2B-1-1	33.30	52.80	11.83	0.36	0.02	0.00	0.99
GC-3-2B-2-1	32.94	53.43	11.75	0.48	0.02	0.00	0.69
GC-3-2B-3-1	33.07	56.97	8.66	0.40	0.01	0.00	0.19
GC-3-2B-3-2	32.98	53.23	11.96	0.41	0.02	0.00	0.95
GC-4-3A-1-1	33.30	53.28	11.93	0.46	0.00	0.01	0.82
GC-4-3A-2-1	33.29	52.85	11.58	0.49	0.03	0.05	0.92
GC-4-3A-3-1	33.47	52.47	12.00	0.47	0.02	0.00	0.89
GC-4-3A-3-2	33.29	53.23	11.23	0.53	0.01	0.00	0.54
GC-5-1-1-1	33.01	53.06	12.15	0.50	0.02	0.00	0.68
GC-5-1-1-2	32.82	53.83	11.42	0.54	0.02	0.00	0.54
GC-5-1-2-1	33.18	53.76	11.58	0.54	0.03	0.00	0.65
GC-5-1-3-1	33.19	53.20	11.64	0.50	0.05	0.00	0.64
GC-5-1-4-1	33.16	52.94	11.64	0.52	0.05	0.00	0.61
Max	33.57	61.30	12.78	0.71	0.09	0.08	1.97
Min	32.55	51.69	4.40	0.36	0.00	0.00	0.11
Median	33.12	53.35	11.58	0.53	0.02	0.00	0.62
Mean	33.13	54.13	10.95	0.52	0.03	0.01	0.64
Sp3 n=32							
GC-1-2A-1-1	33.04	55.08	10.74	0.60	0.04	0.00	0.28
GC-1-2A-1-2	33.15	52.45	12.80	0.54	0.09	0.00	1.15
GC-1-2A-2-1	33.28	52.94	12.39	0.56	0.02	0.00	0.92
GC-1-2A-2-2	33.38	55.15	9.94	0.56	0.02	0.00	0.18
GC-1-2A-2-3	33.11	52.76	12.20	0.58	0.04	0.00	0.67

GC-1-2A-3-1	33.23	52.66	12.02	0.59	0.05	0.00	0.83
GC-1-2A-3-2	32.26	54.03	10.93	0.64	0.02	0.03	0.12
GC-1-2A-4-1	33.43	51.46	12.78	0.51	0.09	0.01	1.31
GC-1-2A-4-2	33.40	54.92	11.11	0.57	0.05	0.00	0.41
GC-2-2B-1-1	32.90	59.62	5.73	0.47	0.00	0.00	0.43
GC-2-2B-1-2	33.19	51.39	12.08	0.35	0.01	0.01	2.10
GC-2-2B-2-2	32.70	56.20	8.62	0.46	0.01	0.00	0.89
GC-2-2B-3-1	33.45	51.12	12.18	0.38	0.00	0.00	2.17
GC-2-2D-1-1	33.03	51.46	12.01	0.35	0.00	0.08	2.14
GC-2-2D-1-2	33.00	53.75	10.39	0.39	0.00	0.00	0.70
GC-2-2D-2-1	33.24	54.95	9.82	0.48	0.00	0.01	1.16
GC-2-2D-3-1	33.49	50.33	12.82	0.37	0.00	0.00	2.47
GC-2-3B-1-1	33.23	50.45	13.19	0.41	0.00	0.00	2.21
GC-2-3B-1-2	33.12	55.49	9.61	0.42	0.02	0.00	0.51
GC-2-3B-2-1	33.48	54.54	9.80	0.40	0.00	0.00	0.58
GC-2-3B-2-2	32.72	52.11	12.21	0.40	0.01	0.00	1.10
GC-2-3B-3-1	33.33	50.84	12.69	0.38	0.00	0.00	1.97
GC-2-4-1-2	32.90	51.07	11.87	0.41	0.00	0.00	1.89
GC-2-4-2-1	33.27	50.26	13.15	0.40	0.00	0.02	2.30
GC-2-4-3-1	33.04	61.32	4.91	0.33	0.03	0.00	0.34
GC-2-1A-1-2	32.88	57.20	8.03	0.46	0.02	0.00	0.53
GC-2-1A-1-1	33.08	57.73	8.22	0.48	0.01	0.00	0.66
GC-2-1A-1-3	33.32	54.46	10.44	0.45	0.00	0.00	0.94
GC-2-1A-2-1	33.39	52.35	11.35	0.43	0.01	0.00	1.76
GC-2-1A-2-2	33.45	50.92	12.03	0.40	0.01	0.01	2.13
GC-2-1A-3-1	32.87	55.69	9.42	0.53	0.00	0.00	0.87

GC-2-1A-3-2	33.03	61.76	3.58	0.40	0.01	0.00	0.28
Max	33.49	61.76	13.19	0.64	0.09	0.08	2.47
Min	32.26	50.26	3.58	0.33	0.00	0.00	0.12
Median	33.14	53.95	10.59	0.46	0.02	0.01	1.12
Mean	33.14	53.95	10.59	0.46	0.02	0.01	1.12

Table 2. Summary statistics of LA-ICP-MS element content and estimated temperature for the Gaoc

Samples No.	Mn55	Fe57	Cu65
Gaocheng deposit			
Sp1 n=56			
GC-1-1A@1	12127.01	125188.08	2343.68
GC-1-1A@2	10962.80	123034.44	1619.38
GC-1-1A@3	11619.68	124602.78	1160.62
GC-1-1A@4	11587.28	120245.59	4781.40
GC-1-1A@5	6365.07	118853.60	2572.54
GC-1-1A@6	9540.97	117631.73	2446.37
GC-1-1A@7	11938.20	124584.87	355.62
GC-1-1A@8	3145.02	98467.25	7007.60
GC-1-1C@1	1881.81	90414.68	1615.68

GC-1-1C@2	11506.59	122580.01	2123.8
GC-1-1C@3	13791.65	123734.45	2822.74
GC-1-1C@4	12423.24	128095.89	1130.10
GC-1-1C@5	4472.13	114133.49	349.73
GC-1-1C@6	12041.98	127243.91	1336.40
GC-1-1C@7	10587.18	105122.93	690.85
GC-1-1C@8	12069.16	101745.06	2656.75
GC-1-1C@9	9460.81	103176.95	1153.35
GC-1-1C@10	12525.88	124876.19	1520.30
GC-1-2A@1	12031.19	125299.18	1482.10
GC-1-2A@2	13591.89	125727.49	1451.54
GC-1-2A@3	20880.89	123038.98	1691.27
GC-1-2A@4	10280.43	118802.83	894.86
GC-1-2A@5	11084.11	105241.28	2454.00
GC-1-2A@6	4869.85	100554.17	6446.43
GC-1-2A@8	15101.43	106383.47	1327.03
GC-1-2C@1	2696.78	101978.45	4569.03
GC-1-2C@2	9278.13	115328.83	3701.33
GC-1-2C@3	1876.02	95992.87	827.76
GC-1-2C@4	5316.70	111137.19	542.74
GC-1-2C@5	9429.84	104175.05	239.94
GC-1-2C@6	13569.50	100927.91	4029.4
GC-1-2C@7	13185.57	103197.30	1146.17
GC-1-2C@8	6653.90	101007.10	1830.28
GC-1-1B@1	10609.14	127221.10	399.13
GC-1-1B@2	2857.57	108237.96	3755.33

GC-1-1B@3	9137.72	118613.70	1339.92
GC-1-1B@4	3511.82	113361.93	1845.31
GC-1-1B@5	11464.78	123292.34	1103.55
GC-1-1B@6	13562.15	124823.46	3901.05
GC-1-1B@7	9399.59	138890.85	7088.04
GC-1-1B@8	6306.71	139279.05	5413.11
GC-1-1B@9	9449.26	125124.33	547.97
GC-1-2B@1	1825.53	221150.83	1151.95
GC-1-2B@2	5733.45	92150.80	970.99
GC-1-2B@3	3542.31	108114.81	4888.45
GC-1-2B@4	4155.89	110367.14	6382.70
GC-1-2B@5	7557.07	102535.18	650.66
GC-1-2B@6	5621.35	110295.32	359.11
GC-1-3D@1	5217.08	116406.49	6012.93
GC-1-3D@2	5037.04	108158.34	521.33
GC-1-3D@3	3099.57	102461.84	2952.93
GC-1-3D@4	8852.83	101184.97	427.73
GC-1-3D@5	8449.95	101993.76	1531.13
GC-1-3D@6	10369.58	100710.82	210.18
GC-1-3D@7	11687.24	117669.97	757.59
GC-1-3D@8	8650.20	108116.63	201.26
Max	20880.89	221150.83	7088.04
Min	1825.53	90414.68	201.26
Median	9439.55	113747.71	1501.20
Mean	8821.26	115226.53	2191.67
Sp2 n=52			

GC-1-2D@1	6463.32	99412.36	610.14
GC-1-2D@2	6763.19	105916.51	808.19
GC-1-2D@3	5878.94	144552.84	555.47
GC-1-2D@4	7355.51	120120.17	1921.52
GC-1-3A@1	10557.26	127453.05	1165.13
GC-1-3A@2	10649.58	126651.20	948.30
GC-1-3A@3	6412.34	117267.47	836.53
GC-1-3A@4	6469.39	117956.51	795.82
GC-1-3A@5	11666.36	94243.00	643.39
GC-1-3A@6	11756.74	92198.29	1694.40
GC-1-3A@7	6987.91	115953.12	5057.71
GC-1-3B@1	10131.19	91961.00	577.40
GC-1-3B@2	7679.46	92906.49	629.55
GC-1-3B@3	6160.40	119670.41	774.03
GC-1-3B@4	11615.29	124252.27	1687.62
GC-1-3B@5	9679.96	91840.60	465.30
GC-1-3B@6	11289.88	119842.37	653.21
GC-1-3B@7	6734.28	113558.47	9177.28
GC-1-3B@8	6736.29	97683.18	819.69
GC-1-3B@9	6692.38	91598.32	943.02
GC-3-1B@1	5835.11	95353.23	7815.58
GC-3-1B@6	9957.92	98799.58	439.96
GC-3-1B@7	11136.05	94819.01	466.01
GC-3-1B@8	10334.57	98821.85	1017.77
GC-3-1B@9	9240.04	98496.08	441.73
GC-3-1B@10	7735.42	99104.38	451.98

GC-3-2B@1	5837.17	98837.27	8325.58
GC-3-2B@2	7578.33	96945.74	588.44
GC-3-2B@3	5921.23	176968.92	518.89
GC-3-2B@4	5595.63	93673.90	5604.5
GC-3-2B@5	7421.57	95053.37	1244.54
GC-3-2B@6	5804.21	97435.89	7484.95
GC-3-2B@7	8581.91	100576.40	4714.5
GC-4-3A@1	8132.63	102373.35	817.83
GC-4-3A@2	8065.68	104407.75	591.46
GC-4-3A@3	8123.24	104610.65	1223.4
GC-4-3A@4	8786.56	105179.46	509.02
GC-4-3A@5	8696.10	103865.07	1642.0
GC-4-3A@6	7912.46	104490.94	4640.0
GC-4-3A@7	8826.70	107189.19	1002.1
GC-4-3A@8	8253.55	105943.81	5096.4
GC-5-1@1	7100.04	108303.82	608.19
GC-5-1@2	8673.41	107770.83	1018.1
GC-5-1@3	5966.29	103781.04	1813.8
GC-5-1@4	7083.57	110216.46	420.12
GC-5-1@5	8181.61	110198.14	523.93
GC-5-1@6	8339.23	92778.58	948.96
GC-5-1@7	8352.12	112577.24	417.10
GC-5-1@8	8919.90	109763.97	2749.7
GC-5-1@9	7479.63	91640.99	1101.1
GC-5-1@10	6443.92	110151.40	495.00
GC-5-1@11	8551.46	114163.02	478.84

Max	11756.74	176968.92	9177.23
Min	5595.63	91598.32	417.10
Median	7989.07	104449.35	828.11
Mean	8087.44	106910.17	1845.68
Sp3 n=49			
GC-2-2B@1	6094.18	90781.39	2486.27
GC-2-2B@2	23148.69	119783.13	747.49
GC-2-2B@3	21497.53	115513.91	1417.94
GC-2-2B@4	14582.51	108066.67	7730.24
GC-2-2B@5	5607.81	83724.41	9683.82
GC-2-2B@6	7999.99	91343.49	8994.32
GC-2-2B@7	23364.03	120908.85	1777.10
GC-2-2B@8	24473.08	91943.94	197.33
GC-2-2D@1	22963.20	97556.96	1581.94
GC-2-2D@2	21907.72	121817.89	1348.63
GC-2-2D@3	23484.66	121971.60	758.21
GC-2-2D@4	22787.85	123543.95	93.26
GC-2-2D@5	17777.53	95954.50	4521.10
GC-2-2D@6	23418.12	93629.02	135.16
GC-2-2D@7	40371.47	124706.81	96.46
GC-2-2D@8	24390.20	101749.95	99.07
GC-2-1A@1	18199.65	84822.10	6421.87
GC-2-1A@2	10271.90	83311.99	2339.53
GC-2-1A@3	18452.31	107765.76	2408.19
GC-2-1A@6	28770.76	83703.84	8317.14
GC-2-1A@7	21224.70	88983.54	9777.23

GC-2-1B@1	21333.49	107524.01	517.63
GC-2-1B@2	21233.95	109580.68	2443.83
GC-2-1B@3	28080.77	99734.39	688.56
GC-2-1B@4	19119.99	104570.00	2575.86
GC-2-1B@5	5247.27	82608.77	206.25
GC-2-1B@6	23066.83	103404.12	4614.04
GC-2-1C@1	21229.76	107191.70	173.87
GC-2-1C@2	19365.22	98721.29	8875.33
GC-2-1C@3	19628.85	101936.30	194.19
GC-2-1C@4	20220.37	107798.10	1145.97
GC-2-1C@5	20177.27	107861.97	2958.64
GC-2-1C@6	20229.95	107326.94	394.63
GC-2-1C@7	21223.14	90815.66	5078.34
GC-2-3B@1	19949.43	104106.67	2562.02
GC-2-3B@2	13897.84	98067.76	4131.63
GC-2-3B@3	20919.41	85670.19	1316.37
GC-2-3B@4	21517.75	102124.84	121.80
GC-2-3B@5	5983.78	70609.60	2517.17
GC-2-3B@6	9424.33	83135.35	4496.87
GC-2-3B@7	21459.14	95195.73	941.54
GC-2-3B@8	21721.00	100325.82	1404.33
GC-2-4@1	21173.60	102338.42	241.16
GC-2-4@2	22561.40	90109.99	101.80
GC-2-4@3	21994.14	94772.50	231.21
GC-2-4@4	18150.47	79798.73	147.87
GC-2-4@5	19953.11	71601.89	81.20

GC-2-4@6	15259.99	97526.55	705.66
GC-2-4@7	9695.96	85836.96	976.18
Max	40371.47	124706.81	9777.2
Min	5247.27	70609.60	81.20
Median	21173.60	98721.29	1348.6
Mean	19277.68	98813.85	2464.8

Samples No.	Mn55_2SE(int)	Mn55_LOD_Pettke	Fe57_2SE(int)	Fe57_LOD_Pettke	Cu65_2SE(int)	Cu65_LOD_Pettke	Ga71_2SE(int)	Ga71_LOD_Pettke
Gaocheng deposit Sp1 n=56								
GC-1- 1A@1	88.18	1.59	1076.63	24.31	114.82	3.21	3.21	0.47
GC-1- 1A@2	85.50	1.61	948.78	25.29	143.45	2.80	2.65	0.41
GC-1- 1A@3	112.93	1.68	1359.06	19.26	19.05	2.74	3.73	0.39
GC-1- 1A@4	104.13	1.58	1098.60	26.95	525.21	2.72	2.83	0.33
GC-1- 1A@5	45.46	1.62	921.01	25.71	28.72	3.15	1.97	0.28

GC-1-1A@6	80.59	1.59	925.47	22.28	251.38	2.95	2.31	0.23
GC-1-1A@7	123.77	1.53	1124.60	25.43	10.02	2.95	3.42	0.42
GC-1-1A@8	22.72	1.61	1071.97	24.21	855.69	3.43	2.85	0.34
GC-1-1C@1	169.28	1.81	2080.23	28.38	13.48	3.20	0.69	0.44
GC-1-1C@2	39.89	1.64	1318.99	30.61	347.33	3.50	1.95	0.50
GC-1-1C@3	147.95	1.50	1124.58	23.95	55.96	2.38	3.50	0.37
GC-1-1C@4	42.04	1.59	1639.33	24.96	155.69	3.04	4.66	0.26
GC-1-1C@5	90.16	1.58	1073.84	24.58	18.76	3.09	4.09	0.40
GC-1-1C@6	119.36	1.59	1088.49	22.91	533.08	2.77	3.28	0.36
GC-1-1C@7	73.97	1.58	2328.20	25.56	1056.99	2.60	2.51	0.47
GC-1-1C@8	53.62	2.09	1302.87	31.54	457.47	2.75	1.51	0.65
GC-1-1C@9	71.78	1.93	958.01	29.71	27.91	3.63	0.81	0.48
GC-1-1C@10	19.47	3.78	1122.95	50.46	367.49	0.23	1.92	0.89

GC-1- 2A@1	78.69	2.38	902.49	32.25	136.14	3.50	3.68	0.64
GC-1- 2A@2	112.07	1.81	1169.20	26.29	149.48	2.65	2.43	0.51
GC-1- 2A@3	241.05	1.77	2018.86	31.03	150.34	2.79	4.57	0.48
GC-1- 2A@4	121.13	2.46	3083.20	41.27	33.05	5.48	1.51	0.68
GC-1- 2A@5	95.25	1.72	1098.64	26.58	18.91	2.29	4.30	0.38
GC-1- 2A@6	92.04	1.56	1108.54	24.82	11.43	3.08	4.37	0.35
GC-1- 2A@8	89.77	1.68	986.45	24.57	242.40	2.92	3.47	0.37
GC-1- 2C@1	91.64	1.67	1002.53	25.38	23.81	2.85	3.83	0.42
GC-1- 2C@2	97.38	1.74	984.13	23.52	22.97	3.07	3.33	0.32
GC-1- 2C@3	93.63	1.49	1035.85	25.45	26.83	2.80	4.61	0.34
GC-1- 2C@4	102.20	1.63	1043.19	20.90	127.98	2.74	4.17	0.33
GC-1- 2C@5	143.28	1.62	947.67	27.76	30.34	2.85	1.75	0.34
GC-1- 2C@6	99.35	1.53	1057.38	24.93	69.19	2.51	4.73	0.37

GC-1- 2C@7	120.24	1.52	1438.62	24.17	285.79	2.69	4.12	0.32
GC-1- 2C@8	48.30	1.48	1910.52	23.50	1539.36	2.86	3.14	0.37
GC-1- 1B@1	95.47	1.44	1079.32	22.58	14.01	2.48	3.69	0.36
GC-1- 1B@2	108.26	1.51	875.86	22.08	14.97	2.40	4.38	0.40
GC-1- 1B@3	29.33	1.79	18313.98	29.39	201.95	2.87	3.03	0.30
GC-1- 1B@4	30.50	1.68	920.53	24.92	71.12	2.83	2.93	0.35
GC-1- 1B@5	75.42	1.80	1122.14	25.73	749.82	2.98	10.46	0.42
GC-1- 1B@6	76.39	1.79	1055.18	20.39	1047.46	2.53	0.63	0.40
GC-1- 1B@7	115.37	1.41	1822.83	19.51	20.53	2.23	1.06	0.33
GC-1- 1B@8	236.74	1.49	1942.72	27.34	21.80	2.70	5.74	0.35
GC-1- 1B@9	63.03	1.87	1139.06	24.32	364.16	3.23	1.43	0.47
GC-1- 2B@1	72.66	1.78	1047.14	26.78	575.82	2.12	5.52	0.34
GC-1- 2B@2	20.93	1.63	1259.04	29.02	53.89	2.81	0.29	0.40

GC-1- 2B@3	75.93	2.31	2201.08	36.17	195.50	4.48	2.95	0.60
GC-1- 2B@4	181.18	2.29	2828.10	34.94	17.51	4.62	2.88	0.57
GC-1- 2B@5	146.38	1.65	1095.28	25.72	310.41	2.85	15.32	0.43
GC-1- 2B@6	91.69	1.62	900.22	23.79	21.19	2.59	4.32	0.34
GC-1- 3D@1	56.88	1.66	925.22	26.82	286.59	2.46	0.36	0.34
GC-1- 3D@2	14.96	1.48	795.81	20.12	37.31	2.43	1.05	0.33
GC-1- 3D@3	42.14	1.31	1537.15	17.58	67.78	2.30	0.57	0.33
GC-1- 3D@4	602.51	1.57	10870.90	26.49	24.21	2.72	1.77	0.36
GC-1- 3D@5	57.19	1.67	786.08	25.47	228.07	2.94	2.07	0.39
GC-1- 3D@6	227.28	1.43	2008.79	24.00	55.37	2.57	9.40	0.37
GC-1- 3D@7	235.48	1.47	1728.49	24.05	22.77	3.04	8.53	0.38
GC-1- 3D@8	151.97	1.48	1750.44	25.78	20.83	2.71	0.69	0.42

Sp2 n=52

GC-1- 2D@1	148.26	1.51	1395.04	22.22	7.39	2.61	0.97	0.32
GC-1- 2D@2	87.38	1.50	897.09	20.56	7.06	2.59	0.70	0.34
GC-1- 2D@3	99.32	1.72	1075.76	27.80	227.45	2.38	2.89	0.32
GC-1- 2D@4	58.74	1.49	1022.75	26.20	313.30	2.71	2.58	0.30
GC-1- 3A@1	101.73	1.76	1131.98	24.04	18.98	3.14	3.44	0.48
GC-1- 3A@2	99.97	1.78	1706.05	25.50	48.36	3.04	1.30	0.41
GC-1- 3A@3	70.79	2.06	2567.71	28.88	116.84	4.30	1.62	0.58
GC-1- 3A@4	151.72	1.87	2166.34	32.57	114.08	3.44	7.53	0.45
GC-1- 3A@5	95.11	1.59	1148.41	22.60	21.28	2.02	0.64	0.36
GC-1- 3A@6	103.99	1.59	1305.16	22.78	5.91	2.55	0.53	0.37
GC-1- 3A@7	37.71	1.53	1039.43	25.46	863.97	2.77	1.52	0.24
GC-1- 3B@1	72.50	2.11	1121.95	30.45	21.75	3.18	0.71	0.45

GC-1-3B@2	124.90	1.93	14713.77	28.12	29.50	2.79	1.04	0.49
GC-1-3B@3	131.11	1.68	3098.56	46.48	2994.92	2.95	2.60	0.35
GC-1-3B@4	29.74	1.70	2636.99	24.38	55.12	1.99	1.35	0.44
GC-1-3B@5	40.05	1.52	872.30	24.00	574.49	2.68	0.95	0.39
GC-1-3B@6	171.44	1.86	2201.12	30.84	26.77	3.40	2.73	0.50
GC-1-3B@7	29.62	1.77	1525.39	26.96	68.24	3.18	0.50	0.45
GC-1-3B@8	80.11	1.36	1013.63	20.91	6.39	2.50	1.71	0.39
GC-1-3B@9	81.51	1.41	920.52	19.99	27.03	2.19	2.90	0.31
GC-3-1B@1	182.59	1.76	2425.09	24.87	9.21	2.76	0.57	0.39
GC-3-1B@6	101.39	1.70	1634.06	28.33	782.58	2.39	0.40	0.39
GC-3-1B@7	243.63	1.51	1141.44	21.82	90.82	2.17	0.63	0.38
GC-3-1B@8	215.79	1.47	1184.09	19.24	153.14	2.37	1.00	0.32
GC-3-1B@9	162.26	1.45	1118.59	20.17	517.54	2.64	1.40	0.40

GC-3-1B@10	70.44	1.53	1736.31	22.74	919.95	2.51	0.52	0.36
GC-3-2B@1	54.10	1.48	844.78	23.35	917.60	2.05	0.87	0.39
GC-3-2B@2	273.48	1.57	1344.70	18.92	213.26	2.48	0.50	0.33
GC-3-2B@3	274.40	1.51	1473.16	21.53	39.48	2.71	0.35	0.34
GC-3-2B@4	174.42	1.68	1002.88	23.67	48.59	2.21	2.72	0.42
GC-3-2B@5	192.99	1.57	903.58	22.52	70.55	2.34	4.66	0.29
GC-3-2B@6	170.01	1.63	928.19	19.54	38.85	2.82	4.49	0.37
GC-3-2B@7	190.07	1.74	1101.37	18.15	9.68	2.40	0.39	0.41
GC-4-3A@1	138.68	1.88	1123.96	26.18	500.15	3.48	3.81	0.26
GC-4-3A@2	174.70	1.74	953.43	21.77	11.32	2.52	0.36	0.44
GC-4-3A@3	189.48	1.85	896.03	24.58	2.68	2.30	0.39	0.39
GC-4-3A@4	198.29	1.81	925.42	24.94	2.59	2.97	0.39	0.44
GC-4-3A@5	184.39	2.59	1372.74	42.10	537.77	5.03	2.36	0.68

GC-4-								
3A@6	48.84	2.43	551.81	38.51	167.79	4.18	2.38	0.61
GC-4-								
3A@7	387.44	2.77	1563.47	36.15	452.39	5.03	4.05	0.64
GC-4-								
3A@8	341.30	3.29	1795.94	58.73	553.51	6.23	2.43	0.89
GC-5-1@1	502.42	2.66	2083.53	49.50	136.40	5.88	2.08	0.73
GC-5-1@2	32.37	2.35	551.83	32.20	185.39	4.70	0.63	0.50
GC-5-1@3	39.72	2.44	880.81	36.59	429.04	4.17	1.28	0.38
GC-5-1@4	134.89	2.51	703.17	36.91	76.58	5.13	0.22	0.54
GC-5-1@5	171.99	2.62	826.08	39.85	559.73	4.84	0.42	0.51
GC-5-1@6	133.29	2.46	704.76	35.81	73.34	4.47	1.43	0.61
GC-5-1@7	175.14	2.53	907.13	39.93	357.55	4.85	4.20	0.56
GC-5-1@8	132.87	2.80	1852.11	48.52	61.24	5.21	0.57	0.60
GC-5-1@9	193.07	2.36	1277.90	32.54	605.49	5.50	0.37	0.48
GC-5-								
1@10	141.02	2.79	668.52	39.03	32.72	5.12	1.66	0.39
GC-5-								
1@11	191.45	3.10	761.44	38.26	904.91	5.14	4.37	0.58
Sp3 n=49								
GC-2-								
2B@1	185.81	3.08	1008.57	42.42	46.08	5.66	2.17	0.72
GC-2-								
2B@2	512.50	3.55	2429.91	52.58	500.19	7.50	2.54	1.01

GC-2- 2B@3	195.92	2.72	677.28	41.73	437.42	5.53	0.24	0.51
GC-2- 2B@4	87.20	3.37	615.90	54.06	365.06	6.45	2.53	0.54
GC-2- 2B@5	44.88	2.83	1200.42	43.26	247.92	6.08	0.35	0.72
GC-2- 2B@6	166.51	2.67	663.63	43.41	17.86	4.43	0.23	0.66
GC-2- 2B@7	116.17	3.41	1452.20	51.14	159.61	5.99	3.18	0.67
GC-2- 2B@8	75.52	2.67	916.47	41.07	242.02	4.24	2.15	0.61
GC-2- 2D@1	135.67	2.83	1087.74	48.03	115.33	5.19	0.33	0.55
GC-2- 2D@2	153.57	2.73	755.00	37.44	146.62	5.09	0.22	0.71
GC-2- 2D@3	509.24	3.48	2205.45	58.59	58.07	6.19	0.89	0.79
GC-2- 2D@4	441.17	3.51	1725.50	49.67	21.86	7.33	0.98	0.54
GC-2- 2D@5	229.94	2.63	817.11	35.91	9.04	5.27	3.69	0.65
GC-2- 2D@6	118.48	2.99	1181.29	44.65	15.11	6.13	1.54	0.55
GC-2- 2D@7	271.40	3.65	1347.05	59.50	8.62	8.13	0.23	0.80

GC-2-2D@8	160.81	2.39	733.70	33.41	42.90	4.29	0.19	0.50
GC-2-1A@1	41.23	2.95	506.77	50.33	115.01	5.04	1.38	0.52
GC-2-1A@2	44.66	2.96	594.35	41.22	290.08	6.49	2.16	0.43
GC-2-1A@3	3809.33	26.08	3190.14	422.97	5.97	47.82	0.49	4.42
GC-2-1A@6	2085.92	25.95	1285.45	418.27	6.43	42.67	0.56	4.98
GC-2-1A@7	2324.11	32.02	1700.11	423.50	6.97	62.56	0.57	6.08
GC-2-1B@1	3084.49	27.44	2771.70	382.73	5.43	44.85	0.55	4.88
GC-2-1B@2	59.96	2.98	586.70	41.26	25.48	5.88	1.58	0.79
GC-2-1B@3	70.90	3.02	653.07	45.83	54.90	5.40	1.10	0.57
GC-2-1B@4	63.49	3.11	601.84	43.88	139.66	6.73	1.74	0.55
GC-2-1B@5	104.85	3.06	575.55	44.47	6.33	6.20	1.52	0.62
GC-2-1B@6	61.64	2.98	567.82	42.13	5.35	5.08	2.37	0.74
GC-2-1C@1	178.73	2.48	1159.93	37.30	1179.94	5.48	0.96	0.75

GC-2-1C@2	176.18	2.53	921.42	41.65	125.54	4.46	0.83	0.71
GC-2-1C@3	248.99	2.93	14101.10	42.54	38.06	6.32	1.61	0.62
GC-2-1C@4	82.51	2.47	916.06	37.80	411.64	4.99	1.01	0.53
GC-2-1C@5	188.75	2.70	964.33	42.39	154.55	5.12	1.13	0.64
GC-2-1C@6	104.04	2.49	887.37	36.82	460.28	5.17	2.93	0.60
GC-2-1C@7	170.72	2.60	921.12	36.13	785.84	4.96	0.87	0.56
GC-2-3B@1	62.30	2.47	790.26	34.13	27.27	4.91	4.07	0.68
GC-2-3B@2	60.59	2.68	733.35	41.80	16.05	4.74	4.31	0.62
GC-2-3B@3	64.70	2.62	760.21	32.31	65.55	5.06	3.45	0.59
GC-2-3B@4	59.90	2.48	747.03	34.82	21.71	5.37	5.99	0.58
GC-2-3B@5	91.48	2.41	1237.03	38.64	29.92	6.19	6.78	0.63
GC-2-3B@6	100.73	2.66	1217.14	33.09	139.38	5.19	8.19	0.62
GC-2-3B@7	72.00	2.98	756.67	39.42	42.56	5.99	4.06	0.80

GC-2-3B@8	87.59	2.92	800.15	38.34	496.24	4.99	4.41	0.69
GC-2-4@1	48.38	3.05	628.91	41.61	52.66	7.11	2.83	0.80
GC-2-4@2	71.89	3.16	681.89	38.75	84.83	6.83	1.02	0.64
GC-2-4@3	36.57	3.03	669.71	39.95	232.24	7.04	2.51	0.70
GC-2-4@4	48.01	3.13	738.07	52.52	7.94	6.24	2.31	0.77
GC-2-4@5	47.62	3.14	658.31	48.62	13.02	6.67	3.28	0.68
GC-2-4@6	45.02	2.85	660.06	58.18	68.80	6.62	0.27	0.54
GC-2-4@7	43.48	3.79	632.40	61.76	64.84	8.34	0.41	0.85

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Agreement

We, all authors, hereby declare that the manuscript titled:

In-Cd-Ga enrichment in sphalerite from the Gaocheng skarn Pb-Zn-Ag deposit, Yunkai Domain, South China

has been reviewed and approved by all authors listed below, and we acknowledge our responsibilities as follows:

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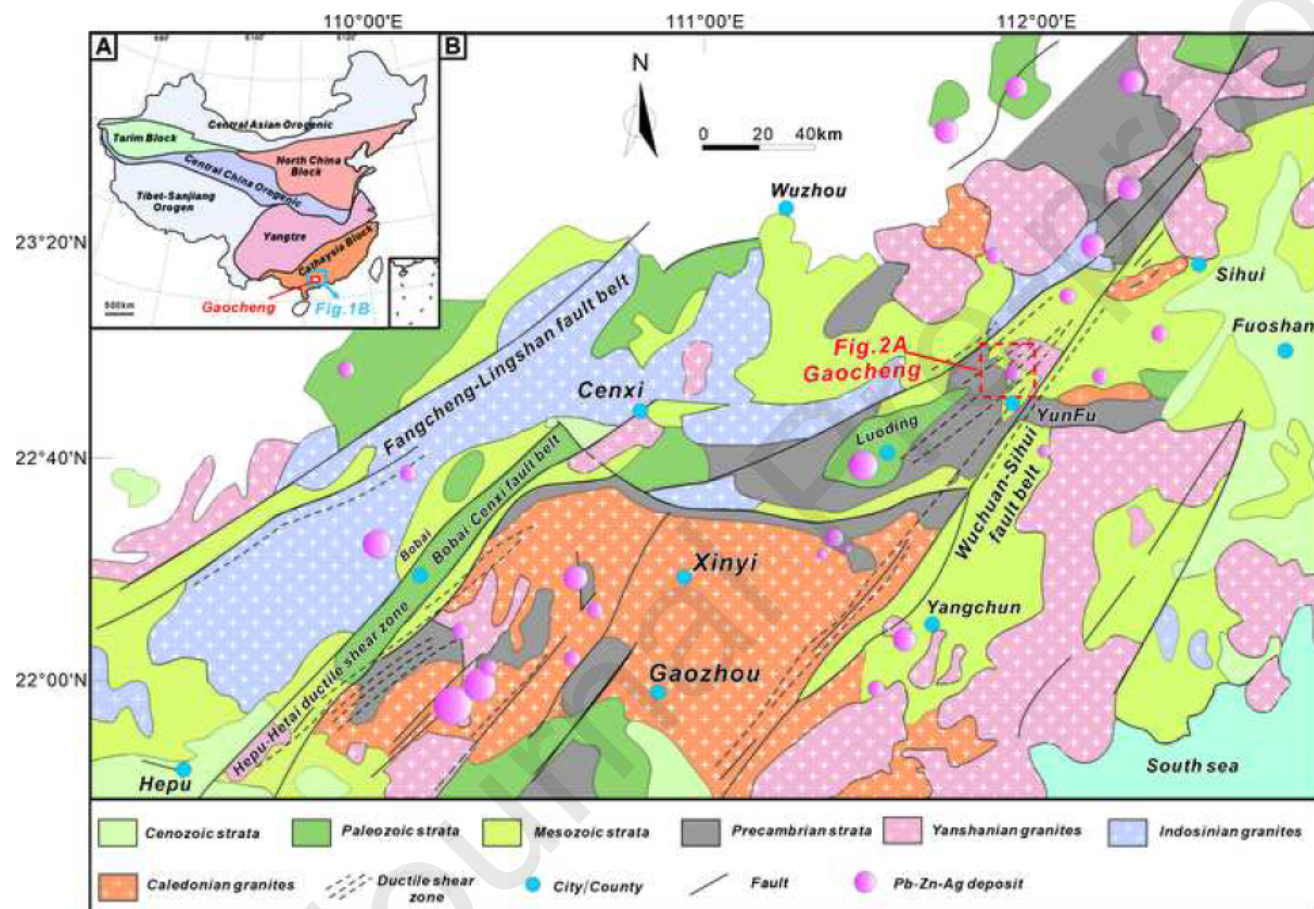


Figure 2

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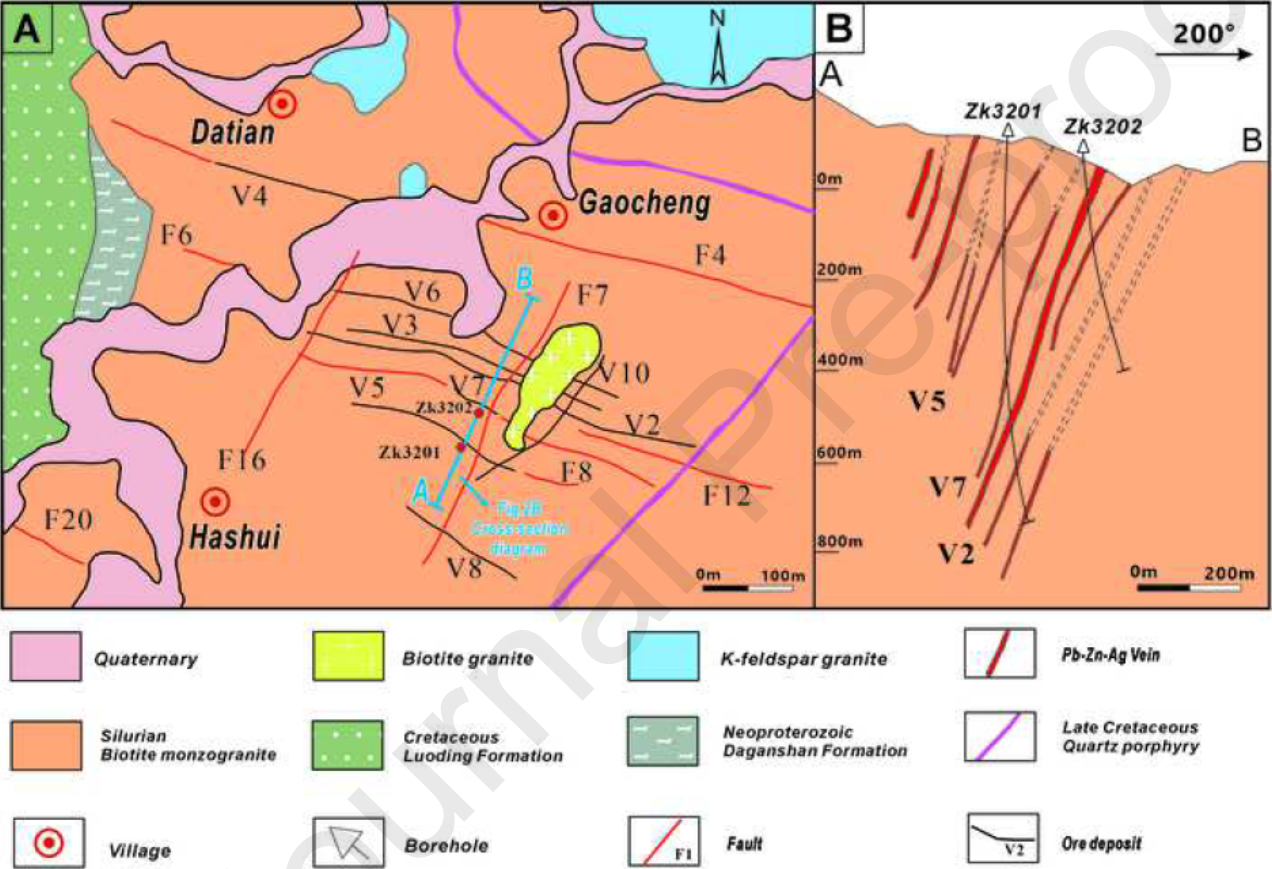


Figure 3

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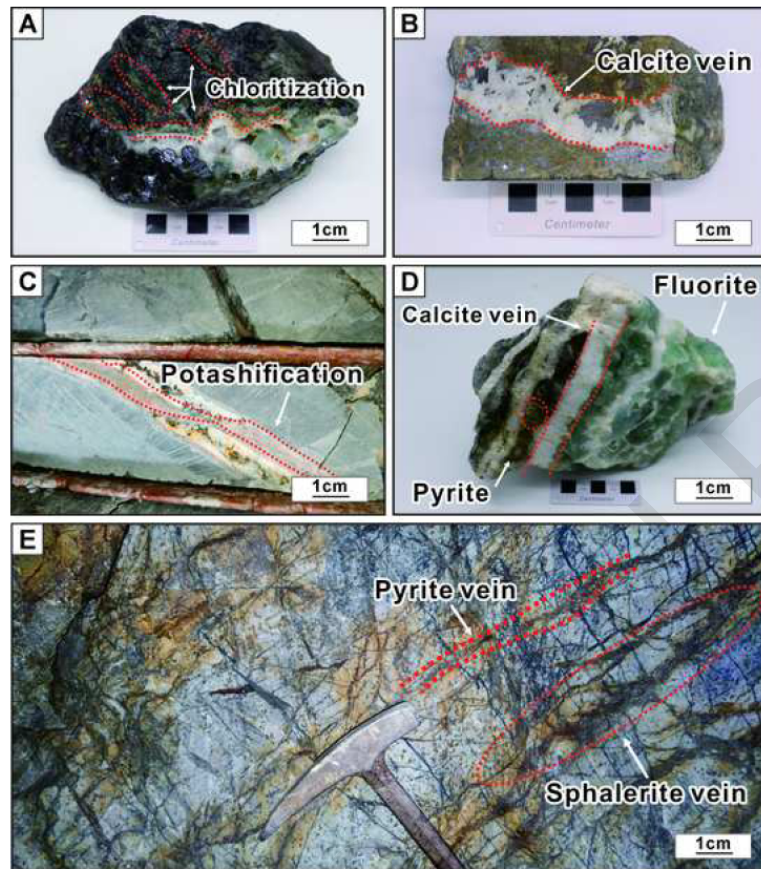


Figure 4

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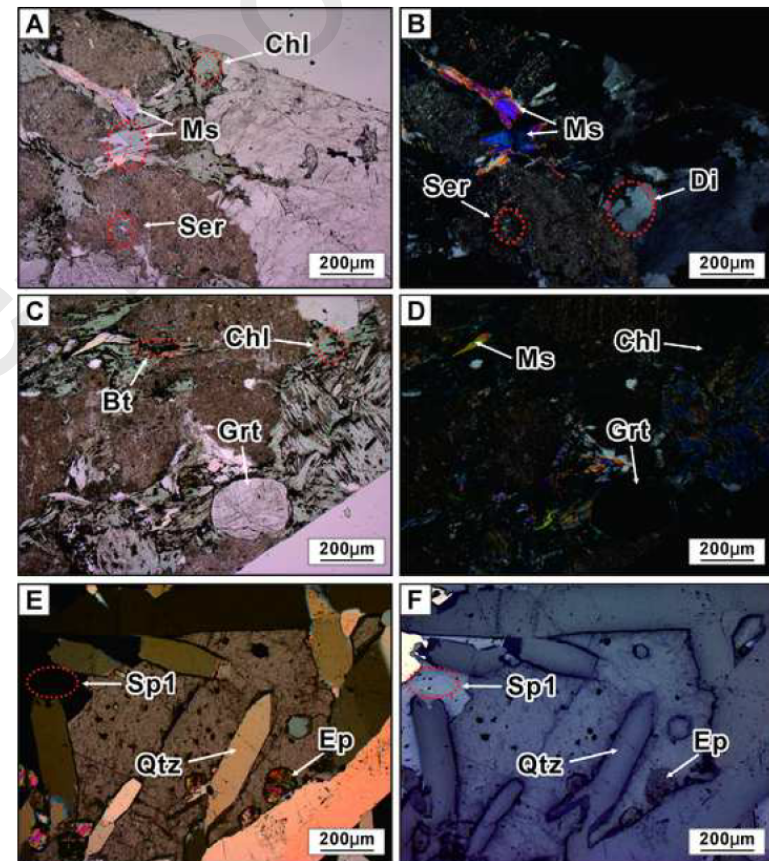


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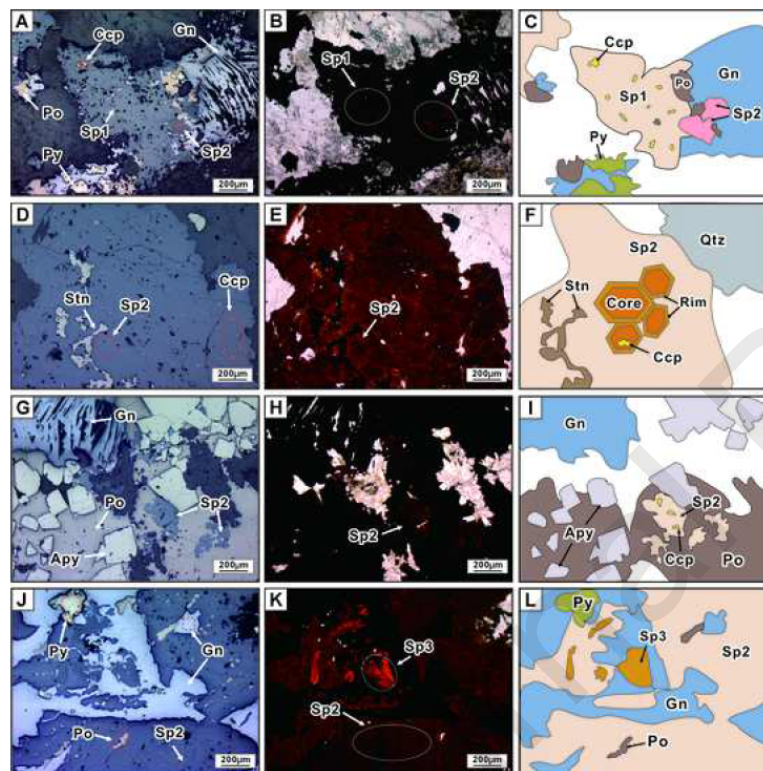


Figure 6

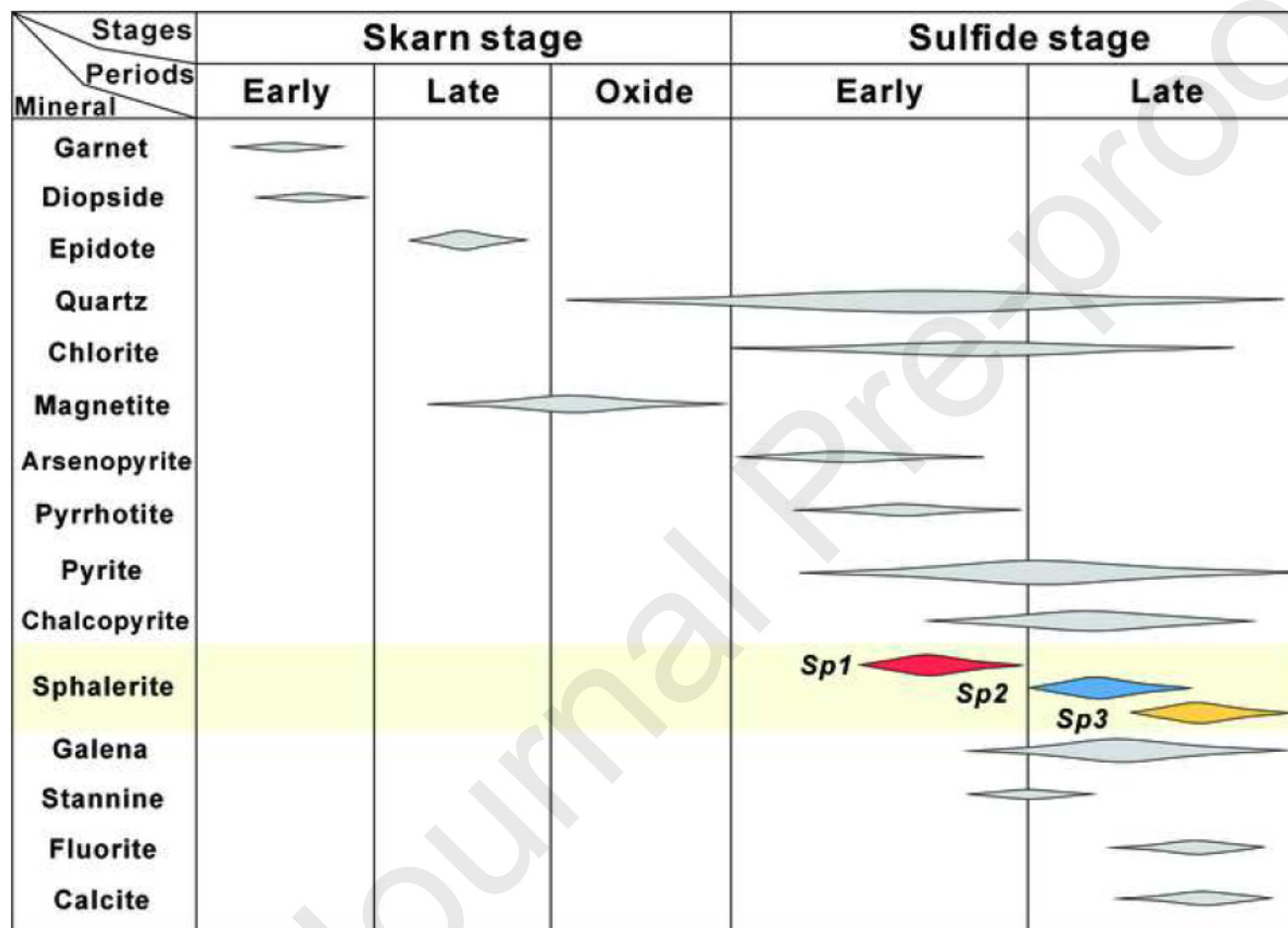
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Figure 7

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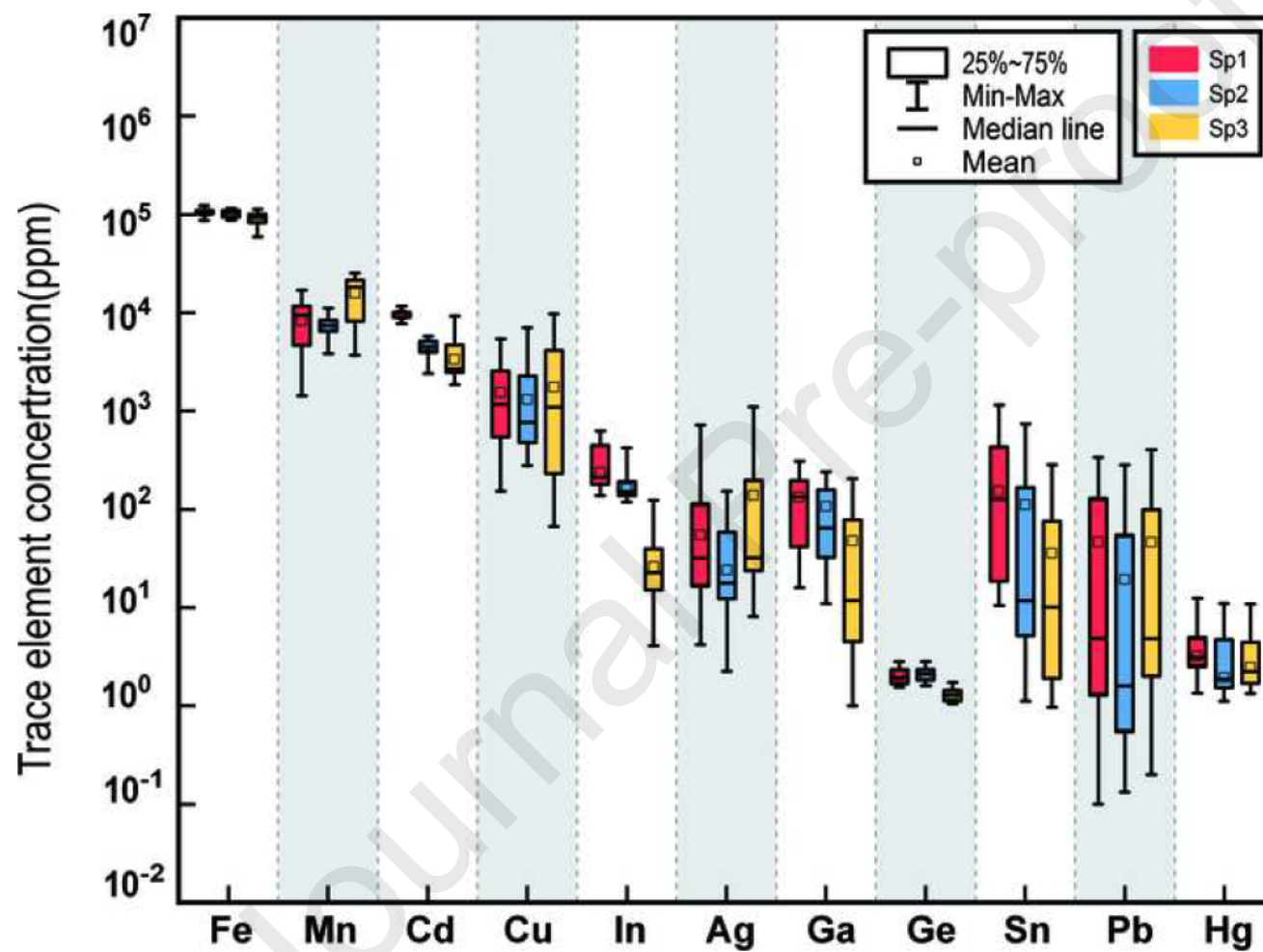


Figure 8

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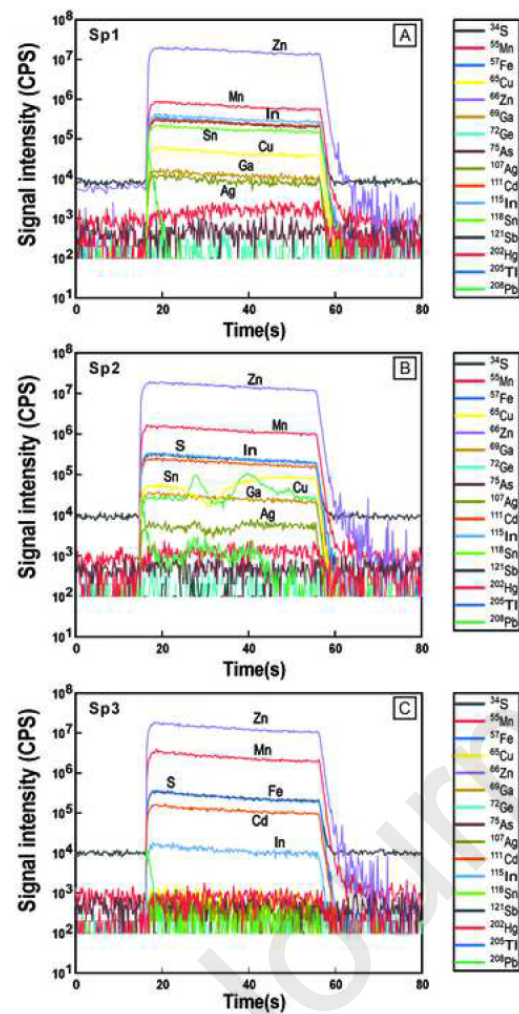


Figure 9

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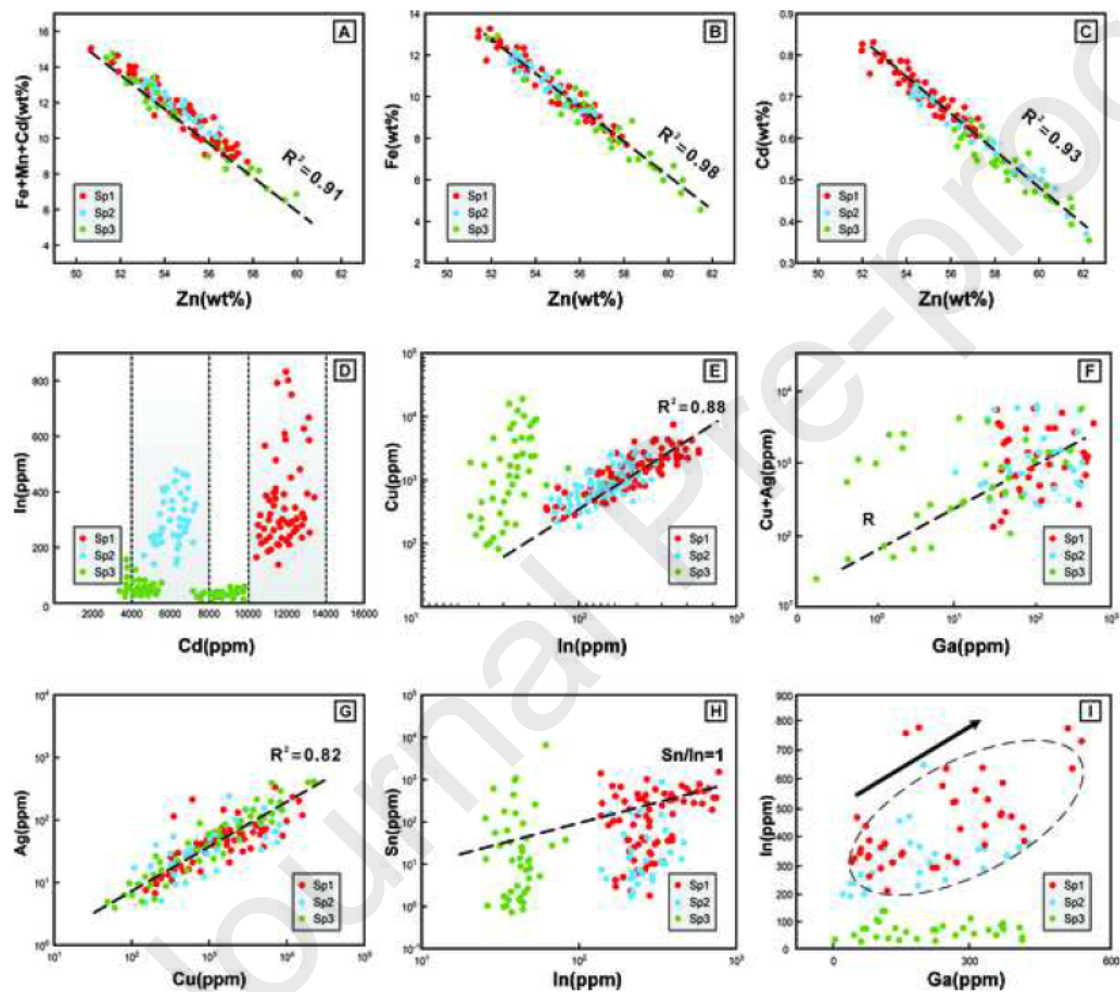


Figure 10

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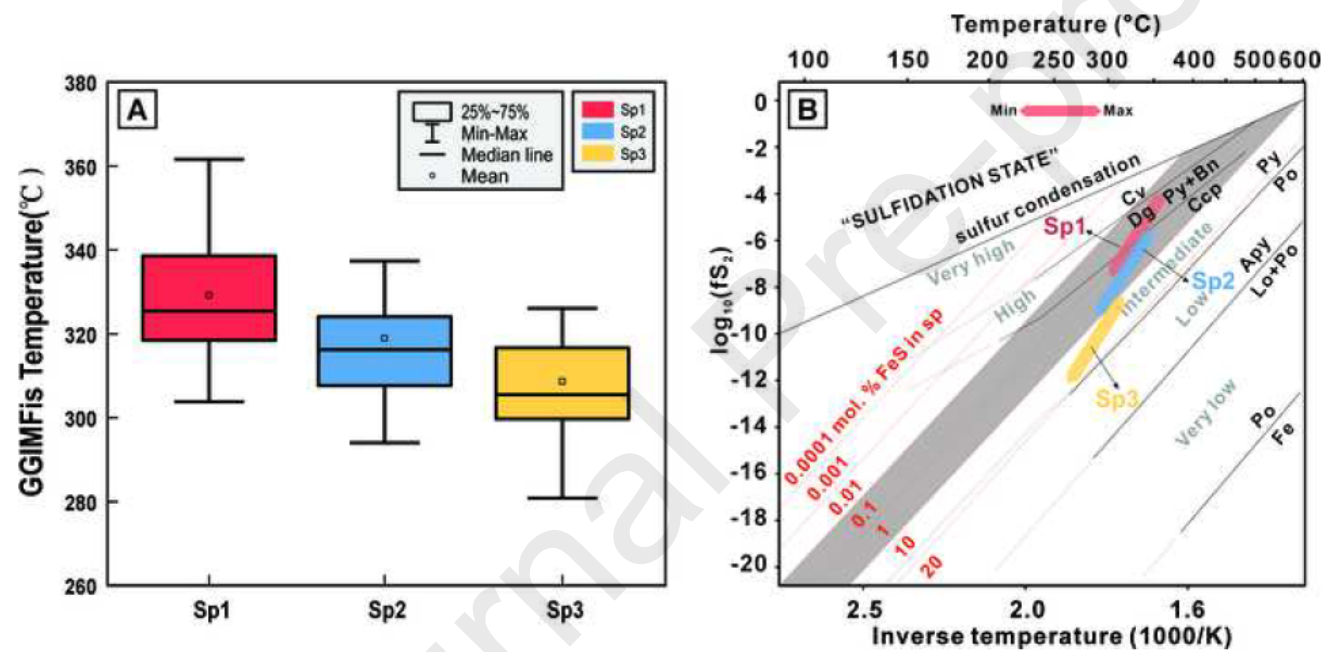


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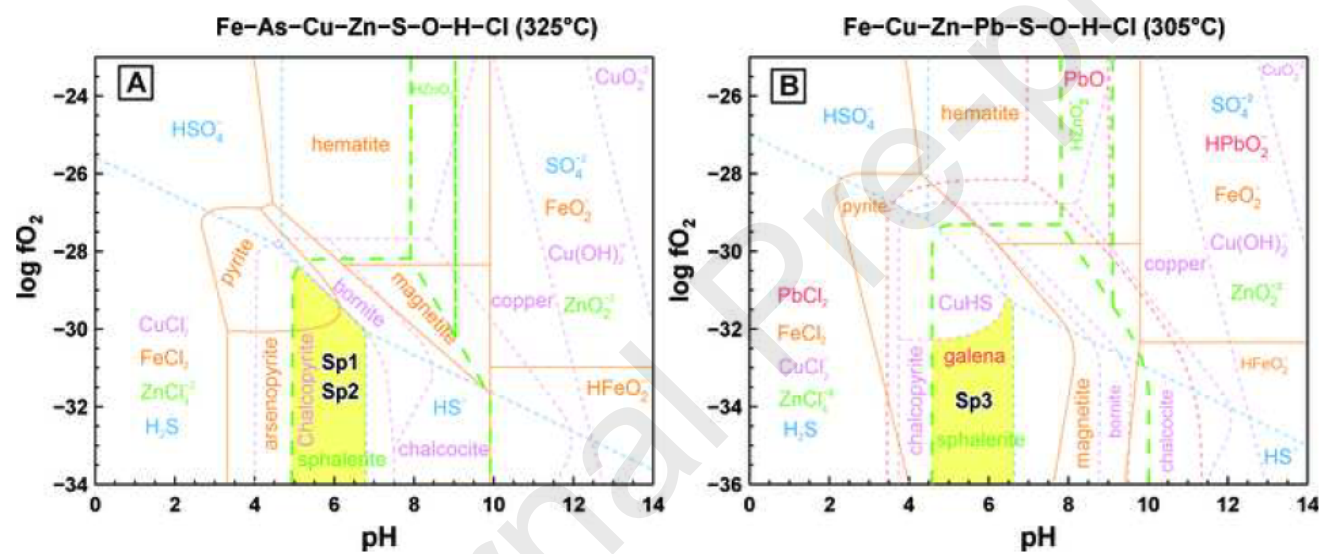


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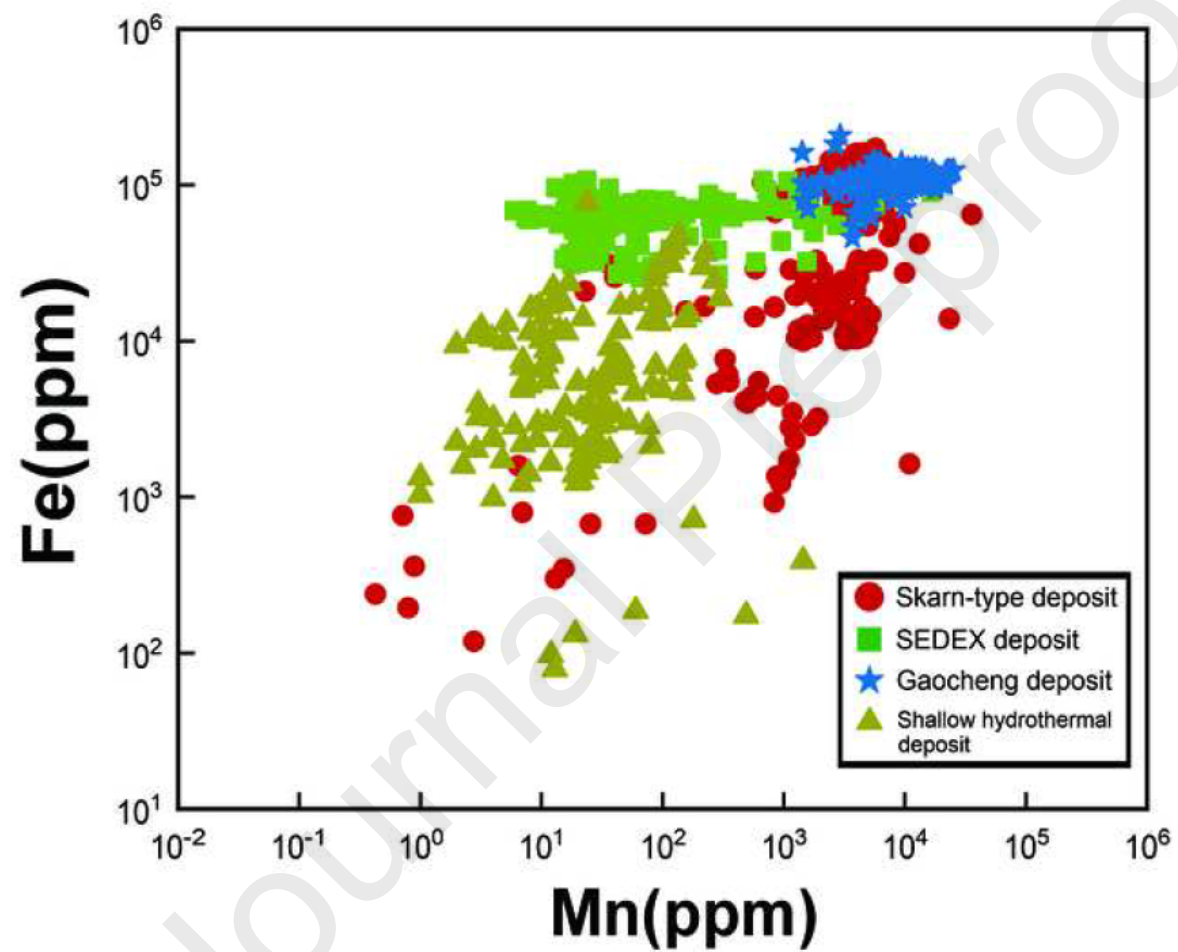


Table1 . EPMA data

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EPMA data.xlsx

Table 1. Summary statistics of electron probe micro-analyzer (EPMA) data for sph

Samples No.	S	Zn	Fe	Cd	In	Sn	Mn
Gaocheng deposit							
Sp1 n=41							
GC-1-1A-1-1	33.02	52.46	12.57	0.59	0.04	0.12	1.10
GC-1-1A-1-2	33.07	52.23	12.44	0.60	0.06	0.03	0.98
GC-1-1A-1-3	33.04	52.52	12.00	0.62	0.03	0.01	1.08
GC-1-1A-1-4	32.74	56.57	8.80	0.64	0.02	0.00	0.27
GC-1-1A-1-5	32.98	52.21	12.35	0.61	0.02	0.02	0.98
GC-1-1A-2-1	32.75	52.99	11.93	0.64	0.06	0.18	0.52
GC-1-1A-2-2	33.05	52.19	12.20	0.60	0.03	0.00	1.06
GC-1-1A-2-3	32.36	55.11	9.60	0.62	0.02	0.00	0.23
GC-1-1A-2-4	33.08	54.25	10.73	0.72	0.02	0.00	0.31
GC-1-1B-1-3	33.33	52.23	12.30	0.56	0.05	0.00	0.97
GC-1-1B-1-4	33.18	56.17	9.01	0.81	0.08	0.01	0.10
GC-1-1B-2-1	33.17	52.27	12.36	0.54	0.03	0.20	0.96
GC-1-1B-2-2	33.17	55.62	8.54	0.64	0.05	0.25	0.15
GC-1-1B-2-3	33.41	55.25	9.52	0.60	0.04	0.01	0.21
GC-1-1B-2-4	33.24	53.59	11.47	0.62	0.02	0.04	0.37
GC-1-1B-2-5	33.32	54.36	11.03	0.59	0.04	0.04	0.50
GC-1-1B-3-1	33.56	52.56	12.27	0.59	0.00	0.00	0.91
GC-1-1B-4-1	33.22	53.49	12.02	0.59	0.03	0.00	0.66
GC-1-2C-1-1	33.18	56.86	8.93	0.61	0.02	0.03	0.15
GC-1-2C-1-3	32.83	55.49	10.54	0.58	0.01	0.00	0.13
GC-1-2C-2-1N	33.18	53.33	10.50	0.57	0.03	0.00	0.46
GC-1-2C-3-1	32.92	56.00	10.70	0.61	0.04	0.00	0.11
GC-1-2C-3-3	32.84	52.12	12.52	0.54	0.06	0.02	1.11
GC-1-2C-3-4	33.30	52.75	12.56	0.57	0.01	0.00	0.64
GC-1-2D-1-2	32.82	56.23	9.62	0.68	0.05	0.01	0.13
GC-1-2D-2-1	32.83	54.50	10.94	0.63	0.03	0.00	0.27
GC-1-2D-3-3	33.30	53.40	12.63	0.61	0.03	0.01	0.42
GC-1-2D-3-4	33.11	54.56	10.66	0.65	0.06	0.02	0.37
GC-1-3A-1-1	33.12	51.93	12.25	0.55	0.05	0.04	1.13
GC-1-3A-2-1	33.19	53.93	11.44	0.51	0.01	0.00	0.50
GC-1-3A-2-2	32.92	53.37	12.06	0.53	0.02	0.01	0.65
GC-1-3A-2-3	33.15	52.25	12.39	0.51	0.03	0.02	0.83
GC-1-3A-3-2	32.65	54.78	10.84	0.55	0.03	0.00	0.41
GC-1-3A-3-3	33.31	51.31	12.60	0.55	0.01	0.28	0.98
GC-1-3B-1-2	33.20	53.81	11.11	0.62	0.03	0.00	0.39
GC-1-3B-1-3	32.72	54.31	11.42	0.58	0.03	0.00	0.23
GC-1-3B-2-2N	32.80	53.03	11.58	0.58	0.05	0.01	0.44
GC-1-3B-2-3	33.17	51.88	12.30	0.57	0.03	0.01	0.93
GC-1-3B-3-1	33.25	52.45	12.59	0.51	0.01	0.00	0.88
GC-1-3B-4-1	33.11	54.16	11.49	0.67	0.02	0.00	0.39
GC-1-3B-4-2	32.95	55.18	10.17	0.55	0.03	0.00	0.17
Max	33.56	56.86	12.63	0.81	0.08	0.28	1.13
Min	32.36	51.31	8.54	0.51	0.00	0.00	0.10
Median	33.11	53.49	11.49	0.59	0.03	0.01	0.46
Mean	33.06	53.75	11.29	0.60	0.03	0.03	0.56
Sp2 n=38							
GC-1-1C-1-1	33.11	52.76	12.64	0.61	0.01	0.00	0.75
GC-1-1C-1-2	32.77	55.53	9.30	0.71	0.02	0.00	0.18
GC-1-1C-1-3	33.42	54.15	11.06	0.63	0.04	0.01	0.41

GC-1-1C-1-4	33.16	51.72	12.71	0.52	0.04	0.08	1.29
GC-1-1C-2-1	33.18	51.69	12.32	0.54	0.04	0.00	1.14
GC-1-1C-3-1	33.50	51.80	12.72	0.61	0.06	0.05	1.07
GC-1-1C-4-1	33.08	52.76	12.78	0.61	0.02	0.00	0.65
GC-1-1C-4-2	32.91	55.02	11.08	0.57	0.03	0.00	0.11
GC-1-1C-4-3	33.35	52.76	11.97	0.54	0.09	0.00	1.15
GC-1-2B-1-1	32.95	55.59	10.48	0.60	0.01	0.00	0.15
GC-1-2B-2-1	32.78	57.05	9.23	0.53	0.03	0.00	0.14
GC-1-2B-2-3	33.23	54.33	11.39	0.63	0.01	0.00	0.37
GC-1-2B-3-2	33.00	52.97	11.62	0.60	0.03	0.00	0.56
GC-1-3D-1-2	33.08	54.05	10.91	0.59	0.03	0.00	0.31
GC-1-3D-1-4	33.10	55.26	10.43	0.60	0.02	0.00	0.16
GC-1-3D-1-5	32.55	53.95	11.33	0.57	0.02	0.00	0.36
GC-1-3D-2-2	33.34	54.63	10.02	0.64	0.05	0.00	0.11
GC-1-3D-3-2	33.57	52.68	12.25	0.55	0.03	0.03	1.08
GC-2-1B-1-2	33.24	52.22	11.61	0.36	0.05	0.00	1.97
GC-2-1B-1-3	33.03	60.09	5.46	0.43	0.01	0.00	0.36
GC-2-1B-2-1	33.13	56.91	7.83	0.44	0.00	0.00	0.38
GC-2-1B-2-3	33.06	61.30	4.40	0.45	0.01	0.01	0.33
GC-3-1B-1-1	33.11	54.56	10.93	0.47	0.00	0.00	0.62
GC-3-1B-3-1	33.20	52.92	11.85	0.44	0.04	0.00	1.12
GC-3-1B-4-1	33.04	55.11	10.37	0.49	0.00	0.00	0.31
GC-3-2B-1-1	33.30	52.80	11.83	0.36	0.02	0.00	0.99
GC-3-2B-2-1	32.94	53.43	11.75	0.48	0.02	0.00	0.69
GC-3-2B-3-1	33.07	56.97	8.66	0.40	0.01	0.00	0.19
GC-3-2B-3-2	32.98	53.23	11.96	0.41	0.02	0.00	0.95
GC-4-3A-1-1	33.30	53.28	11.93	0.46	0.00	0.01	0.82
GC-4-3A-2-1	33.29	52.85	11.58	0.49	0.03	0.05	0.92
GC-4-3A-3-1	33.47	52.47	12.00	0.47	0.02	0.00	0.89
GC-4-3A-3-2	33.29	53.23	11.23	0.53	0.01	0.00	0.54
GC-5-1-1-1	33.01	53.06	12.15	0.50	0.02	0.00	0.68
GC-5-1-1-2	32.82	53.83	11.42	0.54	0.02	0.00	0.54
GC-5-1-2-1	33.18	53.76	11.58	0.54	0.03	0.00	0.65
GC-5-1-3-1	33.19	53.20	11.64	0.50	0.05	0.00	0.64
GC-5-1-4-1	33.16	52.94	11.64	0.52	0.05	0.00	0.61
Max	33.57	61.30	12.78	0.71	0.09	0.08	1.97
Min	32.55	51.69	4.40	0.36	0.00	0.00	0.11
Median	33.12	53.35	11.58	0.53	0.02	0.00	0.62
Mean	33.13	54.13	10.95	0.52	0.03	0.01	0.64
Sp3 n=32							
GC-1-2A-1-1	33.04	55.08	10.74	0.60	0.04	0.00	0.28
GC-1-2A-1-2	33.15	52.45	12.80	0.54	0.09	0.00	1.15
GC-1-2A-2-1	33.28	52.94	12.39	0.56	0.02	0.00	0.92
GC-1-2A-2-2	33.38	55.15	9.94	0.56	0.02	0.00	0.18
GC-1-2A-2-3	33.11	52.76	12.20	0.58	0.04	0.00	0.67
GC-1-2A-3-1	33.23	52.66	12.02	0.59	0.05	0.00	0.83
GC-1-2A-3-2	32.26	54.03	10.93	0.64	0.02	0.03	0.12
GC-1-2A-4-1	33.43	51.46	12.78	0.51	0.09	0.01	1.31
GC-1-2A-4-2	33.40	54.92	11.11	0.57	0.05	0.00	0.41
GC-2-2B-1-1	32.90	59.62	5.73	0.47	0.00	0.00	0.43
GC-2-2B-1-2	33.19	51.39	12.08	0.35	0.01	0.01	2.10
GC-2-2B-2-2	32.70	56.20	8.62	0.46	0.01	0.00	0.89

GC-2-2B-3-1	33.45	51.12	12.18	0.38	0.00	0.00	2.17
GC-2-2D-1-1	33.03	51.46	12.01	0.35	0.00	0.08	2.14
GC-2-2D-1-2	33.00	53.75	10.39	0.39	0.00	0.00	0.70
GC-2-2D-2-1	33.24	54.95	9.82	0.48	0.00	0.01	1.16
GC-2-2D-3-1	33.49	50.33	12.82	0.37	0.00	0.00	2.47
GC-2-3B-1-1	33.23	50.45	13.19	0.41	0.00	0.00	2.21
GC-2-3B-1-2	33.12	55.49	9.61	0.42	0.02	0.00	0.51
GC-2-3B-2-1	33.48	54.54	9.80	0.40	0.00	0.00	0.58
GC-2-3B-2-2	32.72	52.11	12.21	0.40	0.01	0.00	1.10
GC-2-3B-3-1	33.33	50.84	12.69	0.38	0.00	0.00	1.97
GC-2-4-1-2	32.90	51.07	11.87	0.41	0.00	0.00	1.89
GC-2-4-2-1	33.27	50.26	13.15	0.40	0.00	0.02	2.30
GC-2-4-3-1	33.04	61.32	4.91	0.33	0.03	0.00	0.34
GC-2-1A-1-2	32.88	57.20	8.03	0.46	0.02	0.00	0.53
GC-2-1A-1-1	33.08	57.73	8.22	0.48	0.01	0.00	0.66
GC-2-1A-1-3	33.32	54.46	10.44	0.45	0.00	0.00	0.94
GC-2-1A-2-1	33.39	52.35	11.35	0.43	0.01	0.00	1.76
GC-2-1A-2-2	33.45	50.92	12.03	0.40	0.01	0.01	2.13
GC-2-1A-3-1	32.87	55.69	9.42	0.53	0.00	0.00	0.87
GC-2-1A-3-2	33.03	61.76	3.58	0.40	0.01	0.00	0.28
Max	33.49	61.76	13.19	0.64	0.09	0.08	2.47
Min	32.26	50.26	3.58	0.33	0.00	0.00	0.12
Median	33.14	53.95	10.59	0.46	0.02	0.01	1.12
Mean	33.14	53.95	10.59	0.46	0.02	0.01	1.12

valerite (wt.%)	
Cu	Total
0.25	100.15
0.06	99.46
0.03	99.32
0.09	99.13
0.06	99.24
0.27	99.34
0.02	99.16
0.64	98.57
0.42	99.53
0.00	99.43
0.04	99.40
0.28	99.80
0.02	98.42
0.07	99.11
0.01	99.36
0.11	99.98
0.05	99.93
0.16	100.17
0.08	99.85
0.12	99.70
0.05	98.12
0.04	100.41
0.11	99.32
0.01	99.84
0.20	99.74
0.04	99.23
0.03	100.42
0.15	99.57
0.08	99.13
0.05	99.63
0.01	99.56
0.04	99.23
0.00	99.25
0.36	99.41
0.06	99.21
0.01	99.30
0.00	98.48
0.15	99.04
0.02	99.70
0.07	99.91
0.01	99.06
0.64	100.42
0.00	98.12
0.06	99.40
0.10	99.43
0.00	99.88
0.05	98.55
0.05	99.77

0.07	99.58
0.01	98.92
0.09	99.90
0.00	99.90
0.00	99.72
0.06	99.92
0.11	99.89
0.01	99.77
0.08	100.04
0.02	98.78
0.02	98.98
0.02	99.58
0.03	98.80
0.02	98.82
0.11	100.30
0.04	99.49
0.33	99.70
0.17	98.86
0.06	99.60
0.18	99.86
0.04	99.60
0.06	99.37
0.01	99.32
0.04	99.34
0.04	99.34
0.02	99.57
0.05	99.85
0.11	99.32
0.09	99.41
0.02	98.85
0.07	99.49
0.09	99.27
0.05	99.80
0.05	99.27
0.01	98.93
0.33	100.30
0.00	98.55
0.05	99.53
0.06	99.46
0.10	99.88
0.03	100.20
0.05	100.15
0.00	99.24
0.09	99.44
0.08	99.45
0.01	98.04
0.10	99.69
0.05	100.50
0.14	99.29
0.02	99.15
0.19	99.07

0.03	99.32
0.16	99.23
0.09	98.33
0.16	99.82
0.00	99.48
0.00	99.50
0.04	99.20
0.03	98.82
0.01	98.55
0.02	99.22
0.01	98.14
0.00	99.38
0.02	99.99
0.07	99.18
0.01	100.19
0.19	99.81
0.04	99.32
0.03	98.98
0.14	99.52
0.07	99.13
0.19	100.50
0.00	98.04
0.06	99.35
0.06	99.35

Table2 . LA-ICP-MS data

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Table 2. Summary statistics of LA-ICP-MS element content and estimated temperature for the

Samples No.	Mn55	Fe57	Cu65	Ga71	Ge72	Ag107	Cd111	In115
Gaocheng deposit								
Sp1 n=56								
GC-1-1A@1	12127.01	125188.08	2343.68	221.68	2.24	131.36	9681.24	485.22
GC-1-1A@2	10962.80	123034.44	1619.38	181.80	BDL	55.74	11904.75	609.86
GC-1-1A@3	11619.68	124602.78	1160.62	224.99	2.07	54.94	11450.03	513.85
GC-1-1A@4	11587.28	120245.59	4781.46	188.35	2.77	74.09	11340.17	349.98
GC-1-1A@5	6365.07	118853.60	2572.54	124.12	BDL	95.02	12084.13	801.54
GC-1-1A@6	9540.97	117631.73	2446.37	180.70	2.15	63.73	10886.47	565.67
GC-1-1A@7	11938.20	124584.87	355.62	202.70	3.03	24.09	10454.73	664.07
GC-1-1A@8	3145.02	98467.25	7007.66	32.55	BDL	118.91	11126.06	369.26
GC-1-1C@1	1881.81	90414.68	1615.68	53.06	BDL	961.51	8958.84	417.52
GC-1-1C@2	11506.59	122580.01	2123.81	298.39	BDL	47.24	10942.45	386.67
GC-1-1C@3	13791.65	123734.45	2822.74	178.60	BDL	48.97	10375.75	251.35
GC-1-1C@4	12423.24	128095.89	1130.10	197.80	BDL	54.27	11052.10	284.69
GC-1-1C@5	4472.13	114133.49	349.73	25.05	BDL	114.12	11163.31	319.46
GC-1-1C@6	12041.98	127243.91	1336.40	306.28	BDL	62.75	12891.45	626.56
GC-1-1C@7	10587.18	105122.93	690.85	377.71	BDL	42.04	13072.14	393.45
GC-1-1C@8	12069.16	101745.06	2656.75	273.36	BDL	60.04	13149.42	667.12
GC-1-1C@9	9460.81	103176.95	1153.39	302.03	BDL	42.91	13233.96	387.94
GC-1-1C@10	12525.88	124876.19	1520.36	248.04	BDL	64.54	13168.82	585.62
GC-1-2A@1	12031.19	125299.18	1482.16	521.29	2.57	74.87	12250.29	749.94
GC-1-2A@2	13591.89	125727.49	1451.54	409.24	2.35	60.86	11894.58	338.24
GC-1-2A@3	20880.89	123038.98	1691.27	90.77	BDL	54.35	11520.85	792.21
GC-1-2A@4	10280.43	118802.83	894.86	191.96	BDL	30.24	12294.46	292.47
GC-1-2A@5	11084.11	105241.28	2454.00	251.90	BDL	49.81	12701.33	480.64
GC-1-2A@6	4869.85	100554.17	6446.43	139.67	2.08	201.13	13191.48	352.73
GC-1-2A@8	15101.43	106383.47	1327.01	477.67	2.58	66.34	11971.35	831.97
GC-1-2C@1	2696.78	101978.45	4569.03	50.51	2.94	58.42	12809.22	280.36
GC-1-2C@2	9278.13	115328.83	3701.35	261.47	2.71	63.57	12230.99	451.15
GC-1-2C@3	1876.02	95992.87	827.76	200.15	BDL	36.83	12141.70	282.67
GC-1-2C@4	5316.70	111137.19	542.74	44.88	BDL	16.32	10043.79	372.47
GC-1-2C@5	9429.84	104175.05	239.94	55.17	3.34	11.72	10926.90	224.31
GC-1-2C@6	13569.50	100927.91	4029.41	512.51	2.38	81.75	11945.75	588.48
GC-1-2C@7	13185.57	103197.30	1146.17	411.26	BDL	43.35	11442.07	412.09
GC-1-2C@8	6653.90	101007.10	1830.28	30.16	3.14	20.83	11617.69	290.42
GC-1-1B@1	10609.14	127221.10	399.13	40.38	3.39	29.81	9220.45	485.89
GC-1-1B@2	2857.57	108237.96	3755.33	39.75	3.61	31.44	11424.74	400.46
GC-1-1B@3	9137.72	118613.70	1339.92	256.76	BDL	55.61	9589.98	307.15
GC-1-1B@4	3511.82	113361.93	1845.31	290.31	BDL	68.49	9862.12	306.75
GC-1-1B@5	11464.78	123292.34	1103.59	292.16	BDL	40.37	9546.13	413.97
GC-1-1B@6	13562.15	124823.46	3901.05	245.21	BDL	83.93	9675.42	463.10
GC-1-1B@7	9399.39	138890.85	7088.04	77.54	2.35	334.72	9754.33	268.58

GC-1-1B@8	6306.71	139279.05	5413.11	44.67	BDL	71.88	10538.98	380.16
GC-1-1B@9	9449.26	125124.33	547.97	41.07	BDL	16.09	10160.74	339.53
GC-1-2B@1	1825.53	221150.83	1151.99	51.48	3.51	61.96	11137.22	337.52
GC-1-2B@2	5733.45	92150.80	970.99	37.91	2.44	50.44	11331.38	261.37
GC-1-2B@3	3542.31	108114.81	4888.45	134.21	BDL	64.42	12495.97	270.92
GC-1-2B@4	4155.89	110367.14	6382.70	44.60	BDL	69.41	12519.79	264.11
GC-1-2B@5	7557.07	102535.18	650.66	56.21	3.42	24.36	11991.07	423.92
GC-1-2B@6	5621.35	110295.32	359.11	69.62	2.87	15.80	12037.14	321.99
GC-1-3D@1	5217.08	116406.49	6012.93	31.94	2.41	233.29	12689.17	266.62
GC-1-3D@2	5037.04	108158.34	521.33	42.17	2.24	25.77	11584.57	281.87
GC-1-3D@3	3099.57	102461.84	2952.93	61.90	BDL	28.28	12028.24	287.71
GC-1-3D@4	8852.83	101184.97	427.73	48.74	BDL	18.51	11868.05	390.88
GC-1-3D@5	8449.95	101993.76	1531.13	200.04	2.17	144.25	11790.48	301.31
GC-1-3D@6	10369.58	100710.82	210.18	32.33	2.82	9.40	12796.59	334.48
GC-1-3D@7	11687.24	117669.97	757.59	223.63	2.23	36.65	12390.49	355.95
GC-1-3D@8	8650.20	108116.63	201.26	300.46	BDL	6.70	12613.79	281.92
Max	20880.89	221150.83	7088.04	521.29	3.61	961.51	13233.96	831.97
Min	1825.53	90414.68	201.26	25.05	2.07	6.70	8958.84	224.31
Median	9439.55	113747.71	1501.26	181.25	2.58	55.28	11704.09	370.87
Mean	8821.26	115226.53	2191.67	177.30	2.69	80.50	11517.23	413.68
Sp2 n=52								
GC-1-2D@1	6463.32	99412.36	610.14	23.69	BDL	211.23	6917.55	304.65
GC-1-2D@2	6763.19	105916.51	808.19	65.51	2.24	41.49	5541.08	303.49
GC-1-2D@3	5878.94	144552.84	555.47	66.09	2.83	12.00	6645.49	316.13
GC-1-2D@4	7355.51	120120.17	1921.52	138.06	BDL	32.45	5452.95	343.04
GC-1-3A@1	10557.26	127453.05	1165.15	74.68	2.15	48.20	6428.51	374.79
GC-1-3A@2	10649.58	126651.20	948.30	77.02	3.16	48.34	6335.52	395.81
GC-1-3A@3	6412.34	117267.47	836.53	67.03	BDL	24.98	6613.99	207.75
GC-1-3A@4	6469.39	117956.51	795.82	11.93	3.68	11.34	5759.72	232.63
GC-1-3A@5	11666.36	94243.00	643.39	21.15	BDL	13.05	6678.38	272.60
GC-1-3A@6	11756.74	92198.29	1694.40	133.99	BDL	46.03	6580.97	337.76
GC-1-3A@7	6987.91	115953.12	5057.71	157.98	BDL	36.98	5178.14	241.10
GC-1-3B@1	10131.19	91961.00	577.40	84.52	2.25	27.81	6523.49	210.58
GC-1-3B@2	7679.46	92906.49	629.55	20.23	3.05	10.68	6718.11	214.54
GC-1-3B@3	6160.40	119670.41	774.03	20.35	BDL	14.85	6590.40	239.79
GC-1-3B@4	11615.29	124252.27	1687.62	390.01	BDL	39.89	7507.31	272.01
GC-1-3B@5	9679.96	91840.60	465.30	100.72	2.11	15.09	6195.09	191.60
GC-1-3B@6	11289.88	119842.37	653.21	170.67	BDL	7.49	6407.83	171.10
GC-1-3B@7	6734.28	113558.47	9177.28	73.23	2.05	40.37	7141.87	209.68
GC-1-3B@8	6736.29	97683.18	819.69	12.96	BDL	25.14	7524.76	203.82
GC-1-3B@9	6692.38	91598.32	943.02	80.13	3.72	27.74	7320.47	171.81
GC-3-1B@1	5835.11	95353.23	7815.58	56.30	BDL	31.90	5473.21	243.83
GC-3-1B@6	9957.92	98799.58	439.96	180.69	BDL	8.50	4979.24	221.34
GC-3-1B@7	11136.05	94819.01	466.01	50.40	BDL	3.93	5208.44	213.60

GC-3-1B@8	10334.57	98821.85	1017.77	91.22	BDL	11.36	5180.87	242.70
GC-3-1B@9	9240.04	98496.08	441.73	89.05	BDL	5.08	4991.06	236.17
GC-3-1B@10	7735.42	99104.38	451.98	60.32	BDL	4.77	5177.58	139.88
GC-3-2B@1	5837.17	98837.27	8325.58	33.52	BDL	60.44	5192.66	215.89
GC-3-2B@2	7578.33	96945.74	588.44	32.00	BDL	8.98	5123.61	215.42
GC-3-2B@3	5921.23	176968.92	518.89	82.41	BDL	29.18	3811.02	249.51
GC-3-2B@4	5595.63	93673.90	5604.51	26.04	BDL	33.67	5553.07	218.38
GC-3-2B@5	7421.57	95053.37	1244.54	53.04	BDL	15.36	5236.05	219.65
GC-3-2B@6	5804.21	97435.89	7484.99	64.64	BDL	47.05	5280.09	208.64
GC-3-2B@7	8581.91	100576.40	4714.57	28.78	BDL	33.18	5096.53	226.27
GC-4-3A@1	8132.63	102373.35	817.83	351.92	BDL	32.61	6307.08	241.00
GC-4-3A@2	8065.68	104407.75	591.46	335.99	BDL	28.78	6377.26	193.95
GC-4-3A@3	8123.24	104610.65	1223.46	288.92	BDL	58.69	6497.06	275.81
GC-4-3A@4	8786.56	105179.46	509.02	203.14	BDL	29.82	6364.53	258.15
GC-4-3A@5	8696.10	103865.07	1642.07	440.79	BDL	84.10	6131.69	337.49
GC-4-3A@6	7912.46	104490.94	4640.00	162.48	BDL	87.10	6461.72	211.09
GC-4-3A@7	8826.70	107189.19	1002.10	313.77	BDL	39.51	6434.79	236.70
GC-4-3A@8	8253.55	105943.81	5096.40	259.31	BDL	155.32	6496.23	243.66
GC-5-1@1	7100.04	108303.82	608.19	165.91	BDL	33.31	6770.68	267.68
GC-5-1@2	8673.41	107770.83	1018.19	36.09	BDL	50.39	7179.29	332.40
GC-5-1@3	5966.29	103781.04	1813.83	114.41	BDL	240.95	7298.66	354.32
GC-5-1@4	7083.57	110216.46	420.12	66.64	BDL	19.63	6754.48	333.86
GC-5-1@5	8181.61	110198.14	523.93	171.74	BDL	31.48	6829.91	272.28
GC-5-1@6	8339.23	92778.58	948.96	200.95	BDL	28.19	6752.00	295.78
GC-5-1@7	8352.12	112577.24	417.10	50.82	BDL	12.14	6773.17	255.03
GC-5-1@8	8919.90	109763.97	2749.71	30.86	BDL	57.35	6956.01	294.68
GC-5-1@9	7479.63	91640.99	1101.11	150.10	BDL	60.06	6496.94	662.24
GC-5-1@10	6443.92	110151.40	495.00	101.20	BDL	27.65	6640.41	466.20
GC-5-1@11	8551.46	114163.02	478.84	115.03	BDL	22.98	6304.97	477.11
Max	11756.74	176968.92	9177.28	440.79	3.72	240.95	7524.76	662.24
Min	5595.63	91598.32	417.10	11.93	2.05	3.93	3811.02	139.88
Median	7989.07	104449.35	828.11	81.27	2.54	30.65	6431.65	243.18
Mean	8087.44	106910.17	1845.68	119.20	2.72	40.93	6196.00	270.68
Sp3 n=49								
GC-2-2B@1	6094.18	90781.39	2486.27	2.12	BDL	411.77	8135.92	42.66
GC-2-2B@2	23148.69	119783.13	747.49	11.53	BDL	22.06	8107.91	39.22
GC-2-2B@3	21497.53	115513.91	1417.94	30.39	1.87	53.38	7619.97	23.42
GC-2-2B@4	14582.51	108066.67	7730.24	44.70	BDL	946.77	9162.66	38.36
GC-2-2B@5	5607.81	83724.41	9683.82	9.52	BDL	1312.60	9209.25	39.62
GC-2-2B@6	7999.99	91343.49	8994.32	27.40	BDL	398.73	9442.91	35.03
GC-2-2B@7	23364.03	120908.85	1777.10	6.60	BDL	29.07	7745.63	37.03
GC-2-2B@8	24473.08	91943.94	197.33	2.89	BDL	53.42	8146.59	35.74
GC-2-2D@1	22963.20	97556.96	1581.94	222.78	1.93	83.86	7597.06	7.82
GC-2-2D@2	21907.72	121817.89	1348.65	373.72	2.99	67.85	8339.54	19.66

GC-2-2D@3	23484.66	121971.60	758.21	373.84	2.35	88.32	8154.84	6.15
GC-2-2D@4	22787.85	123543.95	93.26	5.13	2.10	92.73	8418.89	40.58
GC-2-2D@5	17777.53	95954.50	4521.16	313.75	2.48	107.04	8956.64	28.67
GC-2-2D@6	23418.12	93629.02	135.16	5.15	BDL	42.96	8428.98	34.69
GC-2-2D@7	40371.47	124706.81	96.46	4.13	BDL	41.06	8417.39	34.17
GC-2-2D@8	24390.20	101749.95	99.07	5.21	BDL	20.79	8244.79	33.41
GC-2-1A@1	18199.65	84822.10	6421.87	29.78	BDL	389.23	5432.43	36.58
GC-2-1A@2	10271.90	83311.99	2339.51	107.47	BDL	733.11	5220.46	39.93
GC-2-1A@3	18452.31	107765.76	2408.19	173.06	BDL	79.60	4725.54	43.85
GC-2-1A@6	28770.76	83703.84	8317.14	10.86	BDL	191.88	5128.03	67.62
GC-2-1A@7	21224.70	88983.54	9777.21	15.73	BDL	967.47	5539.25	71.73
GC-2-1B@1	21333.49	107524.01	517.63	2.12	BDL	15.59	3931.29	41.85
GC-2-1B@2	21233.95	109580.68	2443.83	6.43	BDL	49.04	4399.56	46.15
GC-2-1B@3	28080.77	99734.39	688.56	74.47	BDL	33.44	4233.33	177.15
GC-2-1B@4	19119.99	104570.00	2575.86	250.48	BDL	72.23	4174.04	85.98
GC-2-1B@5	5247.27	82608.77	206.25	4.85	BDL	26.45	4534.43	49.03
GC-2-1B@6	23066.83	103404.12	4614.04	3.38	BDL	111.00	5018.86	43.32
GC-2-1C@1	21229.76	107191.70	173.87	22.51	BDL	8.93	4866.62	37.75
GC-2-1C@2	19365.22	98721.29	8875.35	368.60	BDL	698.17	5000.87	61.60
GC-2-1C@3	19628.85	101936.30	194.19	83.72	BDL	13.25	4180.48	118.97
GC-2-1C@4	20220.37	107798.10	1145.97	61.71	BDL	23.96	3763.16	56.58
GC-2-1C@5	20177.27	107861.97	2958.64	71.14	BDL	145.59	4399.63	82.52
GC-2-1C@6	20229.95	107326.94	394.63	85.18	BDL	21.39	4469.13	81.65
GC-2-1C@7	21223.14	90815.66	5078.34	56.75	BDL	107.39	3731.16	56.62
GC-2-3B@1	19949.43	104106.67	2562.02	2.14	BDL	95.18	4903.34	44.15
GC-2-3B@2	13897.84	98067.76	4131.63	200.66	BDL	212.03	5101.63	39.13
GC-2-3B@3	20919.41	85670.19	1316.37	4.20	BDL	40.31	3898.22	47.68
GC-2-3B@4	21517.75	102124.84	121.80	1.61	BDL	29.39	4496.81	44.88
GC-2-3B@5	5983.78	70609.60	2517.17	130.40	BDL	93.12	3667.92	43.29
GC-2-3B@6	9424.33	83135.35	4496.87	70.80	BDL	90.58	4641.38	48.38
GC-2-3B@7	21459.14	95195.73	941.54	2.32	BDL	38.40	4411.99	47.29
GC-2-3B@8	21721.00	100325.82	1404.35	1.86	BDL	52.84	4699.47	42.92
GC-2-4@1	21173.60	102338.42	241.16	8.06	BDL	18.75	3687.73	37.48
GC-2-4@2	22561.40	90109.99	101.80	6.04	BDL	16.25	3379.76	42.59
GC-2-4@3	21994.14	94772.50	231.21	229.44	BDL	21.24	4499.70	26.25
GC-2-4@4	18150.47	79798.73	147.87	10.65	BDL	17.18	4093.86	46.32
GC-2-4@5	19953.11	71601.89	81.20	1.10	BDL	16.10	3800.13	48.19
GC-2-4@6	15259.99	97526.55	705.66	1.73	BDL	32.49	4753.59	44.34
GC-2-4@7	9695.96	85836.96	976.18	21.69	BDL	59.61	3972.69	42.26
Max	40371.47	124706.81	9777.21	373.84	2.99	1312.60	9442.91	177.15
Min	5247.27	70609.60	81.20	1.10	1.87	8.93	3379.76	6.15
Median	21173.60	98721.29	1348.65	15.73	2.23	53.42	4866.62	42.66
Mean	19277.68	98813.85	2464.82	72.65	2.29	169.26	5732.36	47.76

1 Gaocheng sphalerite (ppm)

Sn118	Hg202	Pb208	T (°C)
617.55	10.96	45.39	323.88
267.89	6.02	0.32	325.89
478.26	4.92	0.67	324.43
461.45	5.75	4.96	320.03
1693.01	6.06	0.16	347.83
523.40	6.32	0.42	323.89
29.27	5.27	6.46	314.60
35.42	6.19	17.16	346.46
34.46	3.99	10.76	352.01
327.70	7.26	10.50	313.84
327.49	6.30	0.86	320.79
672.24	5.35	115.90	352.79
28.36	19.04	178.47	351.11
532.90	5.60	0.63	337.41
36.01	5.39	0.61	339.85
600.90	5.70	2.66	322.36
442.60	5.30	1.57	317.12
773.73	5.50	0.48	324.28
370.60	4.55	0.62	314.55
242.09	4.72	0.64	315.14
56.85	5.19	18.32	341.55
27.30	4.52	1.36	321.54
50.15	3.13	1.54	322.92
41.69	5.30	377.29	315.73
193.00	4.07	0.62	318.83
193.74	5.09	453.20	314.58
720.80	4.51	1.58	314.65
108.67	5.82	33.12	352.75
21.94	11.88	596.16	336.87
12.14	5.92	384.90	328.23
763.36	4.95	0.92	314.80
351.81	4.76	1.49	317.28
110.24	5.75	20.82	360.41
38.15	5.56	55.07	361.30
31.51	7.86	242.68	318.94
587.69	6.27	3.85	331.59
197.45	6.39	244.24	347.15
348.04	6.13	0.95	342.17
698.24	5.64	1.71	333.36
136.03	7.38	317.28	331.66

76.39	8.87	156.15	334.62
37.19	8.02	28.35	348.21
144.08	3.07	171.50	317.50
37.23	7.46	18.99	313.01
1013.39	5.68	202.97	332.91
27.30	4.71	6.53	354.79
24.98	3.31	2.87	339.94
84.74	3.70	8.87	319.21
430.98	2.61	14.11	332.00
104.13	5.47	8.34	318.26
17.12	4.50	26.17	303.17
45.66	3.23	21.23	352.53
29.00	8.84	55.54	350.31
17.01	2.88	0.23	337.40
347.64	3.19	0.72	318.37
18.78	2.41	1.52	345.89
1693.01	19.04	596.16	361.30
12.14	2.41	0.16	303.17
140.06	5.49	7.44	327.06
279.28	5.79	69.29	330.94
29.54	3.57	277.77	314.26
10.09	4.18	40.60	307.27
33.32	5.91	0.30	325.24
11.33	4.24	0.62	326.57
327.30	3.29	1.06	328.67
151.83	3.04	0.51	293.16
37.32	3.14	4.01	329.66
5.86	3.05	6.15	314.85
9.33	3.74	9.30	325.90
385.97	4.30	1.07	323.25
348.07	4.64	24.42	333.18
196.66	3.53	0.89	313.20
2.33	4.22	64.35	322.61
29.06	6.01	226.79	318.92
270.36	3.40	4.02	299.75
4.90	3.83	0.42	316.32
4.19	4.21	0.90	325.32
20.04	3.77	7.91	334.33
37.03	12.73	18.19	318.66
2.92	2.74	32.67	318.66
176.43	3.34	7.43	322.53
2.08	3.33	0.94	319.27
4.44	3.37	0.86	324.11

7.34	3.47	5.58	324.39
3.34	3.28	0.72	316.68
7.43	3.54	0.36	316.08
20.38	3.78	472.30	316.22
1.68	3.64	1.34	310.73
4.41	1.32	39.14	336.33
6.44	3.82	2.00	312.16
2.26	3.51	1.70	317.14
14.37	3.94	9.57	329.43
5.36	3.24	2.41	313.40
273.02	2.92	1.11	324.21
51.20	3.14	0.53	313.54
675.19	3.00	0.80	332.45
158.37	3.34	0.59	334.33
881.58	2.64	4.19	302.54
178.17	2.48	21.25	316.13
453.37	3.49	0.75	312.70
245.12	3.31	1.95	313.41
11.99	3.82	2.95	316.67
11.68	4.42	12.71	306.26
64.47	4.06	102.18	338.28
10.71	3.70	0.26	333.41
42.42	3.68	0.26	317.42
4.07	3.39	BDL	331.23
6.06	3.91	0.22	334.59
5.30	3.97	0.63	317.28
420.80	5.16	2.38	312.21
32.14	3.74	0.28	316.64
5.03	3.66	3.37	316.89
881.58	12.73	472.30	338.28
1.68	1.32	0.22	293.16
17.21	3.61	2.00	318.04
109.69	3.83	27.90	319.97
8.87	2.82	64.65	319.26
0.97	3.27	3.63	321.02
54.75	3.33	1.62	317.98
434.03	3.36	12.77	318.67
159.23	4.58	96.35	297.61
62.53	4.04	454.68	320.34
2.22	3.23	2.86	318.65
1.43	3.38	2.28	303.25
88.62	3.65	6.30	317.82
90.14	3.52	0.79	304.03

225.21	4.02	8.46	304.22
3.01	3.60	6.17	328.19
304.44	4.12	1.92	305.13
1.02	4.14	6.32	305.14
1.64	4.13	BDL	320.71
1.72	4.12	14.23	317.63
33.34	4.64	398.46	321.56
72.75	5.60	55.86	316.03
20.24	2.19	6.27	317.37
27.45	5.25	662.36	288.03
476.33	4.96	99.09	316.15
1.42	4.86	6.27	321.99
7.04	3.10	3.28	304.29
152.60	4.00	6.29	317.41
230.69	3.74	12.62	318.87
8.03	8.56	127.08	320.60
1.30	2.29	4.09	305.54
12.12	3.68	2.55	318.07
14.20	3.62	2.78	295.22
9.22	2.85	1.14	318.63
5.14	5.70	87.24	303.66
37.25	4.08	9.79	290.09
3.02	4.00	12.27	302.98
36.79	6.04	44.35	318.39
3.84	3.00	6.60	294.67
165.18	3.07	2.17	296.03
11.47	11.56	27.84	295.15
2.09	3.04	1.06	291.64
24.98	12.35	341.76	304.91
12.78	3.30	8.71	280.99
2.61	4.81	3.00	297.24
BDL	2.71	0.32	317.52
2.72	3.78	29.53	303.78
1.56	5.26	52.19	301.23
1.04	2.93	5.37	299.20
40.60	9.96	36.23	317.04
4.57	12.64	9.86	292.99
1.21	2.77	1.69	319.90
13.76	5.83	565.44	304.62
476.33	12.64	662.36	328.19
0.97	2.19	0.32	280.99
11.80	4.00	7.53	305.54
59.94	4.60	69.10	308.80

Table3 . Error range and detection limits of LA-ICP-MS data

[Click here to access/download;Table \(Editable version\);Table3 .](#) [Error range and detection limits of LA-ICP-MS data.xlsx](#)

Table 3. Er

Samples No.	Mn55_2S E(Int)	Mn55_L OD_Pett ke	Fe57_2S E(Int)	Fe57_LO D_Pettke	Cu65_2S E(Int)	Cu65_L OD_Pett ke	Ga71_2S E(Int)	Ga71_L OD_Pett ke
Gaocheng deposit								
Sp1 n=56								
GC-1-1A@1	88.18	1.59	1076.63	24.31	114.82	3.21	3.21	0.47
GC-1-1A@2	85.50	1.61	948.78	25.29	143.45	2.80	2.65	0.41
GC-1-1A@3	112.93	1.68	1359.06	19.26	19.05	2.74	3.73	0.39
GC-1-1A@4	104.13	1.58	1098.60	26.95	525.21	2.72	2.83	0.33
GC-1-1A@5	45.46	1.62	921.01	25.71	28.72	3.15	1.97	0.28
GC-1-1A@6	80.59	1.59	925.47	22.28	251.38	2.95	2.31	0.23
GC-1-1A@7	123.77	1.53	1124.60	25.43	10.02	2.95	3.42	0.42
GC-1-1A@8	22.72	1.61	1071.97	24.21	855.69	3.43	2.85	0.34
GC-1-1C@1	169.28	1.81	2080.23	28.38	13.48	3.20	0.69	0.44
GC-1-1C@2	39.89	1.64	1318.99	30.61	347.33	3.50	1.95	0.50
GC-1-1C@3	147.95	1.50	1124.58	23.95	55.96	2.38	3.50	0.37
GC-1-1C@4	42.04	1.59	1639.33	24.96	155.69	3.04	4.66	0.26
GC-1-1C@5	90.16	1.58	1073.84	24.58	18.76	3.09	4.09	0.40
GC-1-1C@6	119.36	1.59	1088.49	22.91	533.08	2.77	3.28	0.36
GC-1-1C@7	73.97	1.58	2328.20	25.56	1056.99	2.60	2.51	0.47
GC-1-1C@8	53.62	2.09	1302.87	31.54	457.47	2.75	1.51	0.65
GC-1-1C@9	71.78	1.93	958.01	29.71	27.91	3.63	0.81	0.48
GC-1-1C@10	19.47	3.78	1122.95	50.46	367.49	0.23	1.92	0.89
GC-1-2A@1	78.69	2.38	902.49	32.25	136.14	3.50	3.68	0.64
GC-1-2A@2	112.07	1.81	1169.20	26.29	149.48	2.65	2.43	0.51
GC-1-2A@3	241.05	1.77	2018.86	31.03	150.34	2.79	4.57	0.48
GC-1-2A@4	121.13	2.46	3083.20	41.27	33.05	5.48	1.51	0.68
GC-1-2A@5	95.25	1.72	1098.64	26.58	18.91	2.29	4.30	0.38
GC-1-2A@6	92.04	1.56	1108.54	24.82	11.43	3.08	4.37	0.35
GC-1-2A@8	89.77	1.68	986.45	24.57	242.40	2.92	3.47	0.37
GC-1-2C@1	91.64	1.67	1002.53	25.38	23.81	2.85	3.83	0.42
GC-1-2C@2	97.38	1.74	984.13	23.52	22.97	3.07	3.33	0.32
GC-1-2C@3	93.63	1.49	1035.85	25.45	26.83	2.80	4.61	0.34
GC-1-2C@4	102.20	1.63	1043.19	20.90	127.98	2.74	4.17	0.33
GC-1-2C@5	143.28	1.62	947.67	27.76	30.34	2.85	1.75	0.34
GC-1-2C@6	99.35	1.53	1057.38	24.93	69.19	2.51	4.73	0.37
GC-1-2C@7	120.24	1.52	1438.62	24.17	285.79	2.69	4.12	0.32
GC-1-2C@8	48.30	1.48	1910.52	23.50	1539.36	2.86	3.14	0.37
GC-1-1B@1	95.47	1.44	1079.32	22.58	14.01	2.48	3.69	0.36
GC-1-1B@2	108.26	1.51	875.86	22.08	14.97	2.40	4.38	0.40
GC-1-1B@3	29.33	1.79	18313.98	29.39	201.95	2.87	3.03	0.30
GC-1-1B@4	30.50	1.68	920.53	24.92	71.12	2.83	2.93	0.35
GC-1-1B@5	75.42	1.80	1122.14	25.73	749.82	2.98	10.46	0.42
GC-1-1B@6	76.39	1.79	1055.18	20.39	1047.46	2.53	0.63	0.40
GC-1-1B@7	115.37	1.41	1822.83	19.51	20.53	2.23	1.06	0.33
GC-1-1B@8	236.74	1.49	1942.72	27.34	21.80	2.70	5.74	0.35
GC-1-1B@9	63.03	1.87	1139.06	24.32	364.16	3.23	1.43	0.47

GC-1-2B@1	72.66	1.78	1047.14	26.78	575.82	2.12	5.52	0.34
GC-1-2B@2	20.93	1.63	1259.04	29.02	53.89	2.81	0.29	0.40
GC-1-2B@3	75.93	2.31	2201.08	36.17	195.50	4.48	2.95	0.60
GC-1-2B@4	181.18	2.29	2828.10	34.94	17.51	4.62	2.88	0.57
GC-1-2B@5	146.38	1.65	1095.28	25.72	310.41	2.85	15.32	0.43
GC-1-2B@6	91.69	1.62	900.22	23.79	21.19	2.59	4.32	0.34
GC-1-3D@1	56.88	1.66	925.22	26.82	286.59	2.46	0.36	0.34
GC-1-3D@2	14.96	1.48	795.81	20.12	37.31	2.43	1.05	0.33
GC-1-3D@3	42.14	1.31	1537.15	17.58	67.78	2.30	0.57	0.33
GC-1-3D@4	602.51	1.57	10870.90	26.49	24.21	2.72	1.77	0.36
GC-1-3D@5	57.19	1.67	786.08	25.47	228.07	2.94	2.07	0.39
GC-1-3D@6	227.28	1.43	2008.79	24.00	55.37	2.57	9.40	0.37
GC-1-3D@7	235.48	1.47	1728.49	24.05	22.77	3.04	8.53	0.38
GC-1-3D@8	151.97	1.48	1750.44	25.78	20.83	2.71	0.69	0.42

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GC-1-2D@1	148.26	1.51	1395.04	22.22	7.39	2.61	0.97	0.32
GC-1-2D@2	87.38	1.50	897.09	20.56	7.06	2.59	0.70	0.34
GC-1-2D@3	99.32	1.72	1075.76	27.80	227.45	2.38	2.89	0.32
GC-1-2D@4	58.74	1.49	1022.75	26.20	313.30	2.71	2.58	0.30
GC-1-3A@1	101.73	1.76	1131.98	24.04	18.98	3.14	3.44	0.48
GC-1-3A@2	99.97	1.78	1706.05	25.50	48.36	3.04	1.30	0.41
GC-1-3A@3	70.79	2.06	2567.71	28.88	116.84	4.30	1.62	0.58
GC-1-3A@4	151.72	1.87	2166.34	32.57	114.08	3.44	7.53	0.45
GC-1-3A@5	95.11	1.59	1148.41	22.60	21.28	2.02	0.64	0.36
GC-1-3A@6	103.99	1.59	1305.16	22.78	5.91	2.55	0.53	0.37
GC-1-3A@7	37.71	1.53	1039.43	25.46	863.97	2.77	1.52	0.24
GC-1-3B@1	72.50	2.11	1121.95	30.45	21.75	3.18	0.71	0.45
GC-1-3B@2	124.90	1.93	14713.77	28.12	29.50	2.79	1.04	0.49
GC-1-3B@3	131.11	1.68	3098.56	46.48	2994.92	2.95	2.60	0.35
GC-1-3B@4	29.74	1.70	2636.99	24.38	55.12	1.99	1.35	0.44
GC-1-3B@5	40.05	1.52	872.30	24.00	574.49	2.68	0.95	0.39
GC-1-3B@6	171.44	1.86	2201.12	30.84	26.77	3.40	2.73	0.50
GC-1-3B@7	29.62	1.77	1525.39	26.96	68.24	3.18	0.50	0.45
GC-1-3B@8	80.11	1.36	1013.63	20.91	6.39	2.50	1.71	0.39
GC-1-3B@9	81.51	1.41	920.52	19.99	27.03	2.19	2.90	0.31
GC-3-1B@1	182.59	1.76	2425.09	24.87	9.21	2.76	0.57	0.39
GC-3-1B@6	101.39	1.70	1634.06	28.33	782.58	2.39	0.40	0.39
GC-3-1B@7	243.63	1.51	1141.44	21.82	90.82	2.17	0.63	0.38
GC-3-1B@8	215.79	1.47	1184.09	19.24	153.14	2.37	1.00	0.32
GC-3-1B@9	162.26	1.45	1118.59	20.17	517.54	2.64	1.40	0.40
GC-3-1B@10	70.44	1.53	1736.31	22.74	919.95	2.51	0.52	0.36
GC-3-2B@1	54.10	1.48	844.78	23.35	917.60	2.05	0.87	0.39
GC-3-2B@2	273.48	1.57	1344.70	18.92	213.26	2.48	0.50	0.33
GC-3-2B@3	274.40	1.51	1473.16	21.53	39.48	2.71	0.35	0.34
GC-3-2B@4	174.42	1.68	1002.88	23.67	48.59	2.21	2.72	0.42
GC-3-2B@5	192.99	1.57	903.58	22.52	70.55	2.34	4.66	0.29

GC-3-2B@6	170.01	1.63	928.19	19.54	38.85	2.82	4.49	0.37
GC-3-2B@7	190.07	1.74	1101.37	18.15	9.68	2.40	0.39	0.41
GC-4-3A@1	138.68	1.88	1123.96	26.18	500.15	3.48	3.81	0.26
GC-4-3A@2	174.70	1.74	953.43	21.77	11.32	2.52	0.36	0.44
GC-4-3A@3	189.48	1.85	896.03	24.58	2.68	2.30	0.39	0.39
GC-4-3A@4	198.29	1.81	925.42	24.94	2.59	2.97	0.39	0.44
GC-4-3A@5	184.39	2.59	1372.74	42.10	537.77	5.03	2.36	0.68
GC-4-3A@6	48.84	2.43	551.81	38.51	167.79	4.18	2.38	0.61
GC-4-3A@7	387.44	2.77	1563.47	36.15	452.39	5.03	4.05	0.64
GC-4-3A@8	341.30	3.29	1795.94	58.73	553.51	6.23	2.43	0.89
GC-5-1@1	502.42	2.66	2083.53	49.50	136.40	5.88	2.08	0.73
GC-5-1@2	32.37	2.35	551.83	32.20	185.39	4.70	0.63	0.50
GC-5-1@3	39.72	2.44	880.81	36.59	429.04	4.17	1.28	0.38
GC-5-1@4	134.89	2.51	703.17	36.91	76.58	5.13	0.22	0.54
GC-5-1@5	171.99	2.62	826.08	39.85	559.73	4.84	0.42	0.51
GC-5-1@6	133.29	2.46	704.76	35.81	73.34	4.47	1.43	0.61
GC-5-1@7	175.14	2.53	907.13	39.93	357.55	4.85	4.20	0.56
GC-5-1@8	132.87	2.80	1852.11	48.52	61.24	5.21	0.57	0.60
GC-5-1@9	193.07	2.36	1277.90	32.54	605.49	5.50	0.37	0.48
GC-5-1@10	141.02	2.79	668.52	39.03	32.72	5.12	1.66	0.39
GC-5-1@11	191.45	3.10	761.44	38.26	904.91	5.14	4.37	0.58

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GC-2-2B@1	185.81	3.08	1008.57	42.42	46.08	5.66	2.17	0.72
GC-2-2B@2	512.50	3.55	2429.91	52.58	500.19	7.50	2.54	1.01
GC-2-2B@3	195.92	2.72	677.28	41.73	437.42	5.53	0.24	0.51
GC-2-2B@4	87.20	3.37	615.90	54.06	365.06	6.45	2.53	0.54
GC-2-2B@5	44.88	2.83	1200.42	43.26	247.92	6.08	0.35	0.72
GC-2-2B@6	166.51	2.67	663.63	43.41	17.86	4.43	0.23	0.66
GC-2-2B@7	116.17	3.41	1452.20	51.14	159.61	5.99	3.18	0.67
GC-2-2B@8	75.52	2.67	916.47	41.07	242.02	4.24	2.15	0.61
GC-2-2D@1	135.67	2.83	1087.74	48.03	115.33	5.19	0.33	0.55
GC-2-2D@2	153.57	2.73	755.00	37.44	146.62	5.09	0.22	0.71
GC-2-2D@3	509.24	3.48	2205.45	58.59	58.07	6.19	0.89	0.79
GC-2-2D@4	441.17	3.51	1725.50	49.67	21.86	7.33	0.98	0.54
GC-2-2D@5	229.94	2.63	817.11	35.91	9.04	5.27	3.69	0.65
GC-2-2D@6	118.48	2.99	1181.29	44.65	15.11	6.13	1.54	0.55
GC-2-2D@7	271.40	3.65	1347.05	59.50	8.62	8.13	0.23	0.80
GC-2-2D@8	160.81	2.39	733.70	33.41	42.90	4.29	0.19	0.50
GC-2-1A@1	41.23	2.95	506.77	50.33	115.01	5.04	1.38	0.52
GC-2-1A@2	44.66	2.96	594.35	41.22	290.08	6.49	2.16	0.43
GC-2-1A@3	3809.33	26.08	3190.14	422.97	5.97	47.82	0.49	4.42
GC-2-1A@6	2085.92	25.95	1285.45	418.27	6.43	42.67	0.56	4.98
GC-2-1A@7	2324.11	32.02	1700.11	423.50	6.97	62.56	0.57	6.08
GC-2-1B@1	3084.49	27.44	2771.70	382.73	5.43	44.85	0.55	4.88
GC-2-1B@2	59.96	2.98	586.70	41.26	25.48	5.88	1.58	0.79
GC-2-1B@3	70.90	3.02	653.07	45.83	54.90	5.40	1.10	0.57

GC-2-1B@4	63.49	3.11	601.84	43.88	139.66	6.73	1.74	0.55
GC-2-1B@5	104.85	3.06	575.55	44.47	6.33	6.20	1.52	0.62
GC-2-1B@6	61.64	2.98	567.82	42.13	5.35	5.08	2.37	0.74
GC-2-1C@1	178.73	2.48	1159.93	37.30	1179.94	5.48	0.96	0.75
GC-2-1C@2	176.18	2.53	921.42	41.65	125.54	4.46	0.83	0.71
GC-2-1C@3	248.99	2.93	14101.10	42.54	38.06	6.32	1.61	0.62
GC-2-1C@4	82.51	2.47	916.06	37.80	411.64	4.99	1.01	0.53
GC-2-1C@5	188.75	2.70	964.33	42.39	154.55	5.12	1.13	0.64
GC-2-1C@6	104.04	2.49	887.37	36.82	460.28	5.17	2.93	0.60
GC-2-1C@7	170.72	2.60	921.12	36.13	785.84	4.96	0.87	0.56
GC-2-3B@1	62.30	2.47	790.26	34.13	27.27	4.91	4.07	0.68
GC-2-3B@2	60.59	2.68	733.35	41.80	16.05	4.74	4.31	0.62
GC-2-3B@3	64.70	2.62	760.21	32.31	65.55	5.06	3.45	0.59
GC-2-3B@4	59.90	2.48	747.03	34.82	21.71	5.37	5.99	0.58
GC-2-3B@5	91.48	2.41	1237.03	38.64	29.92	6.19	6.78	0.63
GC-2-3B@6	100.73	2.66	1217.14	33.09	139.38	5.19	8.19	0.62
GC-2-3B@7	72.00	2.98	756.67	39.42	42.56	5.99	4.06	0.80
GC-2-3B@8	87.59	2.92	800.15	38.34	496.24	4.99	4.41	0.69
GC-2-4@1	48.38	3.05	628.91	41.61	52.66	7.11	2.83	0.80
GC-2-4@2	71.89	3.16	681.89	38.75	84.83	6.83	1.02	0.64
GC-2-4@3	36.57	3.03	669.71	39.95	232.24	7.04	2.51	0.70
GC-2-4@4	48.01	3.13	738.07	52.52	7.94	6.24	2.31	0.77
GC-2-4@5	47.62	3.14	658.31	48.62	13.02	6.67	3.28	0.68
GC-2-4@6	45.02	2.85	660.06	58.18	68.80	6.62	0.27	0.54
GC-2-4@7	43.48	3.79	632.40	61.76	64.84	8.34	0.41	0.85

ror range and detection limits of LA-ICP-MS data (ppm)

Ge72_2S E(Int)	Ge72_LO D_Pettke	Ag107_2 SE(Int)	Ag107_L OD_Pett ke	Cd111_2 SE(Int)	Cd111_L OD_Pett ke	In115_2S E(Int)	In115_L OD_Pett ke	Sn118_2 SE(Int)	Sn118_L OD_Pett ke
0.69	2.16	6.94	0.16	66.55	0.45	6.62	0.01	42.36	0.34
0.65	1.95	2.64	0.02	68.14	0.43	6.63	0.04	13.96	0.37
0.81	1.83	1.71	0.03	103.25	0.72	8.76	0.07	6.28	0.25
0.73	2.00	4.51	0.12	79.24	0.71	4.46	0.01	32.73	0.37
0.65	2.30	1.83	0.02	80.93	0.06	7.63	0.01	16.96	0.34
0.82	1.93	4.76	0.02	69.21	0.90	5.34	0.01	29.62	0.27
0.86	2.42	1.07	0.13	90.66	1.16	2.37	0.01	1.54	0.32
0.65	1.93	4.87	0.02	82.54	0.07	5.30	0.01	2.53	0.33
1.39	2.05	1.74	0.21	128.14	0.19	5.80	0.02	0.83	0.35
1.17	2.70	1.97	0.15	219.13	0.56	8.34	0.01	0.47	0.27
0.67	2.34	1.63	0.11	76.08	0.69	3.60	0.01	30.43	0.26
0.75	2.30	5.74	0.12	87.03	0.62	4.94	0.06	617.35	0.44
0.69	2.15	1.08	0.16	75.89	0.06	5.00	0.04	12.25	0.28
0.69	2.13	7.86	0.02	66.89	0.44	2.12	0.04	104.10	0.37
0.77	2.28	43.31	0.12	69.15	0.07	3.40	0.01	15.23	0.28
0.80	2.95	3.30	0.02	89.03	0.57	3.60	0.05	2.14	0.43
0.81	2.44	0.77	0.14	73.90	0.52	2.77	0.01	0.51	0.37
0.82	5.84	40.47	0.04	105.00	0.16	5.10	0.01	3.78	0.52
0.75	3.24	1.52	0.02	69.11	0.07	4.80	0.01	9.09	0.51
0.61	2.20	1.38	0.02	67.54	0.48	2.22	0.01	226.61	0.34
1.04	2.34	5.00	0.24	177.07	0.61	5.15	0.02	129.23	0.52
1.87	3.74	47.49	0.10	205.29	1.36	11.34	0.03	6.17	0.82
0.68	2.12	1.41	0.02	72.15	0.63	6.67	0.04	5.78	0.45
0.72	2.30	1.22	0.02	69.85	0.73	4.54	0.01	1.54	0.33
0.62	2.31	2.36	0.17	66.32	0.07	8.12	0.05	48.57	0.28
0.66	2.01	1.12	0.02	70.29	0.47	3.95	0.05	11.20	0.30
0.66	2.21	1.45	0.13	68.06	0.67	6.13	0.07	8.19	0.40
0.64	1.79	1.61	0.02	64.38	0.43	8.48	0.01	20.00	0.26
0.68	2.27	2.14	0.12	62.35	0.07	3.50	0.01	8.37	0.36
0.70	2.09	1.24	0.02	64.33	0.72	8.76	0.01	2.09	0.28
0.62	1.98	2.66	0.02	69.75	0.72	3.56	0.04	3.50	0.30
0.91	1.98	3.19	0.04	95.42	0.50	9.40	0.01	7.98	0.32
0.86	1.85	18.00	0.03	184.79	0.47	4.08	0.05	4.68	0.32
0.67	1.68	0.99	0.11	67.57	0.42	1.35	0.01	7.97	0.30
0.65	1.69	1.31	0.02	54.36	0.69	8.25	0.05	3.20	0.34
1.24	1.74	6.99	0.05	111.54	0.59	6.28	0.01	13.53	0.34
0.67	2.21	2.03	0.02	80.82	0.66	3.52	0.01	3.98	0.35
0.72	2.07	7.15	0.02	64.38	0.07	3.19	0.06	210.15	0.39
0.71	2.26	5.29	0.14	65.27	0.08	3.71	0.01	0.78	0.29
1.00	1.68	1.05	0.19	129.12	0.72	3.90	0.04	1.53	0.27
1.05	1.50	1.35	0.05	141.37	0.77	4.85	0.04	11.85	0.36
0.67	2.03	2.68	0.21	70.58	1.01	3.60	0.01	31.50	0.23

0.74	2.22	6.66	0.02	70.61	0.50	6.51	0.01	68.94	0.30
0.64	1.87	1.25	0.02	64.27	0.63	3.35	0.04	1.95	0.30
1.05	3.56	2.25	0.10	399.16	0.39	10.08	0.03	19.47	0.60
1.82	2.87	1.55	0.12	347.85	1.18	7.65	0.08	2.36	0.53
0.67	1.90	1.95	0.13	68.00	0.47	8.22	0.01	32.88	0.24
0.65	2.01	1.21	0.17	65.97	0.62	5.83	0.01	10.21	0.38
0.79	1.84	2.27	0.20	65.11	0.07	3.73	0.01	0.92	0.35
0.50	1.58	37.24	0.11	83.55	0.06	3.73	0.00	4.32	0.30
0.82	1.68	1.77	0.12	122.40	0.13	4.59	0.05	0.97	0.36
1.07	1.74	0.78	0.13	168.45	0.11	5.35	0.01	13.53	0.28
0.58	2.09	2.50	0.13	67.85	0.50	3.63	0.01	1.04	0.41
0.84	1.87	1.93	0.04	104.17	0.49	8.18	0.04	34.87	0.33
0.94	1.81	1.71	0.04	110.39	0.63	9.56	0.04	4.17	0.26
0.84	1.76	1.65	0.05	146.73	0.67	4.21	0.01	4.56	0.26

1.18	1.87	0.91	0.13	120.12	0.49	4.65	0.04	0.84	0.32
0.63	1.93	0.62	0.02	71.60	0.41	3.32	0.01	0.55	0.36
0.70	2.01	4.19	0.02	74.27	0.47	2.83	0.04	221.44	0.39
0.68	1.81	1.99	0.12	72.30	0.46	3.32	0.01	5.74	0.32
0.71	1.99	1.11	0.03	82.10	0.67	2.97	0.01	14.65	0.38
0.97	2.62	1.37	0.05	182.22	0.88	4.42	0.05	0.50	0.41
1.32	2.68	1.42	0.07	185.42	0.28	6.34	0.02	3.60	0.57
1.06	2.43	2.07	0.17	120.07	0.65	6.26	0.02	15.83	0.43
0.75	1.88	0.76	0.02	76.40	0.91	3.11	0.04	0.42	0.38
0.79	2.09	0.50	0.03	77.89	0.62	2.72	0.05	0.45	0.27
0.65	1.71	2.89	0.02	68.49	0.69	2.83	0.01	4.26	0.34
0.96	2.43	1.34	0.02	87.13	1.19	3.49	0.05	2.91	0.37
0.98	2.08	1.59	0.23	83.74	0.53	3.68	0.01	0.59	0.36
0.77	1.83	55.02	0.04	99.41	0.52	4.25	0.01	74.83	0.32
0.86	2.06	1.69	0.14	173.90	0.12	4.14	0.01	24.02	0.30
0.69	1.86	2.44	0.12	133.39	0.61	3.58	0.01	1.79	0.24
1.74	2.37	1.58	0.08	153.31	0.94	9.33	0.06	3.12	0.39
0.88	1.84	3.42	0.06	188.43	0.64	5.82	0.02	1.10	0.33
0.64	1.34	0.55	0.02	66.29	0.41	2.68	0.01	0.39	0.27
0.51	1.76	1.12	0.15	61.33	0.56	2.69	0.00	18.72	0.26
1.12	2.10	0.64	0.07	136.61	0.66	6.96	0.02	0.39	0.38
0.85	2.29	14.10	0.05	109.05	0.87	1.40	0.01	1.19	0.31
0.80	1.87	1.97	0.12	52.95	0.58	0.83	0.01	0.17	0.30
0.66	1.80	4.79	0.12	53.88	0.69	0.65	0.01	25.26	0.33
0.66	2.01	92.36	0.03	68.15	0.44	0.84	0.04	40.52	0.30
0.56	1.80	173.33	0.02	75.57	0.58	0.81	0.01	19.53	0.29
0.47	1.79	23.15	0.18	46.73	0.41	0.62	0.00	5.68	0.27
0.69	1.75	2.94	0.03	112.73	0.71	0.98	0.04	0.27	0.33
0.67	1.86	0.63	0.12	65.07	0.92	0.87	0.01	0.22	0.28
0.63	1.72	2.33	0.12	40.80	0.74	0.36	0.01	56.07	0.30
0.64	1.58	3.39	0.16	42.11	0.44	0.55	0.01	9.58	0.20

0.63	1.93	6.67	0.21	44.61	0.47	0.28	0.01	19.28	0.30
0.66	1.87	0.73	0.13	46.02	0.07	0.73	0.01	0.28	0.40
0.72	2.21	13.22	0.19	49.35	0.08	1.07	0.01	157.74	0.43
0.58	1.91	0.33	0.02	47.36	0.66	0.67	0.04	0.19	0.29
0.63	2.37	0.32	0.02	46.80	0.79	0.68	0.04	0.19	0.33
0.59	2.09	0.24	0.02	44.23	0.79	0.62	0.01	0.22	0.29
0.56	3.14	66.86	0.44	40.54	1.50	0.58	0.01	8.36	0.56
0.51	3.28	134.41	0.03	49.09	0.68	0.70	0.01	104.09	0.56
0.83	3.33	12.34	0.07	111.55	0.99	1.16	0.08	1.53	0.73
0.89	4.28	10.40	0.09	94.79	0.34	3.74	0.03	285.19	0.72
0.87	3.45	5.02	0.11	122.83	0.40	5.10	0.08	68.10	0.85
0.51	2.66	8.39	0.04	54.60	0.70	0.97	0.01	1.93	0.57
0.44	2.13	682.17	0.03	44.61	0.10	0.96	0.01	78.21	0.67
0.59	3.36	1.86	0.25	47.78	0.70	0.66	0.01	0.20	0.59
0.55	2.88	9.95	0.27	81.51	0.16	0.87	0.01	0.54	0.59
0.59	3.12	5.47	0.03	51.34	0.69	3.05	0.01	19.24	0.46
0.67	2.79	6.07	0.21	94.91	0.78	1.21	0.02	98.32	0.70
0.86	3.62	1.99	0.10	148.39	1.01	1.40	0.03	1.99	0.76
0.78	2.95	13.05	0.06	74.63	0.21	0.90	0.02	0.27	0.51
0.59	3.37	1.00	0.03	40.89	0.75	0.54	0.01	0.54	0.55
0.52	3.77	631.47	0.22	43.57	1.09	1.80	0.01	396.33	0.77

0.73	3.36	1.73	0.24	124.22	0.89	1.51	0.02	0.38	0.85
1.88	4.64	14.23	0.17	151.65	1.48	3.13	0.05	11.43	1.00
0.56	3.39	15.45	0.03	33.91	1.22	0.66	0.06	0.16	0.58
0.54	4.48	21.63	0.03	38.90	1.57	0.68	0.01	123.27	0.54
0.53	3.48	5.75	0.22	32.34	1.08	0.61	0.06	0.92	0.51
0.60	3.54	0.95	0.21	42.70	0.13	0.63	0.06	0.25	0.42
0.85	4.51	6.40	0.12	70.03	1.56	1.30	0.03	1.98	0.92
0.68	3.47	8.12	0.31	101.66	1.16	1.21	0.09	2.57	0.67
0.71	3.75	5.03	0.24	113.77	0.88	1.07	0.02	0.32	0.64
0.51	2.49	5.74	0.21	39.85	1.05	0.68	0.01	0.14	0.76
1.20	4.20	2.52	0.37	176.80	1.37	1.59	0.11	0.30	0.69
0.98	4.25	0.87	0.16	121.71	1.39	1.70	0.05	0.33	0.77
0.73	3.14	1.13	0.07	91.64	1.30	0.70	0.09	0.24	0.54
0.51	3.13	0.52	0.34	95.69	1.05	0.75	0.01	6.10	0.58
0.84	4.60	0.59	0.27	46.18	1.35	0.97	0.01	0.53	0.63
0.47	2.96	1.68	0.03	48.39	1.10	0.65	0.01	0.17	0.40
0.42	3.44	3.60	0.03	43.31	0.11	0.66	0.08	0.75	0.87
0.52	4.08	1.41	0.03	43.71	1.55	2.23	0.01	68.33	0.62
4.19	31.44	0.00	0.28	0.02	6.91	0.36	0.08	1.70	5.53
3.51	26.20	0.00	0.28	0.07	12.98	0.45	0.57	1.84	4.18
4.40	39.57	0.00	0.28	0.03	1.06	0.42	0.85	3.75	5.54
4.29	26.97	0.01	1.95	0.02	7.22	0.61	0.59	1.63	4.85
0.64	3.29	0.48	0.03	39.16	1.13	1.87	0.07	0.23	0.74
0.57	3.51	0.28	0.03	42.91	0.12	1.87	0.01	0.42	0.53

0.56	3.89	1.18	0.03	41.17	0.86	2.13	0.07	0.91	0.54
0.53	3.74	0.37	0.03	41.46	1.18	2.19	0.10	0.31	0.48
0.58	4.11	0.34	0.03	37.70	1.16	1.41	0.07	0.83	0.60
0.51	3.17	4.97	0.03	47.15	0.73	1.93	0.01	2.72	0.65
0.54	3.53	1.32	0.03	48.28	0.71	2.32	0.06	0.29	0.50
0.84	3.47	2.05	0.05	152.63	0.87	6.64	0.01	0.53	0.53
0.59	3.18	2.42	0.19	61.08	0.72	2.52	0.01	0.51	0.50
0.47	3.01	1.37	0.03	58.20	0.96	2.39	0.06	0.33	0.50
0.53	3.15	2.96	0.03	61.52	0.96	2.39	0.01	1.10	0.48
0.57	3.90	4.63	0.03	52.61	0.74	2.86	0.06	0.81	0.59
0.52	2.97	1.21	0.03	55.87	1.43	1.49	0.01	18.34	0.53
0.53	3.01	0.81	0.03	54.17	0.70	1.69	0.01	2.06	0.56
0.59	2.97	2.72	0.03	55.91	0.97	3.07	0.01	57.28	0.57
0.54	3.14	0.95	0.03	52.59	1.15	1.85	0.06	11.53	0.63
0.92	2.94	2.03	0.21	93.97	0.19	4.99	0.02	16.50	0.57
0.82	2.77	4.42	0.05	101.01	0.82	2.92	0.02	24.53	0.64
0.65	3.75	1.29	0.03	66.01	1.13	2.52	0.07	27.48	0.67
0.59	3.32	13.91	0.03	69.68	1.32	2.81	0.01	446.53	0.45
0.61	4.00	1.23	0.04	53.91	1.63	2.45	0.07	0.62	0.73
0.57	4.13	5.78	0.24	67.32	1.19	3.06	0.01	1.04	0.52
0.65	4.20	36.95	0.24	51.97	1.20	3.34	0.01	7.82	0.49
0.66	4.10	0.75	0.04	50.89	1.20	2.87	0.07	0.52	0.62
0.60	4.58	1.05	0.24	62.23	0.13	2.47	0.01	2.95	0.71
0.65	4.22	2.01	0.23	47.18	1.17	2.64	0.07	0.36	0.55
0.72	4.29	1.00	0.04	55.10	0.14	2.34	0.01	0.41	0.65

Hg202_2 SE(Int)	Hg202_L OD_Pett Lo	Pb208_2 SE(Int)	Pb208_L OD_Pett Lo
0.44	0.19	3.42	0.00
0.35	0.17	0.05	0.00
0.36	0.17	0.12	0.01
0.32	0.16	2.51	0.01
0.34	0.17	0.04	0.00
0.35	0.17	0.05	0.00
0.37	0.18	0.56	0.02
0.36	0.16	1.29	0.01
0.39	0.24	2.42	0.01
1.08	0.15	21.82	0.02
0.36	0.18	0.36	0.00
0.37	0.18	47.08	0.03
0.32	0.17	0.08	0.00
0.33	0.17	0.14	0.00
0.44	0.17	34.59	0.03
0.45	0.23	5.19	0.03
0.43	0.21	3.60	0.00
0.40	0.37	8525.54	0.37
0.40	0.25	0.95	0.15
0.36	0.15	0.10	0.08
0.62	0.23	35.85	0.05
3.58	0.28	4663.08	0.09
0.35	0.16	0.09	0.04
0.34	0.16	0.08	0.04
0.33	0.18	0.27	0.04
0.32	0.18	0.18	0.05
0.31	0.16	0.07	0.04
0.29	0.16	0.07	0.02
0.30	0.15	0.07	0.03
0.30	0.15	0.88	0.01
0.28	0.13	0.12	0.03
0.31	0.19	0.15	0.03
0.82	0.16	161.92	0.03
0.27	0.14	0.19	0.03
0.22	0.18	0.08	0.03
0.35	0.17	36.69	0.02
0.43	0.14	0.95	0.03
0.35	0.19	10.37	0.03
0.27	0.18	0.30	0.00
0.29	0.17	0.20	0.02
0.45	0.16	0.46	0.03
0.30	0.20	2611.36	0.04

0.31	0.17	0.14	0.04
0.31	0.16	7.56	0.02
3.93	0.26	340.59	0.05
1.65	0.25	376.70	0.04
0.28	0.15	0.12	0.03
0.31	0.16	0.13	0.02
0.35	0.15	0.55	0.02
0.28	0.14	17914.69	0.09
0.35	0.16	5.07	0.08
0.87	0.16	0.09	0.07
0.28	0.18	0.07	0.04
0.31	0.17	0.16	0.05
0.33	0.17	0.13	0.05
0.31	0.17	0.58	0.03

0.36	0.17	0.39	0.04
0.25	0.16	1.11	0.03
0.28	0.15	0.10	0.02
0.32	0.18	1.00	0.02
0.32	0.20	0.09	0.04
0.67	0.20	28.71	0.02
0.96	0.24	52.12	0.04
0.38	0.23	0.40	0.03
0.29	0.17	0.06	0.02
0.33	0.15	0.15	0.02
0.23	0.17	1.00	0.02
1.35	0.23	1.40	0.04
0.28	0.18	6.35	0.03
0.28	0.19	2.14	0.03
0.72	0.19	1.06	0.03
0.43	0.17	3.85	0.02
0.44	0.22	5.01	0.03
1.79	0.24	345.13	0.03
0.22	0.15	0.06	0.01
0.22	0.15	0.12	0.02
0.31	0.18	0.73	0.22
0.33	0.21	17.46	0.17
0.30	0.15	1.72	0.11
0.30	0.19	0.57	0.06
0.30	0.15	0.99	0.05
0.31	0.18	4942.36	0.08
0.27	0.15	498.40	0.07
0.41	0.17	1.89	0.05
0.30	0.17	1.74	0.05
0.23	0.16	0.05	0.05
0.23	0.17	0.09	0.06

0.29	0.19	0.05	0.05
0.26	0.14	0.03	0.05
0.31	0.21	0.15	0.04
0.26	0.18	0.04	0.05
0.29	0.19	0.03	0.04
0.29	0.17	0.03	0.03
0.22	0.34	770.36	0.05
0.30	0.28	1123.78	0.07
0.28	0.35	2.30	0.06
0.77	0.43	16.11	0.04
0.81	0.33	16.76	0.03
0.35	0.31	1178.24	0.04
0.31	0.27	552.11	0.04
0.37	0.33	0.45	0.02
0.41	0.31	1.59	0.03
0.34	0.26	0.32	0.04
0.61	0.32	4.57	0.03
2.14	0.35	48.65	0.05
0.26	0.26	2.10	0.03
0.26	0.30	0.04	0.03
0.25	0.34	0.43	0.00

0.57	0.36	4.69	0.04
1.29	0.31	11.63	0.05
0.21	0.29	0.04	0.03
0.20	0.29	0.13	0.00
0.70	0.34	3.38	0.04
0.25	0.30	0.90	0.03
2.27	0.43	145.98	0.04
0.47	0.32	1.81	0.03
0.86	0.32	0.64	0.04
0.21	0.33	0.05	0.00
1.15	0.43	11.74	0.06
1.46	0.35	19.43	0.05
0.39	0.30	1.99	0.04
1.27	0.34	4.83	0.03
1.53	0.43	1.56	0.04
0.19	0.26	0.20	0.03
0.29	0.34	2053.81	0.06
0.23	0.33	0.25	0.04
27.77	2.42	0.13	0.30
26.61	2.33	0.12	0.37
27.14	2.78	0.43	0.03
24.86	2.07	0.10	0.03
0.23	0.31	0.07	0.03
0.24	0.32	0.08	0.05

0.27	0.36	0.51	0.06
0.25	0.32	0.02	0.04
0.24	0.29	0.12	0.04
0.22	0.28	1672.91	0.05
0.24	0.29	0.31	0.00
0.22	0.33	4.43	0.03
0.28	0.27	0.14	0.00
0.22	0.26	0.37	0.00
0.23	0.24	0.83	0.04
0.21	0.27	0.22	0.00
0.19	0.26	0.10	0.03
0.21	0.25	0.05	0.02
0.21	0.27	0.03	0.02
0.23	0.25	0.06	0.00
0.27	0.26	3.23	0.01
0.24	0.25	9.55	0.03
0.24	0.28	0.11	0.04
0.22	0.26	0.85	0.03
0.26	0.32	0.19	0.03
0.29	0.33	0.97	0.00
0.26	0.34	277.91	0.00
0.24	0.32	0.03	0.00
0.24	0.33	0.04	0.03
0.21	0.33	0.01	0.04
0.26	0.39	0.03	0.04